



EHBEA 2022

EUROPEAN HUMAN BEHAVIOUR & EVOLUTION ASSOCIATION



MAX PLANCK INSTITUTE
FOR EVOLUTIONARY ANTHROPOLOGY



Birth
Rites

Cultures of
reproduction



CAMBRIDGE
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PROGRAMME: 19th – 22nd April 2022

All times are in Central European Summer Time (CEST)

Student presentations are marked with *

Day 0 – Tue 19 April

	Start	End	Speaker	Location	Details
Conference opening	13:30	14:00	Colleran; McElreath	Zoom A	
Plenary 1: Jenny Tung	14:00	15:00	Tung	Zoom A	The long arm of early life: biological embedding in wild baboons
Break	15:00	15:20		Gathertown	
Single 1	15:20		Lidborg*	Zoom A	Do more masculine men have better quality partners?
	15:40		Skirgård		Does political complexity predict the number of languages in Remote Oceania?
	16:00		Foster*		Systematic review and meta-analysis of the effects of predictors of cooperation on loan repayment in microfinance groups
	16:20		Jones		The Dark Secret of Our Success
Break	16:40	17:00		Gathertown	
Posters 1	17:00	19:00	Red: Basava*; Duffy*; Galasinska-Grygorczuk*; Jacobs; McAleavey*; Pal*; Ready; Siekiera*; Varas Enríquez*; Orange: Boon-Falleur*; Farkas*; Glynatsi; Lee; Mokos*; Pawlowski; Rossetti*; Stucky*; Vitzthum; Yellow: Chen*; Fišerová*; Goodman*; Lie-Panis*; Motes Rodrigo; Pettay; Teramoto*; Wakes*; Lime: Costello*; Frankowska; Hicks; Malaei*; Onohara; Power; Schleihau; Trzeciakowska*; Youngblood	Gathertown	See Conference booklet for titles
Event: Early career mixers	19:00	21:00		Gathertown	PhD student event coordinated by Jasmine Calladine. Postdoc event coordinated by Elena Miu.

Day 1 – Wed 20 April

	Start	End	Speaker	Location	Details
Plenary 2: Tim Lewens	09:00	10:00	Lewens	Zoom A	The attractions of cultural selection (or: where Paris meets California)
Single 2	10:00		Blake	Zoom A	Incel activity on social media linked to local mating ecologies
	10:20		Rotella		Increasing population densities predict decreasing fertility rates over time: A 174-nation investigation
	10:40		Turecek		A minimal model of subculture emergence
	11:00		Kandza*		Insights into inter-group cooperation and inequality from shotgun hunting among the BaYaka and Yambe in Republic of the Congo
Break	11:20	11:40		Gathertown	
Parallel 1A 'Mating I'	11:40		Arantes	Zoom A	"Time Slows Down Whenever You Are Around" for Women but Not for Men
	12:00		Arslan		Peri-ovulatory changes in sexual behavior and desire — what can different methods tell us?
	12:20		Chegeni		Some for a good time, some for a long time: Heterosexual women's ratings of the mate value of men with different addictions: An experimental study
Parallel 1B 'Cross cultural'	11:40		Chira	Zoom B	A quantitative test of Jared Diamond's axis of orientation hypothesis
	12:00		Garfield		Norm violations and punishments across human societies
	12:20		Lightner		Supernatural theories of disease, useful falsehoods, and a market for religious specialists
Lunch	12:40	13:40		Gathertown	
Single 3	13:40		Haux*	Zoom A	Chimpanzee and human risk preferences show key similarities
	14:00		Fedorova*		Ger or house: Drivers of household shelter choices and investments in Ulaanbaatar, Mongolia
	14:20		Boothroyd		Agent-based models of the cultural evolution of occupational gender roles
	14:40		Seabright		The burdens of leadership: a game theoretical and empirical study
Break	15:00	15:20		Gathertown	
Parallel 2A 'Marriage'	15:20		Minocher*	Zoom A	Sexual conflict and spousal age gaps: a conceptual replication of Lawson et al. (2020) using data from network-structured games in four communities in rural Colombia
	15:40		Prall		Mating market dynamics on partner preference and relationship quality among Namibian agro-pastoralists
	16:00		Scelza		The impact of extra-pair partnerships on men's reproductive success
Parallel 2B 'Risk'	15:20		Heisßler*	Zoom B	Dynamics of risk regulates cooperation in an incomplete-information game with bandit payoff feedback
	15:40		van Leeuwen		Pathogen avoidance and conformity: Salient infectious disease does not increase conformity
	16:00		Zwirner		Interpersonal trust and prosociality in volcanic environments
Break	16:20	16:30		Gathertown	
Plenary 3: Nicole Creanza	16:30	17:30	Creanza	Zoom A	Theoretical and empirical approaches to understanding cultural change
Break	17:30	17:45		Gathertown	
Posters 2	17:45	19:30	Forest: Arini; Dubourg*; Hilbe; Mikac; Pan*; Richards; Sheng*; Thiebaut; Young; Blue: Baumann; Ekström*; Ge*; Kundtová Klocová; Molho; Perez*; Ritov*; Słojewska*; Tubek-Krokosz*; Żyrek*; Navy: Brinkmann; Golden Eddy*; Leslie*; Müller; Pietraszewski; Rostovtseva*; Street; Visser*; Purple: Ciochoń*; Fitouchi*; Hammerl; Mace; Pretelli*; Scaff; Szala; Windhager; Zetzsche*	Gathertown	See Conference Booklet for titles
Event: Meet the editors	19:30	21:00		Zoom A	Increasing your publishing success: a Q&A with Journal Editors, moderated by Helen Eaton, Commissioning Editor for <i>Philosophical Transactions B</i>

Day 2 – Thu 21 April

	Start	End	Speaker	Location	Details
Plenary 4: Nichola Raihani	09:00	10:00	Raihani	Zoom A	Punishment – motives and outcomes
Single 4	10:00		Passmore	Zoom A	Global relationships between musical, linguistic, and genetic diversity
	10:20		Jang		Cooperative foraging networks among BaYaka women in the Republic of Congo
	10:40		Gibson		Abandoning FGM/C increases the cost of marriage in rural Ethiopia
	11:00		Engelbert*		The effect of leaders' charismatic signaling on information processing outcomes: Combining cognitive science and evolutionary leadership theory
Break	11:20	11:40		Gathertown	
Parallel 3A 'Social learning'	11:40		Ammar*	Zoom A	Memory in cultural evolution: Implications for social learning in fluctuating environments
	12:00		Faessler*		Using successful people as examples of how to behave and how not to behave
	12:20		Schimmelpfennig*		Paradox of diversity in the collective brain
Parallel 3B 'Family & fertility'	11:40		Brandl	Zoom B	Fostering, adoption, and parental absence in Vanuatu
	12:00		Sheppard		Identifying the barriers to reproduction in the UK: a qualitative study.
	12:20		Myers		The presence and role of kin in maternal social networks in the UK – evidence from the COVID-19 pandemic
Lunch	12:40	13:40		Gathertown	
Single 5	13:40		Roberts	Zoom A	CHIELD: A database of causal hypotheses for unifying theories of cultural evolution
	14:00		Martin*		Social and ecological effects on the evolvability of human fertility
	14:20		Willard		Testing the functions of supernatural beliefs in 1930s Irish folk medicine
	14:40		Alami*		What motivates intermarriage in small-scale societies, and what consequences for social cohesion?
Break	15:00	15:20		Gathertown	
Speedtalk 1	15:20		Boyette	Zoom A	Teaching strategies shaped by experience with formal education: experimental evidence from caregiver-child dyads in two Tannese communities
	15:26		Jaeger		Cheater detection at a glance? Testing the accuracy of trustworthiness detection from faces
	15:32		Miu		Female friendship and the horizontal transmission of low fertility values
	15:38		Šeja		Macro-innovations create room for micro-innovations: The discovery process in video game speedruns
	15:44		Shilton*		Group singing is globally dominant, but solo singing reflects social complexity
	15:50		Singh		Introducing Anansi, a global story database
	15:56		Helle		Maternal grandmothers act as buffer against emotional and behavioral problems in children with many adverse early life experiences
	16:02		Frankenhuis		What is the expected human childhood? Insights from evolutionary anthropology
	16:08		Hagen*		Understanding women's support for patriarchal norms in East Asia
	16:14		Kilgallen*		Norm internalization or mismeasurement: Are women truly more accepting of intimate partner violence than men?
Break	16:20	16:30		Gathertown	
Plenary 5: Daniel Fessler	16:30	17:30	Fessler	Zoom A	Negatively-biased beliefs and the 21st century information environment: intersections of evolved psychological mechanisms, cultural evolution, and individual difference
Break	17:30	17:45		Gathertown	
Event: Meet the experts	17:45	19:30+		Gathertown	
Event: Early-career / general mixer	19:30	21:00		Gathertown	PhD student event coordinated by Jasmine Calladine. Postdoc event coordinated by Elena Miu.

Day 3 – Fri 22 April

	Start	End	Speaker	Location	Details
New Investigator: Abigail Page	09:00	10:00	Page	Zoom A	Beyond the mother: interdisciplinary approaches to cooperative childrearing
Single 6	10:00		Hillemann	Zoom A	Socioeconomic correlates of Inuit hunting strategies
	10:20		Stevens*		Does haemoglobin modulate the trade-off between current and future reproductive investment in the Bolivian altiplano?
	10:40		Godoy		Disentangling genetic and non-genetic influences on sociality; results and challenges from a longitudinal study on wild capuchins
	11:00		Jaeggi		Life-history tradeoffs in a historical population (1896-1939) undergoing rapid fertility decline: Costs of reproduction?
Break	11:20	11:40		Gathertown	
Parallel 4A 'Game theory'	11:40		Couto*	Zoom A	Introspection dynamics: A simple model of counterfactual learning in asymmetric games
	12:00		Kleshnina		Optimal sharing in social dilemmas
	12:20		Micheletti		Sibling competition and dispersal drive sex differences in religious celibacy
Parallel 4B 'Mating II'	11:40		Botzet	Zoom B	Analyzing psychological effects of hormonal contraception: Causal effects, selection effects, attrition effects, or reverse causality?
	12:00		Stern		Do women's intrasexual social comparison and jealousy change across the ovulatory cycle?
Lunch	12:40	13:40		Gathertown	
Speedtalk 2	13:40		Iredale	Zoom A	The watching eyes effect: Keep Britain tidy "We're watching you"
	13:46		Redhead		The effects of prestige and dominance on advice and friendship relations: A longitudinal study of collaborative teams
	13:52		Kretschmer*		Physical activity and energetic trade-offs in the BaYaka hunter-gatherers
	13:58		Nila		Cultural differences in empathy in Indonesia
	14:04		Heggarty		Inferring cultural evolution from language trees? Cautionary tales from the case of Indo-European
	14:10		Nettle		The folk concept of 'culture': What do people mean when they explain something with reference to culture?
	14:16		Vermeent*		Attention styles in harsh and unpredictable environments
	14:22		Romani*		The art of nesting: ageing nests of Eastern chimpanzee at Bugoma Central Forest Reserve, Uganda
	14:28		Andrews		Measuring market integration and its effects on the life course
	14:34		Gordon		The spiteful origins of conspiratorial beliefs
Break	14:40	15:00		Gathertown	
Single 7	15:00		von Rueden	Zoom A	Patrons and Managers: the rise of political inequality in an Amazonian society.
	15:20		Walasek*		Sensitive periods, but not critical periods, evolve in a fluctuating environment: A model of incremental development
	15:40		Chen*		What other men want: the accuracy of gender norm perception among Tanzanian men.
Break	16:00	16:30		Gathertown	
Plenary 7: Hillard Kaplan	16:30	17:30	Kaplan	Zoom A	The "embarrassment of riches" theory of brain aging with hypothesis tests in two Native South American populations
Closing remarks & Prizes	17:30	17:45		Zoom A	
EHBEA Annual General Meeting	17:45	18:30		Zoom A	

Notes:

All talks are 15 mins + 5 mins Q&A
Speed talks are 5 mins + 1 min Q&A
Plenaries are 45 mins +15 min Q&A

ABSTRACTS

Plenaries

Professor Jenny Tung

Departments of Evolutionary Anthropology and Biology, Duke University & Max Planck Institute for Evolutionary Anthropology

14:00 Tuesday 19.04.2022

The long arm of early life: biological embedding in wild baboons

Environmental adversity is a key predictor of Darwinian fitness in humans and other primates. In some cases, these effects are immediate. In other cases, exposure to adversity can be temporally distinct from its outcomes, creating lagged associations between environmental experience and trait variation that can stretch across years. These observations are likely to be explained, at least in part, by the process of “biological embedding:” differences in life experience that stably affect biological state, which in turn influences trait variation over the life course. Here, I discuss how our long-term research on wild baboons in Kenya has shed light on this phenomenon. By following individually recognized animals from cradle to grave, we show that the relationship between early adversity and mortality risk is strikingly parallel between baboons and humans. By coupling longitudinal observations with genomic data, we also reveal evidence for biological embedding via stable changes in DNA methylation. Our findings thus uncover a molecular signature of early life adversity—especially closely linked to resource deprivation—that helps clarify conflicting evidence from human populations. At the same time, they underscore the need for caution in interpreting the adaptive significance of early life effects, and argue for a higher evidentiary bar for connecting environment to phenotype via epigenetic mechanisms.

Professor Tim Lewens

Department of History and Philosophy of Science, University of Cambridge

09:00 Wednesday 20.04.2022

The attractions of cultural selection (or: where Paris meets California)

It is hard to resist the idea that a form of natural selection is at work in the domain of culture. After all—as theorists have observed ever since Darwin—culture is an area where there is abundant variation, and differential success with respect to the reproduction of these varying cultural items. So, an informal case for cultural selection is irresistible. However, matters become complicated when one tries to think in disciplined ways about cultural selection, and how it produces cultural adaptation. Here I focus on three themes: first, there is a continuing and confusing proliferation of very different accounts of what cultural selection is. Second, there are reasons to be sceptical of some (but not all!) recent efforts to use the Price Equation to rigorously formalise cultural selection. Third, I suggest an important ambiguity in the way cultural selection is thought to explain cultural adaptation. In all three cases it is possible to sharpen our understanding of cultural selection’s scope and limits by noting its complementary relationships to the processes of cultural attraction. This undermines a common (but not universal!) tendency to cast cultural selectionism and cultural attractor theory as conflicting stances on cultural evolution.

Professor Nicole Creanza

Department of Biological Sciences, Vanderbilt University

16:30 Wednesday 20.04.2022

Theoretical and empirical approaches to understanding cultural change

Human culture is broadly characterized by its diversity; different populations have different artifacts and technologies, and these unique toolkits can also differ in size and complexity. Over the past few decades, researchers have collected data and constructed theoretical models that both support and refute the hypothesized connection between population size and cultural accumulation, prompting us to re-examine the question: should we expect larger populations to have larger cultural repertoires? Using both models of cultural evolution and empirical analyses, I will discuss multiple factors that might obscure the putative relationship between population size and cultural accumulation in humans, including cultural niche construction, inter-population connectivity, and cultural specialization. Finally, I will discuss whether the debated link between population size and cultural complexity can spur new insights into cultural evolution in non-human systems as well, using learned birdsong as a focal example.

Professor Nichola Raihani

Department of Psychology, University College London

09:00 Thursday 21.04.2022

Punishment – motives and outcomes

Humans are outstanding in their ability to cooperate with unrelated individuals, and punishment – paying a cost to harm others – is thought to be a key supporting mechanism. According to this view, cooperators punish defectors, who respond by behaving more cooperatively in future interactions. However, evidence from laboratory and real-world settings casts serious doubts on the assumption that the sole function of punishment is to convert cheating individuals into cooperators. In this talk, I will discuss empirical work that has explored the motives

underpinning punishment decisions, highlighting the important contributions of inequity aversion and other competitive motives in prompting punishment decisions. I will also talk about the consequences of punishment, both in terms of the targets' responses and how punishers are viewed by others based on their investments. Finally, I discuss how punishment in the lab differs from punishment we observe in the real world, focussing on gossip and centralised punishment institutions. I end by highlighting the need to consider both legitimacy and rehabilitation if punishers are to safely wield an effective, yet potentially explosive, cooperation-enforcing tool.

Professor Daniel Fessler

Department of Anthropology, University of California, Los Angeles

16:30 Thursday 21.04.2022

Negatively-biased beliefs and the 21st century information environment: Intersections of evolved psychological mechanisms, cultural evolution, & individual differences

Cultural evolution can solve complex adaptive problems. However, the functions of cultural beliefs are often unclear to those who hold the beliefs. To benefit from cultural evolution's ability to provide solutions to problems, learners must therefore be credulous. However, credulity entails costs, including susceptibility to exploitation, and wasted effort due to false beliefs. One factor influencing the optimal level of credulity is the ratio between the costs of two types of errors: erroneous incredulity (failing to believe information that is true) and erroneous credulity (believing information that is false). This ratio will often be asymmetric when information concerns hazards, as the costs of erroneous incredulity will often be larger than the costs of erroneous credulity; no equivalent asymmetry will generally characterize information about benefits. Natural selection can therefore be expected to have crafted learners' minds to be more credulous toward information concerning hazards. At the level of whole societies, negatively-biased credulity should shape cultural evolution via the aggregated effects of learners' differential retention and transmission of information. At the level of individuals, because hazards often co-occur, the extent of this bias should be attuned to perceptions of how dangerous the world is, leading to individual differences in negatively-biased credulity. Lastly, because perceptions regarding the dangerousness of the world differ between social conservatives and social liberals, social conservatives should display greater negatively-biased credulity than social liberals. I will present results from a variety of studies supporting the above predictions.

Dr Abigail Page (New Investigator)

London School of Hygiene and Tropical Medicine

09:00 Friday 22.04.2022

Beyond the mother: interdisciplinary approaches to cooperative childrearing

It is well established that humans evolved a childrearing system which requires help from non-mothers (allomothers). The extent, and our need for cooperation in childrearing is not in doubt, however, where does the field go next? What dimensions have been overlooked, assumptions left unquestioned? The large literature currently struggles to unify the clear diversity in both who acts as allomothers and the associated outcomes. Here, I argue that a consideration of pay-offs, demographics and mechanisms of support is necessary to create a comprehensive, evolutionary, understanding of cooperative childrearing. Using cross-cultural data from Agta foragers and UK mothers, this thesis is made through: 1) an integration with demography, highlighting how mortality, fertility and urbanisation influence diversity in allomothering, impacting individuals' pay-off; 2) moving beyond a focus on practical support (i.e. childcare and provisioning) with an application of public health perspectives, demonstrating that informational and emotional support also impact maternal investment; and 3) the fitness consequences of practical, emotional and informational support depend on the mother's intentions, needs, costs and benefits. An interdisciplinary perspective is necessary to question common biases and assumptions, and in doing so we can better understand the mechanism by which *all types and sources of support* – or 'allosupport' – impact maternal investment and her ability to care for multiple overlapping dependents. Not only does such an approach shed light on human life history, but it also means we are best placed to translate evolutionary approaches into fields seeking to improve maternal and child health.

Professor Hillard Kaplan

Economic Science Institute, Chapman University

16:30 Friday 22.04.2022

The "embarrassment of riches" theory of brain aging with hypothesis tests in two Native South American populations

This talk presents an evolutionary-physiological model of human brain aging that examines the three-way interaction of age, genome, and exposome. We compare the exposome of the human evolutionary past with the modern era, and offer a new theory, "the embarrassment of riches", to explain why brain atrophy may be accelerated in sedentary, well-fed Western populations. The theory predicts that food energy stores and flows were positively associated with brain volume (BV) in the physically active, food-limited past, but now excess body mass and adiposity are associated with reduced BV. Focusing on the supply of resources to the brain, the theory also predicts that atherosclerosis will be a principal determinant of brain atrophy. We test these hypotheses with data on BV across age in two Native South American populations, the Tsimane and Maseten, whose lifestyles and exposomes fall on a continuum between the evolutionary past and those in wealthy nations. Our results reveal that in these two populations: 1) the relationship of BV with measures of energy stores have an increasing or curvilinear (lower among under- and over- nourished, highest at intermediate levels) relationship with BV; 2) the more acculturated Maseten exhibit a steeper decrease in BV with age than Tsimane, but this decrease is still shallower than in populations from wealthy nations; and 3) atherosclerosis is negatively associated with BV. Complemented by findings from the U.S. and Europe, these results are consistent with the embarrassment of riches theory, with implications for medicine and interventions to improve brain health.

Talks

Single 1

Lidborg LH*, Graham R, Brewer G, Boothroyd LG

Durham University, linda.h.lidborg@durham.ac.uk

Single 1: 15:20 Tuesday 19.04.2022

Do more masculine men have better quality partners?

Objective: So-called 'good-genes' theories suggest that men's testosterone-dependent traits (such as facial masculinity and increased strength and musculature) signal good genetic quality. If these theories are correct, women should consider men with exaggerated masculine traits as more attractive, and to favour such men as partners. Thus, more masculine men should be considered of higher mate quality. It can be expected that this should, in turn, afford more masculine men greater choice when selecting a partner: more masculine men might then be able to acquire higher-quality partners.

Methods: Here, we tested whether there was evidence of assortative mating for mate quality in a sample of young heterosexual couples (N=100 couples), recruited from a U.K. student population. Mate quality was indexed by facial and anthropometric traits commonly argued to correlate with either immunocompetence, same-sex competitiveness, and/or fertility (facial traits: attractiveness, dimorphism, perceived health, dominance, and prestige; handgrip strength, height, BMI, shoulder circumference [in men] and WHR [in women]) as well as self-reported incidence of common illnesses and self- and partner-rated mate value.

Results: Men with greater body masculinity had more facially attractive partners, but this association did not survive corrections for multiple comparisons. Except for a marginally significant effect for stature, we found no evidence of assortative mating between partners. No other male traits showed significant correlations with either self- or partner-rated female mate value, or with female health.

Conclusions: In a U.K. student population, we found limited support for the notion that more masculine men have better quality partners."

Skirgård H

Max Planck Institute for Evolutionary Anthropology, hedvig.skirgard@gmail.com

Single 1: 15:40 Tuesday 19.04.2022

Does political complexity predict the number of languages in Remote Oceania?

Objective: There are a lot of languages in the world and they are not evenly distributed over the population. Multiple hypotheses have been presented for this disparity. This paper tests the specific hypothesis by linguists (e.g. Pawley 2007) that political complexity (hierarchical levels of jurisdiction) predicts number of languages, even when time of settlement and other factors are accounted for.

Methods: a negative binomial generalized linear model with cross-cultural data on political complexity, number of languages, time of settlement and environmental factors per island group in Remote Oceania.

Results and Conclusions: Political complexity does significantly predict the number of languages, even when other factors are included in the model. Area and time depth of settlement are also important factors. The impact of political complexity is probably due to the fact that more hierarchical societies lead to more interactions and cohesion over large distances, which reduces language splitting. It is also possible that more politically complex societies can sustain conflict with others for longer, thereby eradicating variation in conquered territory.

References: Pawley, Andrew. 2007. Why do Polynesian island groups have one language and Melanesian island groups have many? Patterns of interaction and diversification in the Austronesian colonisation in Remote Oceania. In Workshop on migration, Ile de Porquerolles, France.

Foster D*, Mesoudi A, Postma E, Lamba S

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Single 1: 16:00 Tuesday 19.04.2022

Systematic review and meta-analysis of the effects of predictors of cooperation on loan repayment in microfinance groups

Objective: Our objective was to test evolutionary theories of cooperation in the context of the real-world cooperative dilemma created by joint liability microfinance. To do so we aimed to conduct the first meta-analyses of predictors of loan repayment to estimate their overall effects, and to assess the uncertainty, heterogeneity and generalizability of these effects.

Methods: We pre-registered our PRISMA-P meta-analytic protocol and hypotheses, and conducted literature searches for loan repayment studies focussing on 12 variables representing proxies of 4 evolutionary mechanisms proposed to support cooperation in humans: relatedness, reciprocity, partner choice, and punishment. Eligible studies were assessed for their risk of bias using ROBOS - a new tool we created - before extracting effect estimates for analysis. We developed Bayesian multilevel models in brms for each variable to estimate posterior distributions of their meta-analytic effects on loan repayment, and to assess effect heterogeneity.

Results: Many of our meta-analytic estimates are consistent with null effects. Most analyses also indicate high levels of effect heterogeneity, raising the possibility of non-generalisable results from individual studies, and/or reproducibility issues in the microfinance literature. Variation in study design, study quality, sample size, and study population characteristics is responsible for the majority of effect heterogeneity.

Conclusions: Our results contribute to multiple debates regarding the effects of features of microfinance groups on loan repayment efficacy, and the generalisability of the effects of evolutionary mechanisms on cooperation in real-world contexts.

Jones JH

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Single 1: 16:20 Tuesday 19.04.2022

The Dark Secret of Our Success

While the capacity for social learning is generally seen as an unmitigated good, allowing humans to swiftly adapt to changing conditions, reliance on social learning entails costs. Social learning induces strong correlations between the behavior of different people within a population, which create risk, suggesting that there are potentially complex trade-offs undergirding the social nature of human adaptability. Humans are adapted to manage the substantial variability – and risk this variability entails – that come from specializing in high-energy-density resources that define the human ecological niche. Human livelihood risk-management strategies typically involve individuals or small groups making subsistence decisions independently of each other and then aggregating/risk-pooling resources at a central place, a process I call Independence with Risk-Pooling (IwRP). Independence is the key to avoiding the deleterious effects of correlated risks. I present a simple model that captures the way that social learning, by inducing strong correlations in behavior within populations, opens up human populations to catastrophic risk. I show that the correlation of failure across individuals dominates the overall risk of catastrophic failure. I conclude with a discussion of implications. Social learning is likely to accelerate with population size. The standard interpretation of this phenomenon is that more people mean more people to imitate so social learning should be more efficient. However, small populations, which are naturally more vulnerable to extinction, should be more averse to the risks entailed by a heavy reliance on social learning.

Single 2

Brooks R, Blake K, Russo-Batterham D

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Single 2: 10:00 Wednesday 20.04.2022

Incel activity on social media linked to local mating ecologies

Young men with few prospects of attracting a mate have historically threatened internal peace and stability of societies. In some contemporary societies, including the U.S.A., such involuntary celibate, or Incel, men are responsible for much online misogyny and real-world violence. Here we test the prediction that online Incel activity arises via local 'real world' mating market forces that affect relationship formation. From a database of 4 billion Twitter posts between 2012 and 2018, we extract tweets using words peculiar to Incels and tweets about Incels, and geolocate them within the mainland U.S. We show that such tweets arise disproportionately within places where mating competition among men is likely to be high due to male-biased sex ratios, low proportions of women who are single, and high income inequality. Our results suggest a role for social media in monitoring and mitigating the factors that lead young men toward antisocial behaviour in real-world societies.

Rotella A, Varnum ME, Sng O, Grossmann I

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Single 2:10:20 Wednesday 20.04.2022

Increasing population densities predict decreasing fertility rates over time: A 174-nation investigation

Fertility rates have been declining worldwide over the past fifty years, part of a phenomenon known as "the demographic transition." Prior work suggests that this decline is related to population density. In the present study, we draw on life history theory to examine the relationship between population density and fertility across 174 countries over 69 years (1950 to 2019). We find a robust association between density and fertility over time, both within- and between-countries. That is, increases in population density are associated with declines in fertility rates, controlling for a variety of socioeconomic, socioecological, geographic, population-based, and female empowerment variables. We also tested predictions about environmental boundary conditions. In harsher living conditions (e.g., higher homicide or pathogen rates), the effect of increased population density on fertility rates was attenuated. The density-fertility association was also moderated by religiousness and strength of social norms, where the relationship between density and fertility was attenuated in countries with high religiosity and strong social norms. We discuss why and when changes in population density may influence fertility rates and the broader implications of this work.

Tureček P, Kozák M, Slavík J

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Single 2: 10:40 Wednesday 20.04.2022

A minimal model of subculture emergence

Spatial separation was frequently included in models of ethnic divergence. Yet, it has been acknowledged that urban subcultures can, and frequently do, emerge in sympatry. Previous works have attributed this phenomenon to the human tendency to imitate self-similar individuals and actively differentiate from individuals recognized as members of the outgroup. The application of such a model in non-human animals was, however, identified as problematic. Here we present a parsimonious model of subculture emergence that does not have to vary the social learning algorithm by the distance between the agents in trait-space.

All it takes is a slight modification of the Galton-Pearson biometric model used previously to approximate cultural inheritance. The novel model includes the proportionality between the variance of inputs (cultural “parents”) and variance of outputs (cultural “offspring”). In such a model, assortment alone can lead to the formation of distinct, cohesive clusters of individuals (subcultures) with low within-group and large between-group variability even in the absence of spatial separation or disruptive natural selection.

According to our computer simulations, subcultures can emerge even without the intentional preferential imitation of self-similar individuals, which makes the process a possible driving force behind adaptive radiations in taxa with a capacity for social transmission, such as cetaceans and songbirds.

Sympatric emergence of arbitrary behavioural varieties preceding ecological divergence might be a norm, not the exception, in all cultural animals.

Kandza VH*, Jang H, Kiabiya-Ntamboundila F, Boyette AH

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Single 2: 11:00 Wednesday 20.04.2022

Insights into inter-group cooperation and inequality from shotgun hunting among the BaYaka and Yambe in Republic of the Congo

Whereas many evolutionary models emphasize within-group cooperation or between-group competition in explaining human large-scale cooperation, recent work highlights a critical role for inter-group cooperation in human adaptation. In the Congo Basin, forest foragers maintain relationships with neighboring farmers based on a division of labor, despite contrasting economies and sharing norms. Costs and benefits of these relationships have varied across time and space depending on resource availability and third party intervention (e.g., contemporary global commerce). Here we focus on for-hire shotgun hunting in the northern Congo where Yambe farmers own guns and have access to markets to buy bullets and sell meat, but BaYaka foragers hunt. We investigated what each group gains from this cooperative practice, and thus how inter-group relations are maintained. We conducted structured interviews with 77 BaYaka hunters and 15 Yambe gun owners to understand how hunts are arranged and what gun owners and hunters each contribute and receive. Hunts are organized in a ritualized manner within a fictive kinship structure. As bushmeat demand is high, Yambe gun-owners can sell for significant profit while paying hunters in items such as alcohol and cigarettes. However, BaYaka hunters reported that they accept hunts to get meat for their families. Specifically, hunters hide kills or bullets from gun owners, somewhat equalizing their relative benefits. Our results illustrate how each group prioritizes different currencies (e.g., cash, meat, family, inter-group relations) and provide insights into how inter-group cooperation is stabilized in a specific setting.

Single 3

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Single 3: 13:40 Wednesday 20.04.2022

Chimpanzee and human risk preferences show key similarities

Risk and uncertainty are central to all adaptive decisions human and nonhuman animals make — including when deciding where to forage, with whom to mate and how to deal with dangerous situations. Risk preference determines how agents decide in face of these hazards and is a crucial determinant of life outcomes such as health, wealth, and wellbeing. Yet, the biological underpinnings of this important building block of behavior remain unclear. Here, we investigated the extent to which chimpanzees’ (Pan troglodytes) exhibit risk preferences that are in important respects isomorphic to human risk preferences. Chimpanzees are an ideal reference: they are one of humans’ two closest living relatives, live in large, mixed sex social groups and have similar developmental stages with a long infant dependency, a period of adolescence and a long life expectancy of up to 50 years. Using a multimethod approach we studied chimpanzee (N=86) risk preference across domains and measurements. Our results show that chimpanzees’ willingness to take risks shares structural similarities with that of humans across four key dimensions: chimpanzee risk preference manifests as a trait-like preference that is consistent across domains and measurements, chimpanzees are ambiguity averse, males more risk prone than females and that the appetite for risk taking follows an inverted U-shaped relation to age. Our findings suggest that key dimensions of risk preference emerge independently of the influence of human cultural evolution and thus may have deeper phylogenetic roots than previously suspected.

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Single 3: 14:00 Wednesday 20.04.2022

Ger or house: Drivers of household shelter choices and investments in Ulaanbaatar, Mongolia

Shelter is a key human technology, and varies extensively throughout the ethnographic record, from low investment foliage lean-tos to the high investment composite multi-story dwellings present in modern urban environments. Like with other technologies, human investment in shelter is strategic and has consequences for the development of settlement landscapes. As such, understanding the determinants of investment in shelter is necessary for the construction of a holistic account of both the variability of human built environments, as well as their tendency towards high-investment fixed settlement throughout the holocene. We present initial findings from a household ethnography and survey study of settlement behaviour in the ger districts of Ulaanbaatar, Mongolia. The ger districts, named after the Mongolian mobile pastoralist dwelling, are idiosyncratic semi-formal settlements surrounding Ulaanbaatar's urban core, that feature self-built housing.

Crucially, the housing takes two distinct forms - mobile gers, as well as fixed, wooden houses. We utilise this local heterogeneity to address how intended stay, and socio-economic conditions affect housing choices and investments. Moreover, as the changing nature of the human life course alters investment pressures, we consider the role of demographics in the strategic landscape of housing investment decisions. We relate our findings to broader questions of human-environment interaction, in particular focusing on how bottom-up behaviour contributes to the construction of a permanent urban landscape which in turn structures settlement behaviour.

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Single 3: 14:20 Wednesday 20.04.2022

Agent-based models of the cultural evolution of occupational gender role

Objective: The causes of sex differences in human behaviour are contested, with 'evolutionary' and 'social' explanations often being pitted against each other in the literature. Recent work showing positive correlations between indices of gender equality and the size of sex differences in behaviour has been claimed to show support for 'evolutionary' over 'social' approaches. Such accounts, however, neglect the potential for social learning in a complex environment to generate arbitrary gender segregation.

Methods: In the current paper we ran agent based models that allowed agents of two group types to select 'occupations'/roles based on the apparent reward for that role (more popular roles yielded lower rewards per person), and knowledge of the current group-type balance in each role. Models were run under conditions with and without actual group differences in how successful each group was at a given role (With resulting moderation of rewards), and where actual differences disappeared after the first few cycles. We also modelled the effects of agents having beliefs (priors) for whether group differences existed.

Results: The mere belief that group differences may exist in 'innate' competence at particular tasks, and the availability of social information regarding the current balance of groups in each task, led agents to self-segregate where real differences in performance did not exist. Belief in 'small' group differences were particularly resilient; facilitating role-changes such that agents should move without cost to the highest-rewards for their observed skills eradicated segregation under the conditions of a prior for large differences but not for small differences.

Conclusions: These models show support for the role of social learning in patterns of gender segregation, and suggest the need for a formal cultural-evolution analysis of sex- and gender-roles in human cultures.

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Single 3: 14:40 Wednesday 20.04.2022

The burdens of leadership: a game theoretical and empirical study

Recent scholarship on the social and evolutionary function of leadership has emphasized that leaders maintain their position in an organizational hierarchy in large part by providing benefits to their followers. Many crucial functions of leadership, including norm enforcement, conflict mediation, and project management, may come at significant personal cost to leaders. Some models of leadership assume that leaders recuperate these costs, typically either in the form of prestige, mating opportunities, or a greater share of collective production. However, this assumption has not been demonstrated, either theoretically or empirically. Here, we develop an evolutionary game theoretical model to explore costly leadership, when it evolves, and when leaders can expect to recuperate the full cost of their investment. We show that when the benefits of leadership are fungible – for example, if leaders directly increase the production of shareable resources – then groups generally do best by fully compensating leaders for the costs they incur. However, leadership can still emerge even if leaders aren't fully compensated, and in some cases, groups may benefit from under-paying leaders in order to select for particularly invested leaders. These results are discussed in light of empirical data from two small-scale horticultural societies in Bolivia.

Single 4

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Single 4: 10:00 Thursday 21.04.2022

Global relationships between musical, linguistic, and genetic diversity

Objective: Music exists in all human societies, with appreciable within-group diversity. However, across the globe we observe characteristic musical styles that persist over time. How do these characteristic styles evolve? Some argue that musical similarities reflect ancient migrations of people and their cultures, with musical history mirroring the branching phylogenetic trees of linguistic and genetic evolution. Others argue that music primarily reflects more recent contact between neighboring societies and local micro-evolution independent of genetic or linguistic histories.

Methods: We use a newly public database of 5,778 traditional songs from 998 societies coded for 37 variables of musical style to analyze global patterns of musical diversity. We then directly compare musical, linguistic, and genetic diversity for a subset of 120 matched societies (representing 120 languages, 801 songs, and genomic profiles of 108 societies).

Results: We identify five main latent dimensions of musical style: Differentiation, Ornamentation, Rhythm, Dynamics, and Tension. We show that across all dimensions there is considerable between-society diversity, and that this diversity shows a stronger relationship with historical (linguistic and genetic) processes than a model of recent geographic diffusion.

Conclusions: These results suggest that music preserves historical relationships among people and their cultures across vast geographic distances, and could be used as a tool to enhance our understanding of human cultural history.

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Single 4: 10:20 Thursday 21.04.2022

Cooperative foraging networks among BaYaka women in the Republic of Congo

Collaborative foraging is a hallmark of the evolution of human cooperation. While many studies have focused on men's cooperative hunting, women's social networks, especially in the context of cooperative foraging, are understudied. Hunter-gatherer women forage in small or large groups, including a mix of adults and children, as well as kin and nonkin. These foraging groups may create crucial opportunities for cooperation across generations as well as opportunities for shared childcare and social learning. Yet, how women choose foraging partners and coordinate childcare with foraging, and whether cultural transmission plays a role in women's foraging group decisions have been largely overlooked. Here, we used focal follow data of 230 foraging trips from five BaYaka women in the Congo, together with daily camp composition and demographic information including age and relatedness. Using recently developed tools in social network analyses, we implemented a joint stochastic block model and social relations model to examine how women's foraging networks were patterned. Our findings suggest that women's choice in foraging partners was driven by close genetic relatedness, age homophily, and gender homophily. Crucially, we found that women go foraging with children more than with other adults, and that the probability of foraging across age classes was highest in early childhood, followed by infancy. These findings support the possibility that women's daily foraging networks facilitate cooperative childcare and cultural transmission. Taken together, our study provides insights into the important role of women's social relationships in understanding human cooperation.

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Single 4: 10:40 Thursday 21.04.2022

Abandoning FGM/C increases the cost of marriage in rural Ethiopia

Female genital mutilation/cutting (FGMC) affecting 200 million women worldwide, has profound negative consequences for women's well-being, yet affected communities often resist efforts to end the practice. Marriage market dynamics have been proposed to be key to this resistance, because where FGMC is normative, parents are incentivized to cut their daughters to improve their marriage prospects. This assumption has been explored using theoretical modelling, but rarely evaluated directly with data on the economic transactions which occur before marriage (bride price and dowry). Here we present novel data from an Ethiopian agropastoral community where FGMC is declining, providing a unique framework to test whether FGC represents a form of pre-marital investment, and if parents must compensate with other forms of investment when they reject cutting practices. We find apparent financial and reproductive incentives for parents to cut their daughters, reflected in lower marriage costs for parents, and earlier marriage for daughters. In contrast, parents who defy social norms by not cutting their daughters, compensate with higher education and endow daughters with higher dowry payments, enhancing their status and bargaining power within marriage. We also find evidence that education is replacing FGMC in the marriage market, as new parental investment norms are adopted within the community. These findings confirm that abandoning FGMC is costly to parents, and that compensating for these costs through alternative investment is key to FGMC abandonment.

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Single 4: 11:00 Thursday 21.04.2022

The effect of leaders' charismatic signaling on information processing outcomes: Combining cognitive science and evolutionary leadership theory

From an evolutionary leadership perspective, following a charismatic leader and uncritically accepting propositions made by the leader is beneficial. We propose the minimal cognitive effort hypothesis that illustrates how followers reduce their effort when processing messages from a charismatic leader.

In a registered report, we combined traditional leadership research with cognitive science methods to investigate the cognitive outcomes of the charismatic signaling process. We conducted two online experiments, using a 2 (charismatic versus neutral) x 2 (female versus male leader) repeated-measures between-subjects design and videos of trained actors delivering a speech. We investigated the effects of charisma on (Study 1, N = 100) followers' ability to detect false information, accuracy to remember factual statements, (Study 2, N = 140) the persuasiveness of information, and prosocial behavior.

Participants recognized less information in the charismatic compared to the neutral leader condition. Exploratory analyses further provided evidence for an interaction effect of leader sex and leader charisma on error detection. Participants tended to identify fewer false statements in the female charismatic than the male charismatic leader condition. We found no support for the effect of leader charisma on the persuasiveness of factual messages, or increased prosocial behavior among followers.

Our studies provide initial insights into the cognitive outcomes of charismatic leader signaling. We find some evidence for our minimal cognitive effort hypothesis. However, our findings suggest that charisma may affect followers differently depending on the sex of the leader. Future research needs to disentangle the role of leader sex in charismatic signaling.

Single 5

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Single 5: 13:40 Thursday 21.04.2022

CHIELD: A database of causal hypotheses for unifying theories of cultural evolution

Objective: An evolutionary approach to human behaviour increasingly depends on synthesising hypotheses and evidence from many different fields. This is difficult due to the complexity of the hypotheses and the variability in terminology between fields. The objective is to create a tool to help researchers connect hypotheses together.

Methods: We created CHIELD: a web database of hypotheses expressed formally as causal graphs (<https://correlation-machine.com/CHIELD>). This is a graphical method that represents variables as nodes and causal effects as directed connections between them to form a graph. Information is stored about each causal effect (type of supporting evidence, whether the hypothesis was supported). Hypotheses with overlapping nodes are joined together automatically. This allows researchers to: integrate hypotheses into a coherent narrative; spot gaps in empirical evidence; identify possible confounds; design statistical models; and plan future research.

Results: Over 400 hypotheses about the cultural and biological evolution of language were coded by hand and entered into the database. In this talk, a case study will be presented on 56 hypotheses of how cultural traits are affected by population size. Using methods from graph theory and online collaboration tools, we discovered 10 different terms for this concept in linguistics, computational modelling and anthropology. We used CHIELD's open-source tools to revise and unify these hypotheses.

Conclusions: CHIELD provides tools for analysis and a space for discussion about how hypotheses connect to each other. We hope it will become a useful tool for researchers working across disciplines to explain human behaviour.

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Single 5: 14:00 Thursday 21.04.2022

Social and ecological effects on the evolvability of human fertility

Objective: Fertility rates are highly variable within and across human societies, due to social and ecological effects on reproductive behavior. Given that fertility is a fundamental fitness component, these environmental effects are expected to play a central role in the adaptive evolution of human populations, both by shaping selection on phenotypes and by modifying heritable effects on fertility across space and time. However, previous studies have not directly estimated how social and ecological factors interact to shape the microevolutionary potential (i.e. evolvability) of human fertility.

Methods: We developed an evolutionary model of human fertility variation in a heterogeneous environment, and we applied this model to extensive demographic, spatial, and genealogical data on Tsimane forager-horticulturalists. Our model operationalized fitness competition or cooperation through negative or positive covariance, respectively, between heritable effects on individual fertility and the fertility of social partners. We allowed this covariance to change nonlinearly in response to age, group density, and environmental quality, providing flexible predictions of how the evolvability of fertility changes in response to social and ecological conditions across the lifespan.

Results: Our formal model and empirical results show how environmental variation can magnify, diminish, or even reverse patterns of adaptive evolution predicted by classical approaches ignoring eco-evolutionary feedback within generations. We also provide some of the first evidence of heritable social effects outside of the family affecting the evolvability of a fitness component.

Conclusion: Our findings suggest that plasticity in response to heterogeneous environments has played a key role in the social evolution of human fertility.

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Single 5: 14:20 Thursday 21.04.2022

Testing the functions of supernatural beliefs in 1930s Irish folk medicine

Objective: Around the world, people routinely employ supernatural and religious beliefs in the treatment of illnesses. Yet the question remains as to when or why these beliefs are used, or if there is a pattern at all. Using existing theories from cognitive and evolutionary research on religion to shape our hypotheses, we looked for possible psychological and social benefits of supernatural treatments across a variety of diseases in a corpus of over 3,000 folk treatments, collected in 1930s Ireland.

Methods: Treatments were coded for supernatural and natural content, and ritualistic aspects. Thirty-five associate illnesses were coded separately by medical professionals for disease severity, opacity, ability to induce anxiety, and need for social support. Treatment codes were used to predict these qualities of the diseases. The coded data set was split into equal halves. Statistical models were built in one half and then tested in the other half.

Results: We found that supernatural and religious treatments were more likely to be used when the origin and mechanism of the disease were causally opaque. Opacity was also associated with increased ritualisation of the treatment. Severity of the disease, anxiety, and need for support were not reliably related to supernatural, religious, and ritualised treatments, respectively.

Conclusions: In this data, supernatural beliefs are more likely to be employed in the treatment of illnesses that are confusing or unclear. Beyond illness, this gives us some insight into how people might use supernatural beliefs more broadly to understand the more uncertain parts of the natural world.

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Single 5: 14:40 Thursday 21.04.2022

What motivates intermarriage in small-scale societies, and what consequences for social cohesion?

Intermarriage between individuals from different linguistic, ethnic or religious groups is often used as an indicator of minority integration in multicultural contemporary societies. For the individuals involved, however, intermarriage is often fraught with various hardships and is more likely to end in divorce. What motivates individuals to consider intermarriage? Should intermarriage be simply regarded as a sign that group boundaries are fading, or are individuals strategically motivated to marry out despite the risks involved? And what consequences does intermarriage have on intergroup relations? Here, we address the above questions in the context of a multiethnic village located at the intersection of the Amazon and the Andean highlands of Bolivia. We find (1) intermarriage is driven by resource exchanges, whereby indigenous Mosesten women with privileged access to arable land in tribal territory tend to marry wealthier Aymara-Quechua immigrant men; (2) Intermarriage may be a risk management strategy resulting in greater resilience to localized shocks to production for mixed households; and (3) analysis of social networks shows intermarried individuals act as bridges between their respective ethnic groups. Taken together, these results provide important insights for understanding how multicultural societies take shape, and contribute to debates in archaeology and genetics about the nature of intergroup contacts.

Single 6

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Single 6: 10:00 Friday 22.04.2022

Socioeconomic correlates of Inuit hunting strategies

Different harvest activities differ in the required efforts and their success rates. Previous ethnographic research has shown that the probability of an unsuccessful harvest affects foragers' prey and patch choice decisions, with gender and socio-political objectives often affecting risk-sensitivity. We study drivers of Inuit harvesters' food production choices in the Canadian Arctic, where harvesting and sharing of country foods is important for food security and cultural identity, but individuals differ in their involvement with traditional subsistence activities. We analysed 243 foraging trips involving 25 Inuit hunters from Kangiqsuaq, Nunavik, using a Bayesian approach that models both patch choice (where to hunt and what species to target) and within-patch success, while estimating individual, social, and environmental determinants of production. We present how and why hunters vary in their contribution to food production, and discuss potential implications for the long-term resilience of Arctic food systems to ongoing social, economic, and climate change.

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Single 6: 10:20 Friday 22.04.2022

Does haemoglobin modulate the trade-off between current and future reproductive investment in the Bolivian altiplano?

Objective: Female reproductive function flexibly responds to ecological variation in energetics, but the role of other ecologically limited resources, such as iron, remain poorly understood. We investigated the role of haemoglobin in modulating the trade-off between current and future reproduction in a rural population following a mixed subsistence strategy. Specifically, does haemoglobin concentration covary with ecological variation in economic resources and climatic conditions, and does variation in haemoglobin concentration predict markers of investment in current reproduction (ovulation and peak-luteal progesterone)?

Methods: We analysed biomarker and sociodemographic data from the Bolivian altiplano (152 menstrual cycles from 96 women).

Multivariable analyses were conducted using multilevel regression models. Adjustment factors were determined using a directed acyclic graph (DAG) of hypothesised causal relationships.

Results: Haemoglobin concentration was lower in more arduous seasons (higher physical workload, lower food availability) ($p=0.005$). During these arduous seasons, haemoglobin concentration decreased with increasing dependency on agropastoral economic strategies ($p=0.020$).

Anovulation was borderline associated with decreasing haemoglobin during arduous seasons, when it may benefit women to delay reproduction ($p=0.076$). No evidence was found for a relationship between haemoglobin and peak-luteal progesterone concentration.

Conclusions: Women's evolved physiology may flexibly reduce investment in current reproductive function when haemoglobin levels are low but can be expected to improve. These findings add to the literature on how iron bioavailability, here measured through haemoglobin, may be crucial to understanding the scheduling of life history events in women. Funding: US National Science Foundation (SBR 9506107, 9507320).

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Single 6: 10:40 Friday 22.04.2022

Disentangling genetic and non-genetic influences on sociality; results and challenges from a longitudinal study on wild capuchins

In the last two decades various studies have demonstrated links between sociality (e.g., connectedness) and survival/reproduction in mammals. How within- and between-individual variation in social behaviour arise in uncontrolled populations can be challenging to determine but is important for understanding constraints on the evolution of sociality. Using a statistical method known as an 'animal model', we use spatial association data from a wild population of capuchin monkeys (*Cebus capucinus imitator*) to estimate the relative contributions of genetic and non-genetic sources of variation on sociality. Using 20 year of observational data on 376 individuals (range: 6-185 months of data per subject), we find that individuals show moderate long-term (across years) repeatability in the proportion of time spent social (posterior mode: 0.238, 95HPDI: 0.184, 0.284). Conditioned on the effects of age, sex, group size, seasonality, and El Niño-Southern Oscillation phases, we find that repeatability is largely explained by additive genetic variance (h^2 : 0.166, 95HPDI: 0.106, 0.223) and to a minimal extent by maternal effects (m^2 : 0.000, 95HPDI: 0.000, 0.045). Our model captures the majority of variance (latent scale) in our behavioral trait, with most variance explained by temporally changing factors, which may indicate environmental constraints on the evolvability of the trait.

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Single 6: 11:00 Friday 22.04.2022

Life-history tradeoffs in a historical population (1896-1939) undergoing rapid fertility decline: Costs of reproduction?

Objective: Quantity-quality tradeoffs are often invoked to explain reductions in fertility during demographic transitions. The evidence for such tradeoffs in humans has however been mixed. Here we tested for quantity-quality tradeoffs using N=13,663 birth records from the maternity hospital in Basel, Switzerland, 1896-1939, a period characterized by a rapid decline in fertility. Specifically, we predicted that higher parity is associated with worse offspring condition.

Methods: We used Bayesian multivariate, multilevel models to simultaneously analyze multiple related outcomes (probability of live birth, gestational age, birth weight, birth length, head circumference, placenta weight) while accounting for various statistical issues such as missing data, endogeneity and phenotypic correlation. All predictions and the analysis plan were preregistered. We also performed various robustness checks.

Results: We found no evidence for quantity-quality tradeoffs, as offspring condition did not deteriorate at higher parities and there were no residual correlations between parity and offspring condition. Instead, women with better access to resources had fewer children than others living in the same place at the same time.

Conclusions: Barring limitations of the data, these results are consistent with demographic transitions reflecting women's investment in their own embodied capital and/or the adoption of maladaptive low-fertility norms by elites. A preliminary test distinguishing these two hypotheses was inconclusive. In sum, our study contributes to a growing realization that quantity-quality tradeoffs may only play a limited role in demographic transitions to low fertility.

Single 7

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Single 7: 15:00 Friday 22.04.2022

Patrons and Managers: the rise of political inequality in an Amazonian society

Objective: Ethnography in small-scale societies, which are transitioning to market economies, can be a valuable complement to the archaeological record in testing models of the origins of political inequality. In such a society, I test effects on political inequality from increases in paid labor (i.e. "patron-client relationships") and from demand for leaders to mediate conflict (i.e. "managerial mutualism").

Methods: Cross-sectional and longitudinal data were collected over eight years in four Tsimane villages in the Bolivian Amazon. Data include three waves of social network interviews regarding labor exchange, interpersonal conflicts, and conflict mediation. In addition, residents ranked each other on metrics of political influence. Analysis consists of qualitative comparison across villages and longitudinal mixed-models of political influence change within one village.

Results: Villages with greater political inequality do not necessarily have greater income inequality, but they do exhibit a larger fraction of labor assistance that is paid, rather than reciprocated. The most influential individuals tend to be labor patrons. Residents of more politically unequal villages also experience more inter-personal conflict, yet mediation is even more restricted to a few, influential individuals. Longitudinal results within one village are forthcoming.

Conclusions: Different models of the origins of political inequality (e.g. "patron-client", "managerial mutualism") likely operate in tandem, in general and in the Tsimane context. Among extant small-scale societies, increasing exposure to markets generates fleeting opportunities to test in real-time whether and how such models explain changes to political structure.

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Single 7: 15:20 Friday 22.04.2022

Sensitive periods, but not critical periods, evolve in a fluctuating environment: A model of incremental development

Sensitive periods, in which the impact of experience on phenotype is larger than in other periods, exist in all classes of organisms, yet little is known about their evolution. Recent mathematical modeling has explored the conditions in which natural selection favours sensitive periods. These models have assumed that the environment is stable across ontogeny or that organisms can develop phenotypes instantaneously at any

age. Neither assumption generally holds: often, environments fluctuate or phenotypes develop incrementally. Here, we present a model in which organisms gradually tailor their phenotypes to an environment that fluctuates across ontogeny, while receiving cost-free, imperfect cues to the current environmental state. We vary the rate of environmental change, the reliability of cues, and the duration of adulthood relative to ontogeny. We use stochastic dynamic programming to compute optimal policies. From these policies, we simulate levels of plasticity across ontogeny and obtain mature phenotypes. Our results show that sensitive periods occur at the onset, midway through, and even towards the end of ontogeny. In contrast to models of stable environments, organisms always retain residual plasticity late in ontogeny. We conclude that critical periods, after which plasticity is zero, are unlikely to be favoured in environments that fluctuate across ontogeny.

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Single 7: 15:40 Friday 22.04.2022

What other men want: the accuracy of gender norm perception among Tanzanian men

Objective: Effective social learning requires norm estimation, but the estimation may be inaccurate when individuals misrepresent their own beliefs. Here, we explore this idea with respect to gender ideology, hypothesizing that social performance of masculinity will lead men to underestimate peer support for women's empowerment. While prior studies have found some support for this hypothesis, they largely rely on self-report data to measure normative beliefs, which, in the face of social desirability bias, may create a false impression of norm misperception.

Method: We test this hypothesis using data from a semi-representative sample of men living in a single community in Mwanza region, Tanzania (n=590). Moving beyond prior studies of this topic, we estimate local norms using both self-report data and an indirect wife-reported measure of men's attitudes, which we argue more accurately measures true norms.

Results: Data on the discrepancy between men's estimation of the local norm and the actual community norm will be presented for several domains, including authority in decision-making, parental care of sons vs daughters, and women's economic and social position in society. We will also present exploratory analysis on individual variability in the accuracy of estimating local norms (e.g. by socioeconomic status).

Conclusions: We discuss the implications of norm misperception as a potential barrier to positive change in gender norms, and for the evolution of social learning strategies more generally. We also reflect on the methodological difficulties of assessing normative beliefs, particularly with respect to sensitive topics, highlighting the potential avenue of alternative approaches that involve a third-party estimation.

Parallel 1A: 'Mating I'

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Parallel 1A: 11:40 Wednesday 20.04.2022

"Time slows down whenever you are around" for women but not for men

What happens when we unexpectedly see an attractive potential partner? Previous studies in laboratory settings suggest that the visualization of attractive and unattractive photographs influences the perception of time. The major aim of this research is to study time perception and attraction in a realistic social scenario, by investigating if changes in subjective time measured during a speed dating are associated with attraction. The duration of the dates was variable and participants had to estimate the time that passed. Among other measures, participants also rated the potential partners in terms of their physical attractiveness before and after the dates and reported if they would like to exchange contact with them. Results showed that, in a real speed dating situation, when there is a perception of the partner as being physically more attractive, women tend to overestimate the duration of that meeting, whereas men tend to underestimate its duration. Such changes may reflect evolutionary adaptations which make the human cognitive system more responsive in situations related to reproductive fitness.

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Parallel 1A: 12:00 Wednesday 20.04.2022

Peri-ovulatory changes in sexual behavior and desire — what can different methods tell us?

Objective: To compare the informativeness of several methods to investigate ovulatory changes in sexual behavior and desire, as well as individual differences therein.

Methods: I draw on eight published datasets (total N > 1,300) with repeated measures of progesterone and estradiol, two large-scale online diary studies (total N > 2,000) and data from a popular cycle tracking app (N > 100,000) that combine counting methods, fertility awareness methods, and luteinising hormone tests. I compare the validity of different methods to estimate the day of ovulation and ask which conclusions about ovulatory changes could be robust.

Results: Salivary, but not serum, estradiol immunoassays have surprisingly low validity for estimating cycle phase and may suffer from substantial bias, so both negative and positive findings about the cycle on this basis are in doubt. Other methods consistently show small peri-ovulatory increases in sexual desire, whereas effects on sexual behavior are less clear. Individual differences in peri-ovulatory change are estimated to be large, but further work is needed to separate real from measurement-induced differences.

Conclusions: Greater transparency through preregistration and data sharing have moved the field forward, in part by making problems more

obvious. Now, further methodological improvements are necessary to characterize replicable phenomena. We need to walk before we can run, that is, agree on improved methodological standards, before we can answer theoretical questions such as whether ovulatory changes in humans are vestiges or evolutionary adaptations.

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Parallel 1A: 12:20 Wednesday 20.04.2022

Some for a good time, some for a long time: Heterosexual women's ratings of the mate value of men with different addictions: An experimental study

Background: Substance and non-substance addictive behaviors such as alcohol and gaming addictions are now occurring worldwide. Numerous studies suggest an association between addictive behaviors and various adverse effects on individuals' romantic and social life. Given that inherited predispositions play an important role in developing addictive behaviors, choosing a mate with addiction increases likelihood of intergenerational transmission of addictive behaviors. Considering the link between being primed with evolutionary-driven motives and decision making as well as the notion that cognitive control is positively associated with mating decisions, we investigated, once primed with a vignette in which the target is a potential mate with high mating qualities but is diagnosed with an addictive behavior, to what extent heterosexual women consider the target person as a potential short-term and long-term mate.

Methods: 3000 heterosexual females were recruited and were randomly assigned to one of the six conditions (i.e., gambling, gaming, alcohol, cannabis, anabolic androgenic steroids, and healthy). The survey included some demographics questions and a vignette about a male with high qualities for a mate but diagnosed with one type of addiction and a control vignette in which the target was healthy. All the participants answered two single items asking them how they rate the target as a short-term mate and a long-term mate.

Results: There was a significant difference between conditions on the combined dependent variables ($F(10, 6028) = 28.34, p \leq .001$; Pillai's Trace = .91; $\eta^2 = .91$). Results were also significant for each of the dependent variables (perceived attractiveness for short-term mating: $F(5, 3020) = 2.84, p = .01, \eta^2 = .005$; perceived attractiveness for long-term mating: $F(5, 3020) = 46.23, p \leq .001; \eta^2 = .07$).

The post-hoc Bonferroni's test of comparison showed that, compared to the control condition ($M = 4.14, SD = 1.67$), the gambling condition was perceived as the most attractive for short-term mating ($M = 4.48, SD = 1.62, p = .024$). However, the post-hoc Bonferroni's test of comparison for perceived attractiveness for long-term mating showed that the mean score for control condition was significantly higher than the other conditions ($M = 5.74, SD = 1.41, p's \leq .001$). The gaming condition ($M = 4.84, SD = 1.70$) was further perceived as more attractive compared to the gambling ($M = 4.14, SD = 1.68$), alcohol use ($M = 4.21, SD = 1.78$), and cannabis use conditions ($M = 4.34, SD = 1.77$) ($p's \leq .001$). Moreover, the AAS use condition ($M = 4.56, SD = 1.72$) was perceived as more attractive for long-term mating compared to the gambling ($M = 4.14, SD = 1.68$) and alcohol use conditions ($M = 4.21, SD = 1.78$) ($p's \leq .05$).

Discussion: Results from this study will be discussed using an evolutionary perspective.

Parallel 1B: 'Cross cultural'

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Parallel 1B: 11:40 Wednesday 20.04.2022

A quantitative test of Jared Diamond's axis of orientation hypothesis

Objective: Jared Diamond controversially suggested that Eurasia's east-west axis of orientation facilitated the spread of cultural innovations and gave it a substantial political, technological, and military advantage over other continental regions. Qualitative critical appraisals for Diamond's axis of orientation hypothesis are common. But direct quantitative tests of its ecological and geographic assumptions are not. Here we address this critical gap.

Methods: Our analyses leverage D-PLACE, a global cultural, linguistic, and ecological dataset. We first used cultural and ecological data to identify whether environmental parameters influence the ease of cultural diffusion. We then used environmental data to compare the strength of ecologically induced costs to cultural spread between the globe's landmasses - as measured along the corridors for the early diffusion of agriculture, one of the prominent examples in Diamond's work.

Results & Conclusions: We show that although the ecological premise of Diamond's hypothesis is supported, its geographic premise is not. Specifically, while the likelihood of sharing cultural traits increases with the ecological similarity between societies, there is no clear evidence of consistently higher levels of environmental homogeneity in Eurasia. Thus, we do not find support that intercontinental differences in ecological homogeneity predominantly catalysed or fuelled the uneven turn of human fortunes.

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Parallel 1B: 12:00 Wednesday 20.04.2022

Norm violations and punishments across human societies

Punishments for norm violations are hypothesized to be a crucial feature in the maintenance of cooperation in humans. There are limited data, however, on punishment in decentralized societies, including the norm violations that are typically punished, how they are punished, and how these vary with aspects of political and socioecology. Using a sample of 131 largely nonindustrial societies we coded ethnographic

documents for evidence of punishment for norm violations within the domains of adultery, religion, food, rape, and war cowardice. We also coded the type of punishment, whether it included reputational, physical, material, or execution. We tested a suite of hypotheses predicting culture-level covariation in punishment using measures from the Standard Cross-Cultural Sample database using a Bayesian phylogenetic model. We found little evidence of phylogenetic signals in evidence for punishment types, suggesting punishment systems change "quickly" over cultural evolutionary history relative to traits that demonstrate strong phylogenetic signals. We also found evidence that reputational punishment was associated with egalitarianism and the absence of food storage, material punishment was associated with the presence of food storage, physical punishment was moderately associated with greater dependence on hunting, and execution punishment was moderately associated with social stratification. We interpret results within a cultural-ecological framework and emphasize the importance of socio-ecological context for understanding human diversity in punishment systems.

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Parallel 1B: 12:20 Wednesday 20.04.2022

Supernatural theories of disease, useful falsehoods, and a market for religious specialists

Objective: Shamans and other religious specialists invoke the supernatural while healing illnesses. Many current evolutionary perspectives argue that supernatural theories of disease are cognitive byproducts that are subjectively appealing to naive observers. We investigated medicinal specialists and their clientele in two separate studies.

Methods: In study 1, we analyzed cross-cultural descriptions (N=341) of specialists in 47 cultures in the Human Relations Area Files. In study 2, we conducted fieldwork with Tanzanian Maasai pastoralists. We interviewed religious and non-religious medicinal specialists, and their non-specialist clientele (N=84).

Results: In study 1, specialists used supernatural theories of disease in 74% of ethnographic texts, but they also fulfilled practical and transactional roles for their clientele, whose patronage was typically based on expectations for efficacious treatment. In study 2, we found that in the event of illness, clients selected specialists based primarily on efficacy, but also on religious identity and interpersonal trust. We also found that participants explained how medicinal plants work with highly abstract concepts that were not conspicuously supernatural.

Conclusions: Reviewing these findings and some other ethnographic examples, we propose that, in many cases, (1) supernatural theories of disease are useful, abstract falsehoods that help explain complex and uncertain processes, such as illness; (2) specialists use these falsehoods to assist their clientele in a mutually beneficial exchange, typically of services for resources; and (3) "religious" clients defer to specialists in a market, where patronage is conditional on receiving the most effective solutions available for an uncertain and serious problem.

Parallel 2A: 'Marriage'

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Parallel 2A: 15:20 Wednesday 20.04.2022

Sexual conflict and spousal age gaps: a conceptual replication of Lawson et al. (2020) using data from network-structured games in four communities in rural Colombia

Across cultures, women marry older men. This is thought to reflect mutual choice by men and women---research suggests that men's preferences evolved to prioritize youth and fertility, while women's preferences evolved to prioritize wealth and status, which tend to characterize older men. However, recent evolutionary theory has advanced the idea that such preferences are unlikely to be static or universal. Moreover, data shows that spousal age-gaps are larger in regions where gender disparities are higher, and that women married to men much older than themselves suffer poor well-being. To resolve this discrepancy between theory and data, Lawson et al. (2020) proposed that large age-gaps could reflect sexual conflict between men and women. Under this scenario, women marry men older than they prefer, and consequently suffer lowered well-being. However, Lawson et al. find no evidence as such in Tanzania. We conceptually replicate this test in four communities in Colombia, utilizing data from a mixture of experimental games, marriage networks and household interviews. We show that realized age-gaps are not large, and are consistent with the preferences of both men and women, at least during the early reproductive period. We also show that large age-gaps are not associated with fertility or well-being in three of four communities. Our results imply neither a sexual conflict nor a mutual benefits model, and we contribute to a growing body of research in evolutionary anthropology, that shows how seemingly harmful behavioral phenomena can be neutral or beneficial in some cultural contexts, even if detrimental in others.

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Parallel 2A: 15:40 Wednesday 20.04.2022

Mating market dynamics on partner preference and relationship quality among Namibian agro-pastoralists

Mate value is thought to be a crucial component of mate choice in humans, but the majority of research examines reported preference rather than actual behavior. Likewise, little research has been done connecting mate value and partner preferences to relationship quality. Using >10,000 ratings from opposite-sex in-group members in a small scale population of Namibian agro-pastoralists, we examine how mate value discrepancies impact mate preferences and marital and non-marital relationship dynamics. Our results indicate that, while people exhibit aspirational preferences, with inclinations for partners of higher mate value than themselves, real-world relationships are composed of

individuals with similar mate value. This is illustrated through a positive correlation in mate value of individuals in marital and non-marital relationships. Additionally, we find that partners of higher mate value disparity have lower relationship quality, measured through relationship length, frequency of interactions, and reported sexual histories. This study highlights how aspirational preferences, when operationalized in a mating hierarchy, result in assortative partner matching.

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Parallel 2A: 16:00 Wednesday 20.04.2022

The impact of extra-pair partnerships on men's reproductive success

Men's reproductive success is typically enhanced by having multiple partners. Both polygyny and serial monogamy have consistently been shown to correlate with higher male fertility. However, little is known about the effects of extra-marital partnerships on men's reproductive success. A similar pattern might be expected, where marital and extra-marital partnerships have additive effects. But there are other possibilities, including a substitutive relationship between marital and extra-marital fitness, reflecting trade-offs between mating and parenting. A third possibility is that men switch strategies across their lifespans, with the gains and losses of extra-pair partnerships balancing out as they age. Here, we use a detailed demographic database on romantic partnerships among Himba pastoralists to understand the effects of extra-pair mating on reproductive success and, at the population level, on reproductive skew. We find the lifespan perspective to be the best fit for Himba men. We then explore the reasons why gains and losses might balance each other out in this population, and how the role of extra-pair partnerships fits within the broader contexts of marriage and reproduction in this population.

Parallel 2B: 'Risk'

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Parallel 2B: 15:20 Wednesday 20.04.2022

Dynamics of risk regulates cooperation in an incomplete-information game with bandit payoff feedback

Objective: In nature, reward functions associated with social interactions are often opaque to individual decision makers. Also, reward realised may be a subject of noise, making the situation resembling a socially dependent multi-armed bandit problem where expected rewards from behavioural options depend on other players' behaviour. Here we investigated how risk aversion of payoff-based learners can regulate the cooperation dynamics in a noisy, incomplete-information game.

Methods: We used an agent-based simulation of simple reinforcement learners performing a noisy Producer-Scrounger game. When they choose to 'produce' with a small individual cost, reward may be produced with some noise; whereas 'scrounging' is free but generates no reward. The total reward accumulated is shared evenly among the group members. Importantly, the average reward generated by a producing act is marginally diminishing with increasing number of producers, making the ESS based on average rewards a coexistence of both producing and scrounging tactics.

Results and Conclusions: Risk-aversion of reinforcement learning accounted for the group's cooperation-level. We identified two different sources of risk underlying the Producer-Scrounger dynamics: First, producing was riskier than scrounging when other players seldom produced, resulting in self-enhancing of suboptimal risk aversion (i.e., zero cooperation). In contrast, scrounging could become riskier than producing when other players were explorative, resulting in maintenance of the higher cooperation-level. Our result suggests that the profile of risks associated with behavioural options can be dependent upon collective behavioural states, leading to different cooperation-levels due to the hot stove effect.

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Parallel 2B: 15:40 Wednesday 20.04.2022

Pathogen avoidance and conformity: Salient infectious disease does not increase conformity

Objective: By conforming to ingroup norms, individuals coordinate with other group members, preserve cohesion, and avoid costs of exclusion. Experimental studies have shown that increased concerns about infectious disease increases conformity. However, coordination with other group members has myriad benefits, most of which exist independent of pathogenic infection. Accordingly, a strong causal effect of pathogen avoidance motivations on conformity appears unlikely. We performed multiple experiments to estimate the effect of pathogen avoidance motivations on conformity.

Methods: We performed five random-assignment experiments with samples from the Netherlands and US (total N = 1931). All experiments involved a manipulation of pathogen avoidance motivations (e.g., by showing images that evoke pathogen disgust) and measured conformity. **Results:** In line with this theoretical concern, results showed no support for the hypothesis that experimentally increasing perceived infection risk influences conformity.

Conclusions. The findings do not support the hypothesis that the mind contains a fast-acting psychological mechanism that regulates conformity as a function of pathogen avoidance motivations. More broadly, the findings are inconsistent with the notion that such a mechanism explains the associations between the prevalence of infectious disease and conformity observed at the cross-cultural level.

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Parallel 2B: 16:00 Wednesday 20.04.2022

Interpersonal trust and prosociality in volcanic environments

Previous studies have shown considerable plasticity in human sociality depending on the environmental context. For example, environmental adversities such as deprivation have been linked to reduced trust and pro-sociality towards strangers. Nevertheless, this link has not been investigated with adversities deriving from natural threats, where the adversity is not immediate, but in an unpredictable time in the future. With the ongoing climate change and the correlated higher incidence of extreme weather events, it is of primary importance to understand how living with natural threats may influence our behaviour.

We will present results on trust and pro-sociality in populations living under volcanic threat, and compare them to populations of the same culture that are not subjected to volcanic activity. Based on the literature, we hypothesised that proximity to active volcanoes would negatively influence interpersonal trust and, consequently, social tolerance and pro-sociality. We measured the perceived and actual risks associated with volcanic eruptions and investigated their role in a broad spectrum of behaviours. In particular, we used a combination of 1) questionnaires; 2) behavioural experiments; and 3) field observations to measure i) trust and general prosocial tendencies; ii) community identity and parochialism; iii) social capital and iv) risk-taking, in populations subjected to different degrees of volcanic risk. We will discuss how living with unpredictable future adversities impacts our social behaviours.

Parallel 3A: 'Social learning'

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Parallel 3A: 11:40 Thursday 21.04.2022

Memory in cultural evolution: Implications for social learning in fluctuating environments

The ability to remember many cultural traits, and the context in which it is best to use them, is a clear feature of human cognition. However, this ability is rarely considered by models of the evolution of human social learning. Without memory many models of the evolution of social learning have found that the benefit of innovation or asocial learning increases with environmental variability. In other words, social learning is most useful in more stable environments where individuals are likely to learn useful, not outdated, information.

Our agent-based model first considers the role of memory in the evolution of social learning. With memory, agents are able to store information about traits they have previously learned and performed – for example how beneficial the trait was in a particular environmental context – and recall this information at a later stage. Considering the importance of environmental change in previous models, the environment can change probabilistically between time steps and agents can use stored information to assess the likely best trait to use in their current environment.

Drawing on research on the cognitive science of memory, we formulate forgetting information as a probabilistic process whereby rarely used cultural traits are more easily forgotten than those that are frequently used. Here, the ability to store information (or the propensity to forget it) evolves alongside the propensity to socially learn cultural traits or innovate them.

We find that memory affects cultural repertoires, and the propensity to learn and innovate in multifaceted ways. First, large memories limit the need for innovation in rapidly changing environments by increasing the probability that individuals (or others from whom they may learn) carry adaptive information even after an environmental change. This is driven by the presence of large reservoirs of unexpressed but persistent cultural variation. Second, we explore the structure of 'ideal' cultural repertoires that balance the probability of using the wrong cultural trait in a particular environment with retaining previously beneficial traits.

This mechanistic approach to memory in cultural evolution delivers new results and insights into the evolution of social learning in fluctuating environments.

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Parallel 3A: 12:00 Thursday 21.04.2022

Using successful people as examples of how to behave and how not to behave

Objective: Small differences in social learning strategies at the individual level can significantly influence cultural evolutionary dynamics at the population level. Unfortunately, we do not know how complex social learning strategies are and how they vary across individuals and situations. Focusing on success-biased social learning, we studied social learning complexity and heterogeneity at the individual level.

Methods: In an incentivized experiment, we asked participants to make a decision after observing a successful demonstrator's choice, the group affiliation of the demonstrator, and additional information about the meaning of the demonstrator's group affiliation. We tested how all three pieces of information feed into social learning strategies. We built a companion gene-culture coevolutionary model in which cognition could handle the same three types of social information, as well as private information.

Results: Both the model and experiment show that social learners adjust how they respond to a successful demonstrator based on the demonstrator's group affiliation and the meaning of group affiliations. However, these adjustments are symmetric in the model but not the experiment. Holding the value of social information constant, experimental social learners perform best when observing successful ingroup demonstrators when shared group affiliation predicts similar decision-making environments. Further, results show tremendous heterogeneity

at the individual level.

Conclusion: Our results show that people adjust to success-dependent social information in complex and heterogeneous ways that include the use of successful people as negative examples.

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Parallel 3A: 12:20 Thursday 21.04.2022

Paradox of diversity in the collective brain

Objective: Human societies are collective brains. People within every society have cultural brains—brains that have evolved to selectively seek out adaptive knowledge and socially transmit solutions. Innovations emerge at a population level through the transmission of serendipitous mistakes, incremental improvements and novel recombinations. The rate of innovation through these mechanisms is a function of (1) a society's size and interconnectedness (sociality), which affects the number of models available for learning; (2) fidelity of information transmission, which affects how much information is lost during social learning and (3) cultural trait diversity, which affects the range of possible solutions available for recombination. All three levers can increase and harm innovation by creating challenges around coordination, conformity and communication. Cultural trait diversity offers the largest potential for empowering innovation but also poses difficult challenges at the societal and organizational level - 'the paradox of diversity'. We introduce 'cultural evolvability' as a framework for understanding and resolving the paradox.

Methods: We review the literature associated with the paradox in societies and organizations. We present a formal model of the trade-off between increasing specialization and coordination efficiency that arises from the division of labour. While specialisation (diversity) allows for more knowledge and new ideas, the flow of ideas becomes stymied as network efficiency (coordination) decreases.

Conclusions: Using the cultural evolvability framework, we show the implications of the paradox for entrepreneurship, polarization, and cultural clustering. We conclude by presenting ways to resolve the paradox by finding common goals and sources of information; improving communication and translation opportunities; supporting equal access to education; encouraging collaborative working, and managing perceptions of zero-sum competition.

Parallel 3B: 'Family & fertility'

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Parallel 3B: 11:40 Thursday 21.04.2022

Fostering, adoption, and parental absence in Vanuatu

Objective: Alloparenting, where people provide care to children who are not their biological offspring, is a key aspect of human social behaviour. In Pacific Islander societies, many children are adopted or fostered, meaning they are transferred out of the natal home to live with custodial alloparents, even when both biological parents are still alive. While this topic has been debated in the past, little is known about how these arrangements come about today, how they affect children, and who provides care.

Methods: Here, we report findings from a survey carried out with the caregivers of N = 282 children in rural areas of Vanuatu, an island nation in Melanesia.

Results: Children who lived with custodial alloparents were more likely to live with extended kin such as aunts, uncles, and grandparents, usually maternal relatives (83.7%), than unrelated caregivers. Contrary to previous suggestions, we do not find that custodial alloparenting serves to modify large family sizes; instead, the most common reasons were separation of parents (16.7%), parents engaging in labour migration (27.1%), or a combination of both (27.1%). Results for children's outcomes were ambiguous; while fostered and adopted children were less likely to participate in home reading assignments, no differences were found in school performance and truancy.

Conclusions: Our findings suggest that fostering and adoption help families adapt to socio-economic transitions. Living with maternal kin may protect ni-Vanuatu children against some of the adverse outcomes that are associated with parental absence in other settings.

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Parallel 3B: Thursday 21.04.2022

Identifying the barriers to reproduction in the UK: a qualitative study

Evolutionary demographers are still puzzled by the marked low fertility observed in higher income countries. While high-income countries vary in fertility rates and in the potential reasons for the decline, they also report a 'fertility gap'. While people are choosing to have much smaller families, the fertility gap is that proportion of men and women who end up with fewer children than they wanted. Much of this loss is driven by delays to reproduction; leaving it too late. Personal and societal constraints mean that some people aren't able to attain their fertility goals and observational studies have not been able to establish what these barriers are, or how they interact. This paper is the first in a two-part study to identify the barriers to reproduction in low-fertility contexts; what is causing this delay? Here we present findings from initial qualitative data collection from a series of focus groups conducted in the UK among men and women, and stratified by socioeconomic position (measured as university-educated or not). Our results show that all groups except non-university men value social support most highly, although groups differed by whether this was support from partners or parents. University women are uniquely most concerned with career

opportunity costs and non-university men are most concerned with the financial burden. This study constitutes the most up-to-date data we have from the UK and it highlights the value of qualitative data collection when addressing such a nuanced and multi-faceted question as this.

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Parallel 3B: 12:20 Thursday 21.04.2022

The presence and role of kin in maternal social networks in the UK – evidence from the COVID-19 pandemic

Objective: Kin are important supporters in the postnatal period, providing a range of support to mothers across domains. The COVID-19 pandemic in the UK saw widespread disruption to postnatal institutional support services, childcare and educational settings, likely making mothers both more in need of kin support. Here we present findings from an ongoing longitudinal social network study of UK mothers with infants aged under 15 months in late 2020.

Methods: Participants self-reported via online survey up to 25 individuals of importance to them, their relationship, relationships between those named, and the amount of support they provided across multiple domains in late 2020, February 2021, September 2021, and January 2022.

Results: Mothers (n=572) reported an average network size of 9.6, containing 32.3% consanguineal and 15.6% affinal kin, with network structure varying by socioeconomic status. Preliminary results from late 2020 indicate that: partners made up the majority of domestic supporters; whether allofeeders and allocarers were predominantly composed of partners or consanguineal kin depended on socioeconomic status; most emotional supporters were non-kin. Kin were more likely to provide each type of support than non-kin. Number of kin in their network positively predicted hours of childcare support: having one kin member predicted 20 hours, 4 kin (50th percentile) 25 hours, and 9 kin (95th percentile) 34 hours.

Conclusions: New mothers were accessing a range of support from beyond the nuclear family, particularly their own kin, while under socialising restrictions, highlighting the continued cooperative nature of childrearing in a weird WEIRD setting during the pandemic.

Parallel 4A: 'Game theory'

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Parallel 4A: 11:40 Friday 22.04.2022

Introspection dynamics: A simple model of counterfactual learning in asymmetric games

Traditionally, evolutionary game theory captures strategic interactions with symmetric games. In symmetric games, individuals are assumed to have access to the same strategies, and they experience the same payoff consequences. As a result, individuals can learn more profitable strategies by imitating others. However, interactions are often asymmetric. In that case, imitation may be infeasible (because individuals differ in the strategies they can use), or it may be undesirable (because individuals differ in their incentives to use a strategy).

Here, we consider an alternative strategy update process, introspection dynamics, which applies to all asymmetric games. In introspection dynamics, individuals consider an alternative random strategy from their set of possible strategies. Then, they compare this new strategy to their current one. A strategy that yields a payoff advantage will more likely be adopted.

In this work, we formalize introspection dynamics for pairwise games. We derive simple and explicit formulas for the abundance of each strategy over time, and apply this model to well-known social dilemmas. In particular, we analyse the asymmetric volunteer's timing dilemma. In this game, individuals can choose when (earlier or later) to volunteer for a cooperative action. At least one contributor is required to generate a collective benefit. Importantly, this benefit is degrading with time. We show that the player with the lowest contribution cost tends to cooperate without delay.

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Parallel 4A: 12:00 Friday 22.04.2022

Optimal sharing in social dilemmas

When considering biological settings where cooperation is necessary to achieve a goal, the public goods game is frequently used to model the strategic aspects of these interactions. This implies that individuals gain equally from completing the task. For example, in the case of hunters teaming up to form a hunting party, this would imply an equal sharing of the prey independent of individual contributions. However, while providing a benchmark case, such an assumption might be misleading in a joint production setting where flexible sharing is a possibility.

Nature is rich in examples where goods are not shared equally but according to individual contributions: public-goods producers in cancer cells develop more efficient mechanisms of the goods consumption, Capuchin monkeys demand fair payoffs according to the individual effort, successful hunters gain a larger share of the game in many hunter-gatherer communities, and victors obtain higher spoils in conflicts among groups. Yet, by modelling such joint interactions as a public goods game, it is explicitly or implicitly assumed in many evolutionary models that returns are divided equally among all group members. We relax this assumption by considering a linear social dilemma where a publicly provided good can be shared flexibly among contributors. This corresponds to a linear public goods game when sharing is restricted to be equal. We derive exact conditions for when full cooperation is feasible and characterize the corresponding optimal sharing rule. We find that this rule not only makes full cooperation easier, it also allows for cooperation among extremely unequal individuals. In sharp contrast to a pure

public goods setting, inequality of players makes it easier to sustain full cooperation. If cooperation possible at all, there is generally a continuum of sharing rules that can sustain cooperation leaving room for bargaining opportunities.

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Parallel 4A: 12:20 Friday 22.04.2022

Sibling competition and dispersal drive sex differences in religious celibacy

Objective: Recent work has shown that inducing a son to become a celibate monk is adaptive in a Tibetan population, as men with a monk son have more grandchildren through the monk's brothers. A model has elucidated the inclusive fitness costs and benefits associated with this decision. However, in the context of the major world religions, celibates are not always male, with female celibates being numerous in some cases.

Methods: Here, we develop an inclusive fitness model of the evolution of male and female celibacy to investigate what drives variation in their relative frequency.

Results: We find that more monks than nuns are favoured when dispersal is female biased, whereas more nuns are expected under male-biased dispersal. Second, we find that more monks than nuns are favoured when competition over parental resources is more intense between brothers than sisters, that is when wealth is key for male reproductive success. More nuns are expected when sister competition is more intense. Moreover, we find a role for the extent to which brothers and sisters compete over the same resources, with intermediate levels leading to more extreme differences.

Conclusions: These results help us explain variation in the frequency of monks and nuns, with patrilocal bride price societies having almost only monks (e.g. Tibet), patrilineal societies with moderate dowry inflation having roughly equal numbers of monks and nuns (e.g. Medieval England) and societies with severe dowry inflation showing a preponderance of female celibates (e.g. Early Modern Italy).

Parallel 4B: 'Mating II'

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Parallel 4B: 11:40 Friday 22.04.2022

Analyzing psychological effects of hormonal contraception: Causal effects, selection effects, attrition effects, or reverse causality?

Objective. Female reproductive behavior is regulated by hormones, but hormonal contraceptives interfere with these endocrine mechanisms. Scientific evidence on psychological side effects of hormonal contraception is mixed, partly because causal claims are often based on correlational data. In correlational studies, possible causal effects can not be separated from selection effects, attrition effects, and reverse causality. By performing robustness and sensitivity analyses we addressed potential issues of observed and unobserved selection effects in correlational data

Methods. We studied a sample of up to 1,179 women (42% using hormonal contraceptives) to investigate potential effects of contraceptive use on relationship quality and sexual functioning. Outcome measures were based on a 70 day long diary period. To estimate the sensitivity of reported effects in the light of potential unobserved confounders we used a sensitivity analysis that is an extended version of the omitted variable bias framework.

Results. No evidence for substantial effects of hormonal contraception was found except for a positive effect on frequency of vaginal intercourse and a negative effect on frequency of masturbation. These effects were robust to the inclusion of observed confounders, and their sensitivity to unobserved confounders was estimated.

Conclusion. Our correlational study was able to disentangle, to some extent, causal effects of hormonal contraceptives from selection effects. To address attrition effects and reverse causality we present preliminary evidence from a preregistered study based on a longitudinal dataset.

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Parallel 4B: 12:00 Friday 22.04.2022

Do women's intrasexual social comparison and jealousy change across the ovulatory cycle?

Objectives: Intrasexual competition – social status relations among same-sex individuals – is an adaptive behavior for attaining limited social and reproductive resources. Recent research suggests that women's intrasexual competition changes across the ovulatory cycle, potentially to enhance their reproductive success through mate attraction and retention. More precisely, when fertile, women seem to engage in more competitor derogation and report more relationship jealousy. However, due to mixed findings and criticism addressing methodological shortcomings, a clear understanding of ovulatory cycle effects on intrasexual competition remains elusive.

Methods: In this pre-registered, longitudinal study, 257 women self-reported their current feelings of intrasexual social comparison and jealousy in four sessions spaced across their ovulatory cycle. Further, they were asked to rate the attractiveness of 80 facial pictures of other women and men. Conception risk was estimated using luteinizing hormone tests and we assayed salivary hormones in each session.

Results: Multilevel analyses revealed no compelling evidence for enhanced competitor derogation, self-reported intrasexual social comparison, or jealousy when conception risk was high, or in relation to estradiol and progesterone levels. Rather, women higher in intrasexual social comparison seem to rate other women's faces as more attractive, regardless of conception risk or hormone levels.

Discussion: Results from this study did not support a mid-cycle peak in intrasexual competitiveness. We discuss how our results are, however, in line with the female rivalry hypothesis and that women higher in intrasexual social comparison might be more attentive to the attractiveness of potential rivals.

Speedtalks

Speedtalk 1

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Speedtalk 1: 15:20 Thursday 21.04.2022

Teaching strategies are shaped by experience with formal education: Experimental evidence from caregiver-child dyads in two Tannese communities

Objective: Humans uniquely rely on cumulative culture to act upon and make sense of our environment. Teaching is a social learning process thought fundamental to cumulative culture in our species. However, frequencies and forms of teaching vary across socio-cultural contexts. Understanding this variation improves our understanding of the interplay between cognition and culture in shaping learning behavior and contributes to theory around the costs and benefits of different social learning processes in the context of cumulative cultural evolution.

Methods: In this study, we examined how formal education experience relates to the frequency and diversity of teaching behaviors in an experimental paradigm where caregivers were motivated (but not instructed) to teach a simple skill—folding an origami figure—to their child (7-10 years-old). We identified and coded a suite of subtle non-verbal behaviors (e.g. pointing gestures, nodding, visual attention, pausing) construed as facilitating learning. Dyads (n=64) were recruited from two communities on the island of Tanna in the South Pacific nation of Vanuatu that differ in their experience with formal schooling and their acceptance of Western institutions generally.

Results: We found evidence for parallel teaching strategies in both communities. However, the rate and diversity of teaching behaviors were positively associated with caregiver's experience with formal schooling and independently and negatively associated with being from a village that rejects Western-derived institutions.

Conclusions: These results further our understanding of how multiple cultural processes influence variation in social learning, and in particular highlights the powerful influence of formal schooling on the cultural evolution of teaching in human populations.

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Speedtalk 1: 15:26 Thursday 21.04.2022

Cheater detection at a glance? Testing the accuracy of trustworthiness detection from faces

Objective: Placing trust in others can lead to beneficial outcomes, but also to costly exploitation. Forming accurate judgments about the trustworthiness of unknown others is therefore crucial in social interactions. People readily form and rely on trustworthiness judgements based on others' facial appearance, but whether these judgments are accurate remains debated and prior studies have yielded mixed results.

Method: In two fully incentivized studies, examining investment decisions in a trust game (Study 1, n = 131) and predictions of trustworthiness (Study 2, n = 266), we tested whether people can detect the trustworthiness of interaction partners based on their facial appearance. We addressed two key limitations of previous studies (a) by conducting Bayesian analyses that can quantify evidence in favor of the null hypothesis (i.e., chance-level accuracy) and (b) by examining the robustness of proposed moderators, which emerged inconsistently across prior studies (i.e., cropped vs. uncropped faces, male vs. female targets).

Results: Although participants did rely on the facial appearance of interaction partners, their trustworthiness judgments were not accurate: Both incentivized trust decisions (Study 1) and predictions of trustworthiness (Study 2) were unrelated to counterparts' actual trustworthiness. Bayesian analyses yielded strong support for the null. Moreover, accuracy was not moderated by stimulus type (cropped vs. uncropped faces) or counterparts' gender.

Conclusions: Overall, our findings suggest that people are unable to detect the trustworthiness of strangers based on their facial appearance. Even though participants' trustworthiness judgments were not accurate, they relied on them when making trust decisions.

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Speedtalk 1: 15:32 Thursday 21.04.2022

Female friendship and the horizontal transmission of low fertility values

The demographic transition - how and why most populations are having fewer offspring despite resources increasing - remains a mystery. A shift from vertical transmission (learning from parents) to horizontal transmission (learning from peers) is deemed instrumental in the quick spread of low fertility values, but we lack an explanation for how such a shift occurs. Women's personal advice networks are places where ideas about fertility practices are exchanged, therefore investigating ego networks is key to understanding this change. We expect that the expansion of markets and education into rural areas leads to shifts away from valuing kin relationships towards valuing non-kin, and from kin dominated networks to non-kin networks. We expect ego characteristics to influence these friendship dynamics. Here, using comprehensive demographic and relational data from 22 communities in rural Poland, we investigated how the characteristics of women's personal networks change with ego-level and community-level attributes, and how network characteristics affect cultural transmission. We focused on structural features of networks, frequencies and intensities of interaction and bi-lateral social support, and closeness and emotional support. We found that women characterized by market integrated values (i.e. high education, non-farming occupation, status) have less kin and more unrelated friends in their networks, who they rely on for emotional support. Women characterized by traditional rural values rely on genetic kin and

affines, for practical and financial support. These networks of different compositions, characterized by different inter-personal relationships, moderate information transmission and help explain the spread of low fertility values.

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Speedtalk 1: 15:38 Thursday 21.04.2022

Macro-innovations create room for micro-innovations: The discovery process in video game speedruns

Objectives: Innovation processes are generally an understudied area in cultural evolution (Miu et al., 2018; Perry et al., 2021; see Mokyr, 1990 for a precursor). This paper builds upon the distinction between macro-innovations (or “leaps”) and micro-innovations (“tweaks”) to test several hypotheses about the punctuated nature of cultural innovation. For this, we use a large dataset of video game speedruns from www.speedrun.com, including 20,521 games. Speedrunning is a practice of completing a game as quickly as possible and requires exploring and exploiting game logic: e.g. bugs can quicken game completion.

Methods: We operationalized “innovations” as any reduction to the best game time. Macro-innovations were then defined as unusually large innovations, found via kNN outlier detection. We pre-registered and tested three hypotheses. (1) Macro-innovations, being cultural disruptions, are expected to open new opportunities for micro-innovations. As such, micro-innovations should primarily be concentrated following a macro-innovation. (2) Macro-innovations don’t result from larger effort: they are serendipitous and cannot be explained by the amount of preceding attempts. (3) Following the theory of cumulative cultural evolution, we expected larger gamer communities to produce more innovations.

Results: Using mixed-effect modeling with various robustness checks, we found support for all three hypotheses.

Conclusions: This study supports the punctuated equilibria view of cultural innovation. Also, it stresses that datasets of recent culture are suitable for studying cultural change: speedrunning happens in restricted “history sandboxes” illustrating how innovations occur in culture at large.

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Speedtalk 1: 15:44 Thursday 21.04.2022

Group singing is globally dominant, but solo singing reflects social complexity

Objective: Singing is involved in the most crucial events defining social life, including religious ceremonies, rites of passage and funerals. The structure of such group interactions has been hypothesized to reflect how cultures perceive and act in the world. One important feature of music performance is its sociovocal organization, ranging from a single unaccompanied soloist to a multi-part chorus. It was previously hypothesized that this feature is related to social organization, so that emphasis on solo singing is more common in large-scale hierarchical societies, while levelled group singing is more common in small-scale egalitarian ones.

Methods: We analyze two global song corpora: Cantometrics (5,778 coded recordings from 992 societies) and NHS Ethnography (4,709 coded ethnographic texts from 60 societies). We examine the relationship between the proportion of group singing per society and measures of social complexity taken from the Standard Cross-Cultural Sample.

Results: We find that in both corpora, group singing is the most common form of performance overall and in most societies. Social complexity - particularly aspects related to social and technological differentiation - is associated with a greater prevalence of solo singing and more leader-dominant group singing.

Conclusions: These results support previous proposals that music is primarily a communal phenomenon and that sociovocal organization and social complexity are functionally related. Future phylogenetic analyses may help to refine these results while controlling for historical relationships among societies.

Singh M, Zeng TC, Aygen A, Bollée J, Choi J, Kaur N, Kunaparaju A, Mackiel A, Nhuquynh N, Owens A, Quintana S, Sanders S, Stasta C, Xue M, Yin H, Barrett J, Moya C, Baumard N, Blasi D, Chaudhuri P, Crockett M, Mehr S, Tamir D, Tangherlini T, Tehrani J

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Speedtalk 1: 15:50 Thursday 21.04.2022

Introducing Anansi, a global story database

For more than a century, scholars have compared stories with the goal of identifying universal narrative patterns. Despite the overwhelming interest, however, these projects have been stymied by a lack of richly annotated databases of the world’s stories. In this presentation, I introduce Anansi, a global database of more than 1,500 stories and 10,000 characters, named for the West African trickster who delivered literature to humanity. Our aim with Anansi is to produce a resource that can be used both (1) to understand global patterns in narrative and (2) to study psychology and society in contexts that are otherwise nearly impossible to investigate. Including myths, folktales, and legends from 76 diverse cultures, the stories of Anansi are each manually coded for >25 story-level variables and >60 character-level variables, providing rich characterizations of plot, character, emotion, and other content. The manual coding is supplemented by automated natural language processing (NLP) techniques, facilitating comparisons with large literary corpora. Depending on how much we get done in the next four months, this talk will feature one or more of the following ongoing analyses: (A) comparisons of topics, pronoun use, and sentiment between Anansi and Western literary corpora; (B) investigation of the structure underlying variation in the world’s stories; and/or (C) investigation of the features characterizing moral stories.

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Speedtalk 1: 15:56 Thursday 21.04.2022

Maternal grandmothers act as buffer against emotional and behavioral problems in children with many adverse early life experiences

Objective: Grandparents can increase their inclusive fitness by investing in their grandchildren. Although grandparental investment can be regarded as human universal, the outcome of such investment tends to vary from one social environment to another and between grandparent types. We examined whether higher grandparental investment buffered against the negative influence of many adverse early life events experienced by the grandchildren on their emotional and behavioural wellbeing, and whether this varied between different grandparent types.

Methods: We used population-based survey of British adolescents aged 11-16 years old and applied instrumental variable regression approach that provides more causally interpretable results in the case of endogeneity (e.g. measurement error and confounding).

Results: Our findings showed that the investment of maternal grandmothers only modified how badly grandchildren's emotional and behavioural scores were moulded if they had experienced several adverse early life events in their life. The higher the investment from maternal grandmothers, the milder the influence of adverse early on life experiences on grandchildren's emotional and behavioural problems. Conclusions: Our results highlight the importance of maternal grandparents as the main investors in their grandchildren and how that investment can protect their grandchildren from showing emotional and behavioural problems related to harsh developmental environment.

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Speedtalk 1: 16:02 Thursday 21.04.2022

What is the expected human childhood? Insights from evolutionary anthropology

Objective: In psychological research, there are often assumptions about the conditions that children expect to encounter during development. These assumptions shape prevailing ideas about the experiences that children are capable of adjusting to, and whether their responses are viewed as impairments or adaptations. Specifically, the expected childhood is often depicted as nurturing and safe, and characterized by high levels of caregiver investment. Here, we synthesize evidence from history, anthropology, and primatology to challenge this view.

Methods: We review the findings of systematic reviews, meta-analyses, and cross-cultural investigations on forms of threat — infanticide, violent conflict, and predation — and forms of deprivation — social, cognitive, and nutritional — that children have faced throughout human evolution.

Results: Results show that mean levels of threat and deprivation were higher than is typical in industrialized societies, and that our species has experienced much variation in the levels of these adversities across space and time. These conditions likely favored a high degree of phenotypic plasticity, or the ability to tailor development to different conditions.

Conclusions: This body of evidence has implications for recognizing developmental adaptations to adversity, for cultural variation in responses to adverse experiences (e.g., PTSD), and for definitions of adversity and deprivation as deviation from the expected human childhood.

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Speedtalk 1: 16:08 Thursday 21.04.2022

Understanding women's support for patriarchal norms in East Asia

Many studies have found that not only men, but also women sometimes endorse norms that may be considered as patriarchal or sexist and against women's interests. Such observations have been explained as resulting from women's socialization into a patriarchal gender ideology, resulting in their 'false consciousness'. Patriarchy likely has its roots in an evolutionary history of sexual conflict (Smuts 1995), but potential variation among women's interests in norms is often overlooked. This study attempts to critically evaluate the gendered costs and benefits of specific norms. I consider whether women themselves can sometimes benefit from supporting patriarchal norms, such as through gains from alliances with high-status men, or from benefits to male offspring (Kandiyoti 1988; Brooks and Blake 2021). Also, younger women likely experience higher costs from patriarchy than older women. This study uses data from the East Asian Social Survey to test whether these hypothesized interests are reflected in the stated attitudes in of women in China, Taiwan, South Korea and Japan.

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Speedtalk 1: 16:14 Thursday 21.04.2022

Norm internalization or mismeasurement: Are women truly more accepting of intimate partner violence than men?

Objective: A surprisingly consistent finding from the intimate partner violence (IPV) literature indicates that women self-report greater acceptance of IPV compared to men in settings where IPV is particularly common. This pattern has been widely interpreted as evidence of women's internalization of gender inequitable norms. Here, we propose an alternative explanation that this result is an artifact of men disguising true support for IPV in self-report data.

Methods: We utilize data on acceptance of IPV from couples (n=317) in Northwestern, Tanzania to assess whether women are truly more accepting of violence, or if this effect disappears when examining an indirect wife-estimated measure of men's support for IPV; believed to be less susceptible to social desirability bias.

Results: Consistent with prior studies, self-report data show that women report greater support for IPV than men do. However, this difference

is reversed when indirect wife-reported estimates on men's attitudes are considered, with men supporting IPV more than their wives. Conclusions: Our results provide a more parsimonious explanation of gender differences in support for IPV, and more generally highlight the inadequacies of self-report data on sensitive topics. We suggest the true puzzle for future research is to understanding why some women support IPV at all, rather than why they report greater support for IPV than men.

Speedtalk 2

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Speedtalk 2: 13:40 Friday 22.04.2022

The Watching Eyes Effect: Keep Britain tidy "We're watching you"

The Watching Eyes Effect (via the presence of a pair of photographic eyes on information posters), has been shown to reduce anti-social behaviour, such as, reduced cycle-bike thefts, reduced car emissions at train level crossings, and increased handwashing in public bathrooms (Meleady et al, 2017; Nettle et al, 2012; Pfattheicher et al, 2018). Between December 2013 and March 2014, the charity Keep Britain Tidy, across 17 local partner councils, used the Watching Eyes Effect in a "We're Watching You" campaign to deliver a social experiment that tested the effectiveness of watching eyes posters on the prevention of dog fouling. In 2019 we accessed the data from this field study and report the findings of this Keep Britain Tidy dog fouling to show that the Watching Eyes Effect significantly reduced dog fouls during the poster intervention when compared to pre-intervention across the target sites. We highlight the possible reputational concerns of the Watching Eyes Effect that contribute to field studies.

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Speedtalk 2: 13:46 Friday 22.04.2022

The effects of prestige and dominance on advice and friendship relations: A longitudinal study of collaborative teams

Objective: Theory suggests that status relations contribute to how cooperative relationships are established and maintained. Yet, it remains unclear as to how different forms of status pattern the different affective relationships (e.g., friendships) and instrumental (e.g., advice) connections that often mark cooperation in human groups. Here, we examine the interplay between two forms of status—prestige and dominance—and the dynamics of friendship and advice networks.

Methods: We collected data from undergraduate students (n = 401) working in randomly-assigned task groups at a North American University over a 16-week period. Peer-reports of prestige and dominance, and friendships and work-related advice networks, were collected at 4 timepoints from task groups.

Results: Longitudinal network analyses indicate that prestigious individuals were more likely to be considered by others as friends, and to be sought for work-related advice. By contrast, dominant individuals were less likely nominated as friends. Interestingly, our results suggest multiplexity between friendship and advice, with friends being more likely to ask advice from one another, and individuals reciprocating friendship with advice.

Conclusions: Our findings provide some initial evidence on the dynamic interplay between processes of status differentiation and the networks of social relationships that often characterise social groups. More specifically, our results lend support to theory on cultural evolution and the evolutionary origins of different forms of social status, as well as theory that highlights the importance of friendships for mobilising social and informational resources that are embedded within groups, and further emphasises that friendship dynamics are coupled with prestige-based status.

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Speedtalk 2: 13:52 Friday 22.04.2022

Physical activity and energetic trade-offs in the BaYaka hunter-gatherers

Objective: This research aims to contribute to a wider understanding of hunter-gatherer physical activity and energetic trade-offs. Research into hunter-gatherer physical activity to date has shown a large disparity with post-industrial populations. A model of constrained energy expenditure would predict differing trade-offs in energy allocation between different groups within a population but attempts to explore these differences have been limited by their scope and scale.

Methods: In this study physical activity data was collected from a sample of 126 BaYaka hunter-gatherers using wrist-worn accelerometers. From which the role of sex and age in relation to physical activity patterns were explored.

Results: Overall activity levels were high, with adults partaking in over 3hrs/day of moderate-to-vigorous physical activity. For adults (ages 16-50), males were observed to undertake more activity than females, but this difference did not result from different volumes of moderate-to-vigorous physical activity. Additionally, activity was seen to vary between age groups, with movement highest amongst young adults (ages 16-25) and lowest amongst individuals aged 50+.

Conclusions: Like other hunter-gatherer populations, the BaYaka are highly active. A sex difference between adult activity levels may either indicate males increasing provision or females diverting energy away from movement to reproduction. Young adults being the most active fits existing models but the high levels of physical activity observed during childhood may require a trade-off in energy allocation, which may have implications for our understanding of pygmyism if movement is being traded off against growth.

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Speedtalk 2: 13:58 Friday 22.04.2022

Cultural differences in empathy in Indonesia

Empathy is the capacity to share and understand the feelings of others, and to respond to them with care. It has often been described as an individual trait, conceptualizing it as consistent across a person's life course. Some recent research has highlighted geographic variation in empathy within and between countries, suggesting that empathy is, in part, culturally transmitted. However, to date, most studies centre on Western, educated, industrialized, rich, democratic (WEIRD) societies. To better understand this trait more broadly, we conducted an online survey of self-reported empathy across the diverse cultures and backgrounds of Indonesians. We used a combination of two standardized self-reported questionnaires to determine self-reported differences in multiple aspects of empathy Indonesia: the Interpersonal Reactivity Index (IRI) to measure individual empathy and the 2002 General Social Survey (GSS) to measure helping behaviour towards strangers and relatives. We also recorded three aspects of individuals' culture: their religion, ethnicity, and place of residence. Globally (N = 2869), we found sex and age differences in self-reported tendencies to experience feelings concern towards others (Empathic Concern); feelings of discomfort and concern when witnessing others' negative experiences (Personal Distress); and helping behaviour (to strangers and relatives). Using linear regression, we found evidence of one source of cultural influence on differences in empathy: religious beliefs significantly influenced Empathic Concern and Personal Distress in Indonesian societies, but not helping behaviour. We discuss these findings in the context of Indonesian society and the broader implications.

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Speedtalk 2: 14:04 Friday 22.04.2022

Inferring cultural evolution from language trees? Cautionary tales from the case of Indo-European

Not Darwin but Dante might lay claim to first conceiving of 'descent with modification' — albeit applied to how languages relate to each other, not species. By now, however, evolutionary modelling and Bayesian phylogenetics are making the running in linguistic methodology too. Furthermore, such family trees of languages have been imagined as a ready-made evolutionary framework for the populations and societies that spoke them, from which to infer how other aspects of their cultures must also have evolved — even if for sceptics, it all seemed too good to be true. Bayesian phylo-chronology, in particular, has sought to date the origins and divergence phases of major language families, to be set against the archaeological and ancient DNA records. Many such chronologies seem relatively uncontroversial, but not for the language lineage of almost half of the world's population: Indo-European. Alternative analysis assumptions have led to widely different date estimates, and conflicting inferences for prehistory. This paper diagnoses the methodological flaws behind such inconsistent results. I identify misconceptions between linguists and phylogeneticists on how to process language data into binarised form, as input to phylogenetic analyses — with unforeseen consequences. I illustrate with real language data how seemingly innocuous coding decisions led to egregiously inconsistent datasets, and thus wayward results, not least in branch lengths. These artefacts can be banished only by enforcing a radically new methodology for language databases, proposed here. The implications are obvious for past attempts to infer chronologies, but extend also to existing inferences on cultural evolution built upon language trees.

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Speedtalk 2: 14:10 Friday 22.04.2022

The folk concept of 'culture': What do people mean when they explain something with reference to culture?

Social scientists and philosophers have debated numerous definitions of culture. Whatever the merits of these, none of them captures all and only the ways that the term is used in ordinary language. An alternative approach is to investigate empirically what lay people mean when they invoke the culture concept; in other words, the folk concept of culture. I argue that the folk concept of culture is an explanatory stance people use to explain patterns of behaviour or cognitive contents that have enough of a set of hallmark features. These include distinctiveness (something cannot be cultural if all humans do it); diffusion (something is more cultural if it is spread over multiple individuals); persistence (something is more cultural if it is stubbornly difficult to get rid of); redundancy (something is more cultural if it persists even when you remove key individuals); and holism (something is more cultural if it cannot be decomposed into discrete elements). Culture may also operate as a stance of last resort, to be appealed to when more instrumental explanations have failed. I investigate these claims with novel empirical studies of lay people's 'culturality' judgements. I show that the folk concept of culture is both family-resemblance (no unique set of features defines every usage), and graded (something can be more or less cultural). I argue that debates about culture in the literature are often unsatisfactory and inconclusive because they shift back and forth between technical definitions of culture, and the phenomena that fit the folk concept.

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Speedtalk 2: 14:16 Friday 22.04.2022

Attention styles in harsh and unpredictable environments

Objective: Previous research suggests that growing up in adverse conditions might enhance present- and externally-focused cognitive abilities (attention-shifting) but hurt internally-focused abilities (inhibition). However, the mechanisms underlying such effects remain unknown. For

example, the same mechanisms (e.g., vigilance) might enhance attention-shifting and hurt inhibition or different abilities. Here, we test how adversity shapes performance on three types of attention tasks, measuring both externally- and internally-focused abilities. Methods: We tested 550 participants between age 18-30 (49.1% females) from the United States via Prolific. Participants completed three attention tasks: 1) Attention Cueing Task, requiring fast responses to peripheral targets; 2) Flanker Task, requiring inhibiting incongruent targets; 3) Change Detection Task, requiring detection of subtle spatial changes. Next, participants completed self-reported measures of their childhood environment (before age 13), including violence exposure, unpredictability, and poverty. Results: We use Drift Diffusion Modeling (DDM) to decompose performance on all tasks into distinct cognitive processes (e.g., information processing efficiency and strategic trade-offs). We plan to test the effect of dimensions of adversity on DDM parameters using a multiverse approach. Specifically, we will examine main effects of adversity for change detection performance and use mixed models to test adversity X task condition interactions for cued attention and flanker performance (for our detailed preregistration, see https://anonymous.4open.science/r/preregistration1_attention/file/preregistrations) Conclusions: The current research could provide a more mechanistic understanding of associations between experiencing adversity and enhanced and impaired cognitive performance.

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Speedtalk 2: 14:22 Friday 22.04.2022

The art of nesting: ageing nests of Eastern chimpanzee at Bugoma Central Forest Reserve, Uganda

Ground nesting in chimpanzees is an aspect of their behavior which is relevant to our own ancestors' descent to the ground. We currently know very little about variation in this behavior across the chimpanzee range, although it is clearly more common than once thought. Differential decay rates between ground and tree nests may warp our perspective of their occurrence across chimpanzee range. In addition, we need to know local nest decay rates for ground nests as well as tree nests when calculating densities of this endangered species. Decay rates vary between research sites, and are affected by topography, rainfall, nest height, canopy cover, nest construction styles, species of vegetation used and possibly whether built on the ground or in the trees. Here we present preliminary results on the comparison between decay rates between ground and tree nests at the Bugoma Central Forest Reserve in Western Uganda. This represents the first systematic study to our knowledge of the decay rates of chimpanzee ground nests. We followed the decay of 56 ground and 51 tree nests decay from 7 nest groups, over a period of 1 year, collecting detailed information every week on 16 ground and 18 tree nests; we checked the remaining 73 nests 4 times. Out of 107 nests, 51% of tree nests and 64% of ground nests had decayed after 6 months. Vegetation-type appears to affect duration of nest life: tree nests built using the tree species *Omusevera* (local name) lasted more than 1 year (longer than the average), whereas ground nests constructed using saplings of the same species lasted more than 7 months. We discuss the significance of differing decay times for ground nests compared to tree nests and how it may impact our understanding not only of chimpanzee densities, but of their traditions.

Andrews J

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Speedtalk 2: 14:28 Friday 22.04.2022

Measuring market integration and its effects on the life course

With each passing year, the fraction of societies that remain unconnected to the world's vast interconnected market economy shrinks. Evolutionary anthropology can often struggle with accepting, conceptualizing, and measuring this transition. However, grappling with the transition is vital to understanding the contextual realities of the communities we work with, especially in uncovering how societies and individuals adapt to markets. This work has two goals, first is to establish a methodology for measuring the rate of market integration in transitioning societies. To do so, we use a combination of simulations, statistical modelling and ethnographic data. Second, we develop parallel methods for estimating the effect of exposure to markets on life course events, such as age at first marriage/child, total fertility, probability of separation and migration events. These models are then applied to ethnographic data from Pemba Tanzania to provide insights into how the transition towards a market economy has affected local life-course trajectories.

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Speedtalk 2: 14:34 Friday 22.04.2022

The spiteful origins of conspiratorial beliefs

Conspiratorial thinking seem to be a human universal behaviour, but the adherence to conspiracy theories has demonstrably negative consequences for the believer and for society in general. To understand why people fall victim to believing in false information, we propose that such beliefs stem from a foundational social behaviour: spite. In four pre-registered studies (N = 1,227) we test the hypothesis that the predictors of conspiratorial belief established in the literature (epistemic, existential, and social motives) are cues of competitive disadvantage that provokes a common facultative 'spiteful' psychological response; which in turn makes a person more open to conspiratorial ideas. In Study 1, we establish the relationship between spite and four previously established predictors of conspiratorial beliefs: realistic threat and in-group narcissism (social), political powerlessness (existential), and an intolerance for uncertainty (epistemic). In Study 2, we find that spite mediates the relationship between each of these predictors and belief in conspiracy theories, a pattern we replicate in Study 3. In Study 4, we directly manipulate spite and find initial evidence for a causal relationship between spiteful ideation and conspiratorial beliefs. Overall, these studies suggest that spite is a key factor for understanding why people engage with conspiracy theories. The importance of identifying and understanding the underlying foundation of conspiratorial belief is discussed.

Posters

Posters 1

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Posters 1, Red Room: 17:00 to 19:00 Tuesday 19.04.2022

A cross-cultural study of social complexity and levels of violence in war

Objective: Previous research on the relationship between social complexity and warfare indicates both that more indiscriminate violence may occur among small-scale societies due to reliance on raiding and the lack of distinction between combatants and noncombatants, or among politically complex societies engaging in violent conquest with organized militaries. This study examines these relationships through analyzing levels of wartime violence towards enemies and measures of social complexity in a global sample of ethnographic and historical societies.

Methods: I have coded variables on political structures and warfare on 73 societies from ethnographic and secondary historical sources. Using these variables, the likelihood of different enemy individuals of the focal society being targeted/killed during war will be tested for correlations with social complexity, military formalization, and violent political expansion.

Hypothesized results: It is possible that increased social complexity will be positively correlated with the scale of enemies killed by the focal society during war, and that this relationship will be mediated by the presence of formal militaries and political expansion. Alternatively, there may be a negative correlation as fighting becomes distanced from the general population. As there exists considerable variability in how these and other factors interact with warfare patterns, there may also be no discernible relationship. However, collating and examining the available data could help contribute to further research on how political institutions and wartime behavior interact cross-culturally.

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Posters 1, Orange Room: 17:00 to 19:00 Tuesday 19.04.2022

The human capital assessment mechanism and its behavioral implications

We present a theory of optimal resource allocation in humans, a behavior observed across human societies. After examining the ancestral ecology of capital availability, we propose that a capital assessment mechanism allows individuals to (1) assess the amount of capital available to them and to (2) allocate capital in a fitness maximizing manner. We define what we mean by capital and how the amount of capital available is likely to be assessed by individuals. We then show that the optimal way to allocate capital depends on the total amount of capital and describe testable predictions that follow from this theory. We discuss the psychological and behavioral implications of the capital assessment mechanism, consider some evidence for them, and outline ways to empirically validate this theory.

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Posters 1, Yellow Room: 17:00 to 19:00 Tuesday 19.04.2022

Adult Sex Ratio and dispersal state lead to sex inequality in work burden

Heavy workloads can harm health and reduce opportunities for proper childcare, potentially even transmitting harmful effects on to the next generation. This study aims to explore the impact of adult sex ratios (ASRs) and individual dispersal status after marriage on the workload gap between spouses, through using a removable activity tracker. The operational sex ratio in the local mating market, according to our findings, has played a critical role in capturing sex-specific labour patterns. Using a cross-cultural approach, we discover a significant association between female workload and female-biased operational sex ratio in the local marriage market. High sex ratios in some areas increase women's bargaining power in relationships, hence reducing their work burden, but the predominance of celibates plays a crucial role in increasing the labour burden on women by generating low operational sex ratios.

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Posters 1, Lime Room: 17:00 to 19:00 Tuesday 19.04.2022

The Psychological Profile of Incels (involuntary celibates)

Objective: There exists a dearth of primary-data collected from self-identified incels, a sub-culture of men who build their identity around their inability to form sexual or romantic relationships. This study compared a sample (n = 143) of self-identified incels with non-incels (n = 365) across a range of measures, to fill a gap in the literature.

Methods: Incels and non-incels were compared on a range of measures including interpersonal victimhood, self-perceived mate-value, reasons for being single, sociosexual-desire, depression, anxiety, loneliness, and life satisfaction.

Their accuracy in predicting female mate preferences were compared to the real preferences of a sample of women.

Moderation analyses were used to consider interactions between incel-status and sociosexual-desire on wellbeing, while cross tabulation was used to explore demographic differences between the two groups of men in ethnicity, political affiliation, and likelihood of being NEET. Results: Relative to non-incels, incels had higher-levels of interpersonal victimhood, depression, anxiety, loneliness, and life dissatisfaction, with low self-perceived mate-value and an external locus of control relating to singlehood. They also reported lower minimum standards for mate-preferences, and, when predicting female mating preferences, overestimated the importance of physical-attractiveness and underestimated the importance of intelligence, kindness, and humour. While incels scored higher on levels of sociosexual desire, this trait failed to moderate the relationship between incel-status and wellbeing. Incels were ethnically and politically diverse, and likely to be NEET. Conclusion: Incels represent a newly identified “at-risk” group to target for evolutionary-psychology-informed mental-health interventions.

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Posters 1, Red Room: 17:00 to 19:00 Tuesday 19.04.2022

Sound of Success in the Early Paleolithic: Experimental Archaeology of Lithoacoustics

Objective: Anecdotal evidence suggests that among groups of knappers a particularly “good” sounding hit attracts the attention of others, precipitating a chorus of approval. This project explores this phenomenon and its evolutionary implications. How useful was the sound of success in the Paleolithic?

Methods: 1480 recordings and 190 flakes were collected from knapping sessions conducted on basalt, quartzite and flint. Flakes will be morphometrically analysed and recordings analysed in R using the Soundgen package. Acoustic properties chosen to be explored are broadly associated with timbre in musical instruments. Wiener entropy (signal: noise ratio), dominant frequency and resonance will be extracted and compared among raw material/outcome categories. This is complemented by a listening preference test conducted via Gorilla (90 participants). Participants are tasked with rating a selection (N=60) of recordings, asking were they enjoyable and were they attention grabbing.

Results: This work is ongoing, delayed due to COVID and no results have been produced as yet. However, some predictions can be made. It is hypothesised that listeners will prefer the sound of flake-producing strikes on high-quality materials and that this will correlate with acoustic features extracted from the recordings i.e productive strikes on high-quality materials will have a higher signal: noise ratio, dominant frequencies within the human musical range and will be more resonant.

Conclusion: This project aims to quantify and explore the acoustics of stone tool-making. It is hoped that these novel experiments in lithoacoustics might yield insights into the social and technological soundscapes of early hominins and their ramifications for behavioural evolution.

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Posters 1, Orange Room: 17:00 to 19:00 Tuesday 19.04.2022

Investigating the differential role of early life unpredictability and volatility on human psychopathological development

Previous works in behavioural ecology and developmental psychopathology have shown that several dimensions of early life adverse experiences, such as deprivation, threat and uncertainty, could impair physical and mental health later in life. However, the concepts and operationalizations of uncertainty vary greatly, and more precision is needed to characterize how it modulates development. A separate line of work in cognitive neuroscience suggests that human subjects distinctly adapt their learning and decision-making strategy to unpredictable and volatile sources of uncertainty. Combining these two research strands, we aimed to explore how these two uncertainty sources, once scaled up at the time scale of human development, contribute to the emergence of psychopathology. Using data from the Fragile Families and Child Wellbeing Study, we conducted a set of pre-registered linear regressions to predict Internalising and Externalising symptoms assessed at ages 9 and 15, from indicators of Deprivation, Threat, Unpredictability and Volatility assessed at ages 3 and 5. First, comparison analyses penalizing for model complexity showed that models containing Unpredictability and Volatility effects accounted for the data considerably better than models only including Deprivation and Threat. Second, Unpredictability seemed to be more strongly associated with Internalising symptoms, whereas Volatility had a stronger effect on Externalising symptoms. Third, we observed an interaction between the two uncertainty sources, such that one counteracted the effect of the other on symptoms severity, a pattern consistent with a Volatility/Unpredictability trade-off proposed by the human reinforcement learning literature. Extensive control analyses suggest robustness of these effects.

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Posters 1, Yellow Room: 17:00 to 19:00 Tuesday 19.04.2022

Concordance and discordance in intervention behaviour within mate choice reported by parent-offspring dyads

Prior research showed that parental intervention behaviour and offspring’s defensive patterns do occur within offspring’s mate choice. However, it remains unknown whether parent-offspring perspective on own and the other’s real-life behavioural patterns are in accordance. The main aim of the current study was to test the level of concordance between parent-offspring reported acts. In particular, we focused on parental behaviour toward their offspring, their offspring’s partner, and the couple, and acts employed by offspring and their partners toward offspring’s parents.

We conducted semi-structured interviews with total 20 offspring (mean age = 31,4y; SD = 5,46) and their 20 parents (mean age = 58,05y ; SD = 7,63) consisted of 5 son-mother, 5 son-father, 5 daughter-mother, and 5 daughter-father dyads.

Thematic analysis identified 32 parental acts and 18 acts applied by offspring or offspring's partner within current or previous offspring's relationships. Concordance ratio was computed as the number of parent-offspring concordances divided by the sum of concordant and discordant dyads.

Results showed that discordance was more prevalent than concordance across most of reported acts. The highest concordance was found over neutral and helpful parental behaviour, but also when expressing critical opinions toward the offspring's partner. In the case of the offspring's/partner's behaviour, the highest concordance was found in confiding or extreme behaviour, e.g., separation from the family. The lowest concordance was revealed in one's behaviour that was done behind the other's back, e.g., maintaining contact with the offspring's ex-partner, or lying to the parent/withholding information from the parent. Our findings suggest that parental and offspring perception of intervention acts differ to a considerable degree.

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Posters 1, Lime Room: 17:00 to 19:00 Tuesday 19.04.2022

Social distance as a strategy of pathogen avoidance by women in the first trimester of pregnancy - the mediating role of disgust propensity

The behavioral immune system (BIS), represents a set of actions to avoid the spread of infectious diseases (Fincher & Thornhill, 2008; Gelfand, 2011; Murray & Schaller, 2012). Research in the social science stream has repeatedly confirmed the activity of BIS by showing that for instance, prejudice against certain groups may be mediated by disgust propensity (Golec de Zavala et al., 2017; Cottrell & Neuberg, 2005). Pregnant women (especially in the I trimester) experience immunosuppression - a reduction in the effectiveness of the immune system. The compensatory behavioral prophylaxis hypothesis assumes that during this period, women will be motivated to avoid the potential threat of infection, and the main mechanism here is also disgust propensity (Fessler, 2002; Fessler & Navarrete, 2003; Nowak & Pawlowski, 2019). However, there is no research showing the distancing of pregnant women from social stimuli.

In a correlational study conducted on 425 pregnant women (130 in Ist trimester, 191 in IIrd trimester, and 104 in IIIrd trimester) we show the relationship between gestational age (trimester) and greater social distance toward out-groups (as possible pathogen threat) mediated by disgust propensity. Our results may be a piece of initial evidence for the biological basis of prejudiced behaviors.

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Posters 1, Red Room: 17:00 to 19:00 Tuesday 19.04.2022

Creativity across the ovulatory cycle - a within subject study

Physical beauty is not accidental – it evolved through the process of sexual selection to signal a genetic fitness (Miller, 2000; Darwin, 1871). Psychological traits can serve the same function. Creativity - a monomorphic trait among humans, is strongly associated with beauty instinct. Creative potential is considered to be one of the most desirable traits in a partner of either sex (Buss, 1989; Li et al., 2002), and there is some evidence that it also translates into reproductive success (Clegg, Nettle, & Miell, 2011). But if creativity is an evolved signal, it should also be more evident when the probability of conception is high. In two studies we tested creative potential of women in three phases: early follicular (menses), ovulation and luteal (premenstrual). In Study 1 (N = 751; between subject) we found positive correlation between the probability of conception and creative originality, and flexibility of thinking. We also tested arousal as a mediator of this relationship. In Study 2 (N = 72; within subject) we replicated the effect of originality with the use of biological measures of the cycle. We also found positive emotions as a mediator between the cycle phase and flexibility of thinking. The results of our study are discussed in terms of signaling theory, due to which women advertise their fitness with their creativity. Being original can be particularly eye-catching to a potential partner.

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Posters 1, Orange Room: 17:00 to 19:00 Tuesday 19.04.2022

A bibliometric study of research topics, collaboration, and centrality in the iterated prisoner's dilemma

Changes in the publishing world, both technological and economical, have led to an increase of accessible scholarly output. This substantial amount of data offers us opportunities to explore how we conduct science. Here, we investigate the research topics and collaborative behavior of authors who work on the Prisoner's Dilemma. Thus, we ask the following questions: what topics do researchers explore, and who do these researchers write with?

We collect data on 2420 unique articles from five publishers. Initially we apply a topic modelling algorithm to the abstracts of the papers and we identify five main topics that researchers explore. In addition, we use the list of authors to generate the co-authorship network and apply a graph theoretic analysis. We show that the Prisoner's Dilemma field is more collaborative compared to other game theoretic fields. Authors create writing "communities". Each author belongs to at least one community and publishes with people from that community. We find that there are not many collaborations across these communities.

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Posters 1, Yellow Room: 17:00 to 19:00 Tuesday 19.04.2022

Health status moderates preferences for male risk takers as mates

Risk taking is a sexually selected behavioural trait that is more commonly shown by males. Risk taking has a signalling function, serving to advertise one's intrinsic quality to prospective mates of the opposite sex. Previous research has established that male risk takers are judged as more attractive for short-term flings than long-term relationships but the environmental/socioeconomic context surrounding female preferences for male physical risk takers has been overlooked. Using a Qualtrics survey distributed online through Amazon Mechanical Turk, we obtained data on 957 heterosexual females from 47 countries. Self-reported health was strongly associated with preferences for high risk takers as short-term mates but the effect was moderated by country-level health (i.e. life expectancy), such that the association was weaker in countries with lower health. These findings echo recent research on masculinity preferences which were more strongly expressed in countries with higher national health indices. The security provided by better health and access to health care may allow individuals to capitalize on the genetic quality afforded by selecting a risk prone male, whilst concurrently reducing the costs associated with the risk taker's lower paternal investment. Interestingly, the risk of contracting COVID-19 did not predict avoidance of risk takers, perhaps because this environmental cue is too novel to have moulded our behavioural preferences.

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Posters 1, Lime Room: 17:00 to 19:00 Tuesday 19.04.2022

Ground-nesting as a large-scale behavioral phenomenon in the chimpanzees of Northern DRC

Humans at some point after diverging from the genus *Pan* became predominately terrestrial, but little is known of the earliest behavioral adaptations to increased terrestriality, particularly involving sleep. Here we present data on a large and interconnected ground-nesting behavior in chimpanzees of Northern DRC. We present data on the distribution of the behavior, which extends at least to the border with Uganda, and create statistical models to investigate the impact of human hunting, carnivores and large animals, season and habitat-type on ground-nesting as well as nest height across a 50,000 km sq region. In addition, we suggest a methodology to improve classification of terrestrial resting structures, including 'leaf cushions', 'flimsy ground nests', 'complex ground nests', and 'half ground nests'. Finally, we compare the construction-styles of nests, which include different mixes of herbs and detached and attached woody elements, across the region.

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Posters 1, Red Room: 17:00 to 19:00 Tuesday 19.04.2022

Animal pyrocognition and hominin fire use

Fire transformed our evolution by providing numerous benefits – cooking in particular – but little is known about what kindled its early transition. Some animals also possess pyrocognition, the behavioural and cognitive abilities required to harness the potential of fire. These animals can serve as models for how early hominins managed this unique and dangerous phenomenon, and how its use went on to shape the creatures we are today. The meagre literature and novel preliminary results suggest that some animals (1) associate with wildfire, (2) prefer cooked food, (3) explore fire with tools, and (4) use fire as a multifunctional resource. Savannah chimpanzees and vervet monkeys get close to wildfire, and burned landscapes make it easier for them to forage, travel and detect predators. Great apes and ravens prefer cooked food, probably because it is sweeter and softer, despite it not being common in their evolutionary history. Chimpanzees and New Caledonian crows explore heat sources from a safe distance by using tools, and black kites spread wildfire by transporting burning sticks. Finally, the multifunctionality of fire is shown in a captive troop of Japanese macaques that has experienced bonfires for sixty years. They warm themselves by it, compete over access to it, eat charcoal and ash, retrieve burning sticks, and are calm and confident near it. These results support the hypothesis that hominins also possessed these pyrocognitive capacities and benefitted from wildfire long before they first actively controlled or made fire.

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Posters 1, Orange Room: 17:00 to 19:00 Tuesday 19.04.2022

Human fertility after a disaster - Systematic literature review

Objective: Fertility response to acute environmental shocks, such as disaster, is a topic of interest increasingly shared by the public, policy makers, and researchers alike. In evolutionary human science, the fertility-disaster nexus is a context for studying how fertility relates to mortality change, a key dynamic underlying life history evolution. As yet, to the best of our knowledge, no systematic review has been performed to assess the current state of our assessing to what extent disaster affects fertility rate. Moreover, an evolutionary theory of fertility responses to disaster remains relatively undeveloped.

Methods: We are conducting a systematic review of literature on the relationship between disasters (distinguished from conflicts or economic shocks) and the counts of live births and pregnancy terminations (stillbirths, miscarriages, and abortions). Methods for selection of studies and evidence synthesis have been developed before data collection (<https://osf.io/dpgqf/>). We will (1) present one component of our systematic review - the associations between disaster and fertility across populations, (2) compare hypotheses on fertility response to disaster developed in evolutionary biology and in economic and demographic research, aiming for an integrated framework.

Results: As of now, we have completed an initial round of screening, selecting 193 studies by reviewing the titles and abstracts of 1,876 articles identified from a systematic search through multiple databases. We are currently conducting full-text screening to select a final set of studies for review, based on a predefined inclusion criteria.

Conclusions: When finished, this study will provide evidence for evaluating how and why fertility might respond to disasters.

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Posters 1, Yellow Room: 17:00 to 19:00 Tuesday 19.04.2022

Cross-contextual reputation and the extent of human cooperation

Cooperation extends far beyond the confines of the give and take of reciprocity. We trust partners based on a variety of cues, pertaining to their propensity to help us as well as others, and engage in other types of prosocial behavior. This suggests that our cooperative reputation may extend beyond the specifics of a given context, and across various social dilemmas.

Here, we attempt to clarify the logic of cross-contextual reputation. We develop a repeated two-game model, which individuals play throughout their life. Individuals are characterized by a general quality (e.g. their time preferences), which affects the payoffs they derive from both games. To mimic the informational constraints of a large, fluid society, we assume individuals only have access to one instance of a prospective partners' past behavior. We show that individuals reliably infer trustworthy behavior in one context from behavior in another when games are sufficiently similar, and quality sufficiently variable. Under these conditions, reputation is cross-contextual; and cooperation in both games is stabilized.

Understanding when reputation extends across contexts can help us better understand the extent of human cooperation. When they are reliable, cross-contextual inferences allow us to take advantage of the variability of social dilemmas we face, to build-up trust with a privileged partner, or stabilize other prosocial behaviors, including contribution to public goods and prosocial punishment. Yet, reputation and cooperation cannot extend indefinitely. By clarifying the limits of cross-contextual reputation, we may better grasp when and where the range of feasible cooperation should vary.

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Posters 1, Lime Room: 17:00 to 19:00 Tuesday 19.04.2022

An Evolutionary Perspective on Emotion-Cognition Interactions

Based on an evolutionary perspective, the human mind is a collection of cognitive mechanisms that evolved during our evolutionary history to solve problems related to survival and reproduction. Many studies have distinguished between two types of mental processing: emotions and cognition. The building blocks of higher-order cognition (e.g., reasoning, planning, and problem-solving) are suggested to be top-down control processes such as inhibitory control, working memory, and cognitive flexibility. On the other hand, emotions (e.g., disgust, jealousy, fear, anger, etc.) are characterized by autonomic neurophysiological processes which have been primarily related to basic biological demands. Despite the long history of interest in the relationship between emotion and cognition, the interplay between these mental processes has not yet been well clarified from an evolutionary view. In this article, we review previous research on cognitive systems and develop an evolutionary model of emotion-cognition interactions.

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Posters 1, Red Room: 17:00 to 19:00 Tuesday 19.04.2022

Building Better Than We Know: Maintenance, Eye Movements, and Social Trust.

Objective: To determine if remediating poorly maintained residential properties increased levels of social trust and examine which aspects of the built environment participants attended to in the process of making their decision.

Methods: Using a within-subject design and a multilevel approach to analysis, participant's eye movements and attitudinal states were recorded as they were presented with a series of static images of residential properties. These residential properties were chosen from areas which are categorised as being among the most deprived according to the English Indices of Deprivation. The images then had the salient aspects of physical disorder corrected using digital imaging software. The resulting experimental stimuli were presented in both 'original/unmaintained' and 'manipulated/maintained' conditions to each participant along with a question regarding social trust.

Results: Images of residential properties in the maintained condition were associated with higher social trust scores (likelihood ratio test $\chi^2(1) = 548.13, p < .001$). The introduction of the maintenance variable as a random slope also significantly improved model fit. Images of residential properties in the unmaintained condition were associated with longer dwell durations (likelihood ratio test $\chi^2(1) = 329.46, p < .001$) and were visited earlier in the viewing session, $t(3878) = -5.73, p < .001, d = -.18$.

Conclusions: Remediating poorly maintained residential properties significantly increased levels of social trust. Cities should experiment with place-based maintenance-led interventions to develop effective attitudinal and behavioural change strategies.

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Posters 1, Orange Room: 17:00 to 19:00 Tuesday 19.04.2022

Running for reputation - how to use your altruistic act to become cool and sexy

Objective: Helpfulness is cool and sexy: altruism increases reputation and is a desirable attribution in a mating partner. Therefore, humans try to convince friends about altruistic values. As costlier altruism has a higher impact on reputation, it could be hypothesised that emphasizing the cost affects the response of the peers.

Methods: Data of a real-life charity situation where fundraisers enter running races and ask their friends to 'adopt' their kilometres donating were used.

Results: We showed that the way fundraisers communicate the cost of their altruistic act affects how their friends donate. Emphasizing how difficult is running (= subjective cost of altruism) increases the number of donors only if the fundraiser run a marathon (=the hardest distance), however, decreases the average amount donated. Opposite and same-sex friends respond to the communication of the fundraiser in different ways. The number of opposite-sex donors increases as the runner emphasizes the difficulty. Male donors of the male runners donate regardless of the cost, while female donors tend to support more those female runners that communicate less cost.

Conclusion: Humans use their altruistic acts to gain a better reputation, using it as a costly signal. The opposite-sex friends react positively, as altruism is "sexy". Women and men form their friendship in a different way that is noticeable in the results: men have a large male friend-group, where group members help each other establish a good reputation. Women form dyadic relationships with other women and don't support each other.

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Posters 1, Yellow Room: 17:00 to 19:00 Tuesday 19.04.2022

The Method of Local Restriction and its application to animal cultures

Objective: The goal of this project was to infer how much do great apes cultures rely on copying social learning mechanisms based on the spatial distribution of behaviours across populations.

Methods: To this end, we formalized the "Method of Local Restriction". The Method of Local Restriction is based on the premise that if behaviours are acquired via copying social learning, they should become locally restricted over time to a single population, for example as a consequence of the unavoidable copying errors accumulated during their transmission. We applied the Method of Local Restriction across ape species, populations and domains using the published literature and by consulting with great ape experts.

Results: We found that the vast majority of great ape behaviours are not locally restricted to a single population but instead re-appear in unconnected populations. Thus their spatial pattern of distribution is more consistent with repeated, socially-mediated innovations than with copying social learning. However, we found 25 locally restricted behaviours (out of 126), three of which were locally unique to a single population of a single great ape species.

Conclusions: Overall we found indirect evidence that great ape cultures are mainly sustained by learning mechanisms distinct from those underlying modern human cultures. We believe our method can be used to estimate the degree of reliance of other species of primates, including extinct hominins, on copying social learning to evaluate potential evolutionary trajectories of this ability in our lineage.

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Posters 1, Lime Room: 17:00 to 19:00 Tuesday 19.04.2022

Conditions for Japanese lexical-pitch accents to change from a complex to a simpler state

Objective: We examine factors driving cultures from a complex state to a simpler one. As a case study, this research focuses on the geographic variation of Japanese dialectic accents to infer the rates at which an accent loses its complexity, deduce the transmission rate of accents, and construct the accent genealogy over the Japanese mainland.

Methods: Accents were classified into four patterns according to the presence/absence of two characteristics, "tone" and "pitch change", where the presence of each was considered indicative of greater complexity. Generating a mesh network based on the geography of the Japanese mainland, we assigned the accent patterns at 100 locations to construct empirical data for the spatial variation of accents. Simulating the spread of accent variants on the network, we performed Bayesian inference to estimate the posterior distribution of transmission rates, mutation rates between accent patterns, and topology of accent genealogy.

Results: Posterior distributions of mutation rates showed that tone is more apt to be lost than pitch change; in particular, for those accents having tone and lacking pitch change, the loss rate of tone is extremely high.

Conclusion: By assuming a higher loss rate for tone than for pitch change, we can explain the observed distribution of accent patterns, in which the most complex accent pattern is found around Kyoto, the ancient capital, and those lacking tone in the periphery. Future study would incorporate more characteristics of accents to elucidate the detailed process whereby the variety of dialectic accents have descended from their common ancestor.

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Posters 1, Red Room: 17:00 to 19:00 Tuesday 19.04.2022

Reputation facilitates the co-evolution of cooperation and social rewarding

Empirical and theoretical work suggests that reward and punishment can promote cooperation. We focus on cooperation through rewarding when reputations of individuals are at stake. More precisely, we look at the mechanism of peer rewarding where agents incur personal cost to directly reward behaviour of their co-players. We aim to show that reputation does not allow antisocial rewarding to evolve – an observation that has been made in recent theoretical studies on pool rewarding.

Our theoretical model involves interaction between donors and recipients in two consecutive stages (donation followed by rewarding by the recipients). Recipients can gain reputations for their rewarding behaviour. In turn, donors can condition their cooperation with recipients when such reputation information is available.

We analyse these mathematical games by characterising their Nash equilibria and studying them using evolutionary approaches. Using concepts from evolutionary game theory and extensive computer simulations, we investigate the fate of rewarding and cooperation in populations under the mechanisms of imitative learning and random adoption of strategies. Our results show that social rewarding and

cooperation are highly favoured when reputation information is abundant. Evolution disfavours antisocial rewarding in both the two-player game and the public goods game. Only when reputation information is low and players are allowed to assort with similar strategies as themselves, antisocial rewarding can gain some prominence.

Our model offers an explanation as to why individuals may choose to incentivize cooperation with rewards. Furthermore it provides an analytical reasoning into why antisocial rewarding is rare.

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Posters 1, Orange Room: 17:00 to 19:00 Tuesday 19.04.2022

New hypothesis on the evolution of breasts

The possession of permanent, adipose breasts in women is a uniquely human trait that develops during puberty, well in advance of the first pregnancy. The evolutionary origins of this unusual among primates trait have been the focus of many hypotheses e.g. breasts as a product of sexual selection. Here we propose a new hypothesis that breasts appeared as early as in Homo, originally as a by-product of other coincident evolutionary processes of adaptive significance. These included an increase in subcutaneous fat tissue (SFT) in response to the demands of thermoregulatory and energy storage, and of the ontogenetic development of the evolving brain. An increase in SFT triggered an increase in oestradiol levels (E2). An increase in meat in the diet of early Homo allowed for further hormonal changes, such as greater dehydroepiandrosterone (DHEA/S) synthesis, which were also crucial for brain evolution. DHEA/S is also easily converted to E2 in E2-sensitive body parts, such as breasts and gluteo-femoral regions, causing fat accumulation in these regions, enabling the evolution of perennially enlarged breasts. Furthermore, it is also plausible that after enlarged breasts appeared, they were co-opted for other functions, such as attracting mates and indicating biological condition. Finally, we argue that the multifold adaptive benefits of SFT increase and hormonal changes outweighed the possible costs of perennially enlarged breasts, enabling their further development.

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Posters 1, Yellow Room: 17:00 to 19:00 Tuesday 19.04.2022

Parental investment by genetic fathers and stepfathers: roles of mating effort and childhood co-residence duration Inclusive fitness theory predicts higher investment in biological relatives than non-kin. However, families may also include individuals that are not genetically related, but are expected to provide investment in terms of time, money and care. Here we compare the investment of stepfathers, biological fathers who are divorced and who are in a relationship with the child's mother in their children in terms of financial and practical help, emotional support, intimacy, and emotional closeness. In addition, we investigate if the relationship between childhood co-residence duration and parental investment differs between stepfathers and divorced biological fathers. We used data from adolescents and younger adults (aged 17–19, 27–29, and 37–39 years) from the German Panel Analysis of Intimate Relationships and Family Dynamics (Pairfam), wave 2, collected in 2010–2011. We found that biological fathers who were in relationship with the child's mother were investing the most, while stepfathers invested the least. Furthermore, the investment of both divorced fathers and stepfathers increased with the increased childhood co-residence duration. However, childhood co-residence duration increased financial help and intimacy more in stepfathers than in divorced fathers. Our findings support evolutionary theories such as inclusive fitness theory and mating effort in explaining social behavior and family dynamics. Furthermore, social environment, such as childhood co-residence is associated with parental investment among fathers.

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Posters 1, Lime Room: 17:00 to 19:00 Tuesday 19.04.2022

The "ENDOW" Project (Economic Network Dynamics and the Origins of Wealth (Inequality))

Objective: In this presentation, we introduce the "ENDOW" project (Economic Network Dynamics and the Origins of Wealth (Inequality)), a collaborative and interdisciplinary research project of anthropologists and economists, aimed at exploring the effect of social network structure on the degree of inequality in wealth.

Methods: To do this, we are gathering demographic, economic, and social network data from over forty communities in thirty-plus countries around the world, across two points in time (separated by five years). Among other communities, our sample includes wage labourers in India, foragers in the Congo, horticulturalists in Suriname, pastoralists in Namibia, and fishers in Papua New Guinea.

Results: The communities are all relatively small and contained (ranging in size from roughly thirty to two hundred households), but are remarkably and purposefully diverse in their economies, market integration, institutional forms, and cultural practices. In this talk, we will introduce the motivation for the project, its key analytical challenges, and some of our preliminary findings.

Conclusions: The variation in the ENDOW sites allows us to explore the extent to which differences in socioecological setting correlate with local social network structure, and so ultimately influence people's potential for social and economic mobility.

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Posters 1, Red Room: 17:00 to 19:00 Tuesday 19.04.2022

Cooperative resource harvesting among kin, clan, and neighbors in rural China

In rural Yunnan (China), commodification of mushrooms such as matsutake has led to greater participation in global markets and the emergence of new, local institutions for commons governance. We use a multilayer community-detection algorithm to investigate how cooperative ties created through novel governance institutions in Baihua (pseudonym), a rural Yi village, overlap with existing patterns of affiliation, including kinship ties, distance between households, clan affiliation and networks of social support. In this context, cooperative harvesting partnerships are a risky undertaking that requires considerable investment of time and labour; as such, harvesting partners need to trust that their partners will put in the required effort. We find that social support ties in the village, household proximity, and to a lesser extent, clan membership, strongly predict cooperative resource harvesting ties. Adding biological kinship to the analysis does not improve model performance if social support networks are known. Further, kinship networks reduced to only more closely related kin do not improve model performance compared to all known kinship ties. The results suggest that, in this case, although biological kinship plays a structuring role in the establishment of trusting relationships, it is not itself a sufficient criteria for motivating costly cooperation. Further, effects within network clusters may be important in enabling such cooperation.

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Posters 1, Orange Room: 17:00 to 19:00 Tuesday 19.04.2022

Cooperation in Simultaneous Interactions

Most work on direct reciprocity assumes that individuals only engage in one repeated game at a time, or that people treat their different games as independent. However, humans routinely engage in different interactions simultaneously. Here we explore experimentally how people cooperate when they take part in two concurrent repeated games with either the same or different partners in order to see if behaviour in one game is linked to behaviour in the other game. We support our experimental approach with predictions from a simulation. In our experimental design, we focus on two different conditions. In the multichannel condition the players engage in two different versions of a repeated prisoner's dilemma (RPD) simultaneously against the same participant. In crosstalk they also engage in the two RPDs simultaneously but against two different participants. The two RPDs played differ only in their payoffs. One game has a very high benefit of cooperation, while the other one only has a low benefit. We explore the influence from one game on the other in the form of behavioural spillovers.

While previous theoretical research finds that taking part in multiple interactions concurrently increases cooperation, our experimental results show the opposite effect in both of our concurrent games conditions. We also find that subjects are influenced by the behaviour of their partner in the other game. The effect is strongest in the multichannel treatment, indicating that participants are able to link the two games when it is strategically relevant.

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Posters 1, Yellow Room: 17:00 to 19:00 Tuesday 19.04.2022

Individual Differences in Content Biases for Danger and Edibility in the Food Domain

Objective: Research has shown that our learning and generalisation systems appear to be sensitive to certain types of evolutionarily relevant content, such as danger and edibility. The present study examines how individual differences in anxiety—which has been linked to risk taking behaviours—and related personality traits influence these content effects in recall and generalisation.

Method: 180 adult participants see a series of food images (e.g. obscure fruit and vegetables) and are told each has/does not have a property. Participants are in 1 of 3 property conditions where the property is either Edibility (edible/not edible), Danger (toxic/not toxic), or a control property (grows in summer/does not grow in summer). Images are three possible colours and the rate at which items have the property depends on colour (e.g. 25% of red items, 50% of yellow, and 75% of green having the property). Participants are presented with the images again, along with novel images of each colour, and asked if these have or do not have the property, measuring recall and generalisation respectively. Anxiety and personality is measured using the State-Trait Anxiety Inventory and the HEXACO Personality Inventory.

Results: Data collection is expected to be complete by the end of December and provisional results will be presented. It is predicted that both recall and generalisation will be greater for dangerous content. It is further predicted that this effect will be stronger in participants scoring higher on anxiety and emotionality.

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Posters 1, Lime Room: 17:00 to 19:00 Tuesday 19.04.2022

Chimpanzees seek help, but not strategically

Seeking help can be a highly adaptive behavior. From an evolutionary perspective, strategic help-seeking can significantly improve an individual's fitness. However, it has not yet been investigated whether our closest relatives—chimpanzees—seek help strategically.

In Study 1, we investigated whether chimpanzees seek help selectively when they need it. Chimpanzees (N=19) sought help when it was necessary, but not if they could solve the problem themselves ($c2(1)=30.821$, $p<.001$).

In Study 2, we investigated whether chimpanzees seek help strategically: Do they consider action-related costs of potential helpers as much as they consider their own? Chimpanzees (N=14) had a stronger preference for a low-cost option when they had to obtain a reward on their own, but not when they sought help from others ($c2(1)=7.989$, $p=.005$).

These findings imply that chimpanzees seek help when they need it, but they do not strategically consider other's costs when deciding whom to seek help from.

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Posters 1, Red Room: 17:00 to 19:00 Tuesday 19.04.2022

Traces of cooperation in early childhood. Cultural developmental perspective embedded in a processual-relational paradigm

Objective: From an evolutionary perspective, it is our ability to cooperate that co-evolves with culture and together they create who as humans we are (Boyd & Richerson, 2009). The poster shows how the cultural concept of a "developmental niche" (Harkness & Super, 1986) can be applied under the processual-relational paradigm in developmental science (Overtoon, 2013). It proposes to observe the child's daily activities both from a macro and micro perspective in order to better understand and describe the emerging ability to cooperate.

Methods: The ethnographic method was used during the fieldwork carried out in the small-scale society of Yurakare in Bolivia. A pilot study was carried out with the participation of 10 children aged between 2-6 years. The research methods used are: focal follows at regular intervals using ethogram, the child's behavior recorded on video, and fieldnotes.

Results: The poster shows the coding scheme used by the author in her first fieldwork run in 2021. As an example of the implementation of the presented theoretical framework, the results present an ethnographic microanalysis of cooperative behaviors in the child's activity selected on the basis of quantitative data from the ethogram used.

Conclusion: The presented research methods implemented in the process-relational approach are an important complement to research based on laboratory experiments in developmental psychology, but above all, they are a comprehensive way of studying child cooperation.

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Posters 1, Orange Room: 17:00 to 19:00 Tuesday 19.04.2022

The evolution of religiosity by kin selection

Despite religion's apparent ubiquity, hypotheses about the selection pressures that may have shaped its cognitive foundations remain controversial. Here, we develop and analyse a mathematical model inspired by Crespi and Summers' suggestion that parent-offspring conflict has driven the evolution of religious beliefs to explore the causes and consequences of these selection pressures. To this end, we employ kin selection methodology to investigate how selection may mould an individual's propensity for religiosity and corresponding patterns of gene expression, revealing that the evolution of religiosity is modulated by genetic relatedness between social partners, that selection in relation to religiosity may depend on an individual's age and sex, and that religiosity can foment intragenomic conflicts of interest that give rise to parent-of-origin specific patterns of gene expression and concomitant clinical disorders. More generally, we develop a formal, theoretical framework that enables the derivation of testable, comparative predictions about adaptive as well as maladaptive phenotypes.

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Posters 1, Yellow Room: 17:00 to 19:00 Tuesday 19.04.2022

Women's serial monogamy as an adaptive strategy -A case study of a population in Botswana

A recent cross-cultural study showed that women who had more reproductive partners had higher reproductive success than women who had only one reproductive partner.

However, to interpret the positive correlation between the increased number of reproductive partners and reproductive success in women as a result of women's adaptive strategies, it is necessary to test whether a causal relationship exists between those two.

To test the causal relationship between the increased number of female reproductive partners and reproductive success, this study used inter-birth intervals (n=182) as an approximate measure of fertility in a population of Botswana where women had increased reproductive success through serial monogamy (n=69) and used time-series analysis to test whether women's re-partnership improved their future fertility.

To address the issue of fecundity-mediated pseudo-correlation between the number of reproductive partners and reproductive success, I compared the degree of improvement in fertility gained by re-partnership between periods of high (younger) and low fecundity of women. Results showed women who had a high cumulative number of re-partners and were relatively old achieve higher fertility than young women. This indicates that for women in this population, continued re-partnership has a cumulative fitness benefit in women's fertility and that increasing the number of reproductive partners is an adaptive strategy.

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Posters 1, Lime Room: 17:00 to 19:00 Tuesday 19.04.2022

Emergence of word-like phonology from iterated imitations

Objective: Our research contributes to addressing the puzzle of language origins. Existing research (Edmiston et al. 2018) with speakers of English suggests linguistic properties such as phonological words can emerge spontaneously through repeated imitations of environmental sounds. Our study tests this with speakers of languages from related (German) and unrelated (Polish, Chinese) language families.

Methods: Data was collected in iterated learning chains of 8 participants per chain: first participant imitated a 'seed' environmental sound – such as glass, zipper or sneezing – and that participant's output was played back as input to be imitated by the next participant in a chain, etc. (cf. 'Chinese whispers'). We collected imitations from native speakers of Polish (8 chains), German (8 chains), and Chinese (8 chains). If imitations become gradually more language-like with each iteration, then the last (8th) generation imitations should sound more wordlike (thus, easier to transcribe) than first generation imitations. We asked native speakers of each language to convert the 1st and 8th generation

imitations into spelling; we expected more orthographic similarity (computed with the Ratcliff Obershelp algorithm) between the transcriptions of the 8th than 1st generation imitations.

Results: Significant differences were found only in Polish speakers. Importantly, first generation imitator appears to play a vital role in a chain.

Conclusions: Edmiston et al.'s (2018) findings were confirmed in the speakers of Polish but not German or Chinese. Care should be taken when generalizing results obtained from speakers of a single language.

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Posters 1, Red Room: 17:00 to 19:00 Tuesday 19.04.2022

The role of resource dynamics in variations of the life cycle among female humans

The female human life cycle is characterized by a long lifespan, within which there is usually a short reproductive period in between long juvenile and post-reproductive stages. Within those boundaries, there is a high variability in the timing and outcome of longevity and reproduction, with individuals showing short lifespans with high fertility and vice versa. The evolution of the female human life cycle has been related to the surplus of resource production during adulthood and the inter-generational transfers towards juveniles. However, there is no clear understanding whether and how these resource dynamics shape the observed variation in human populations. Here we develop a theoretical framework to describe how different resource dynamics influence the variability of the female human life cycle. For this, an agent-based model is used to structure and describe how resource dynamics (i.e. production, consumption, sharing, and storage) at different developmental stages of individuals can lead to different variability levels within a population. We expect that differences in the ways in which individuals obtain and allocate resources across their lifespan can contribute to understand the boundaries under which the female human life cycle varies, allowing to understand other possible scenarios beyond the common longevity versus fertility trade-off.

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Posters 1, Orange Room: 17:00 to 19:00 Tuesday 19.04.2022

Self-recorded psychosocial/physical wellness in Icelandic women varies by season but does not covary with menstrual cycle hormonal variation

Objective: Dramatic circumpolar seasonal variation in photoperiod and climate could be expected to have a large impact, if any exists, on mood and/or menstrual cycle hormones. We tested the hypotheses that wellness varies seasonally in healthy pre-menopausal women living year-round at high latitude, and that variation in mood is associated with variation in reproductive hormones (estrogens, progesterone, testosterone). Prior studies have been hampered by sparse sampling and/or reliance on assumed hormone profiles and ovulation day rather than measuring hormones. This is to our knowledge the first study to test these hypotheses based on daily prospective self-reported mood/wellness data and concurrent hormone measurements spanning a complete ovulatory cycle in both summer and winter seasons. Methods: Thirty-two healthy Icelandic women reported self-assessments of psychosocial and physical wellness. Ovulation was determined from daily salivary progesterone and estradiol concentrations. A composite daily wellness score and urinary hormone biomarkers (LCMS measured concentrations) were compared by season using mixed effects models adjusted for age and BMI.

Results: Wellness score was significantly higher in summer than in winter ($p_{LT}=0.001$). Within-individual testosterone concentration was significantly higher in summer than in winter ($p_{EQ}=0.038$). No other hormone biomarker varied significantly by season. None of the measured hormones, nor time before/since ovulation, were significant predictors of specific or composite wellness scores.

Conclusions: Season was a significant factor in psychosocial and physical wellness. Although testosterone varied seasonally, other reproductive hormones did not. During ovulatory menstrual cycles, self-recorded psychosocial and physical wellness did not covary with reproductive hormone concentrations. (Funding: NSF-1319663; Fulbright; NIH-P30-CA71789)

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Posters 1, Yellow Room: 17:00 to 19:00 Tuesday 19.04.2022

Heterogeneity of aggression: The evolution of human aggression types

Objective: Aggression can be observed in many social species and may be considered as a uniform or pluriform phenomenon. Recently Gómez et al., (2016) provided evidence of aggression having an evolutionary basis when considering the single facet of lethal aggression as a pooled measure. Our research aimed to explore different aggression types, both lethal and non-lethal, to identify their evolutionary history.

Methods: We constructed a species level comparative database containing data on intergroup aggression, intragroup aggression, intergroup lethal aggression, intragroup lethal aggression, and infanticide. Using the Bayesian brms package (Bürkner, 2017) a multivariate analysis was run on 104 primate species and sub-species to identify any correlations between aggression types and the phylogenetic influence.

Results: We found strong/moderate correlations between intragroup and intergroup killing (0.67; 95% CI = 0.24-0.94), intragroup killing and infanticide (0.61; 95% CI = 0.08-0.93) and intergroup killing and infanticide (0.66; 95% CI = 0.13-0.94). No significant correlations were found between the other aggression types. Further, a substantial amount of the total variability for intragroup (60.6%) and intergroup killing (47.7%) can be explained by phylogeny, with moderate amounts explaining intra-group aggression (26.9%) and infanticide (25.3%).

Conclusions: The findings of our study suggest that aggression forms are distinct rather than a homologous behaviour, with phylogeny explaining high/moderate variance in four or five aggression types. To follow, an investigation is required to identify the additional factors impacting on each aggression type, particularly intergroup aggression.

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Posters 1, Lime Room: 17:00 to 19:00 Tuesday 19.04.2022

Cultural transmission bias in the spread of voter fraud conspiracy theories on Twitter during the 2020 US election

During the 2020 US presidential election, conspiracy theories about large-scale voter fraud were widely circulated on social media platforms. Given their scale, persistence, and impact, it is critically important to understand the mechanisms that caused these theories to spread so rapidly. The aim of this study was to investigate whether retweet frequencies among proponents of voter fraud conspiracy theories on Twitter during the 2020 US election are consistent with frequency bias, demonstrator bias, and/or content bias. To do this, we conducted generative inference using an agent-based model of cultural transmission on Twitter and the VoterFraud2020 dataset. The results show that the observed retweet distribution is consistent with a strong content bias and demonstrator bias, likely targeted towards negative emotion and follower count, respectively. Based on the confounding effects of the timeline algorithm and population structure, we are most confident in concluding that the differential spread of voter fraud claims among proponents of voter fraud conspiracy theories on Twitter during and after the 2020 US election was partly driven by a content bias causing users to preferentially retweet tweets with more negative emotion.

Posters 2

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Posters 2, Forest Room: 17:45 to 19:30 Wednesday 20.04.2022

Gender differences in matrilineal societies: A meta-analysis of behavioural experiments

Research in behavioural economics has revealed that men, on average, are more risk-seeking and competitive than women. Additionally, women tend to be more prosocial than men in mixed-gender interactions, but less prosocial than men in same-gender interactions. An emerging literature has investigated the effects of kinship norms on behavioural gender differences, with behaviours in matrilineal societies compared to those in patrilineal societies. In patrilineal societies, kinship norms prescribe that relational and material resources are transmitted across generations through the male line (from father to son). In matrilineal societies, instead, kinship norms prescribe that intergenerational resource transmission occurs through the female line (from mother to daughter, or from maternal uncle to niece, or a combination of the two). As this strand of the literature has produced rather mixed results, we are conducting a meta-analysis to pinpoint the role of kinship norms in shaping gender differences in risk-taking, competitive and prosocial behaviours. With this meta-analysis, we will address three research questions: Do gender differences in behaviour vary with the society's kinship norms (patrilineal vs. matrilineal societies)? In matrilineal societies, is there a reduction, closure or inversion of the behavioural gender gap? In matrilineal societies, is there an effect of type of behaviour (risk taking vs. competition vs. prosociality)? Our predictions are in line with the argument that matrilineal kinship, in giving women access to relational and material resources in a way that creates common expectations of greater female influence within society, can sustain more progressive gender roles, thus influencing behavioural gender differences.

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Posters 2, Blue Room: 17:45 to 19:30 Wednesday 20.04.2022

Impact of recommender systems on cultural evolution

Digital technologies have not only fundamentally changed the way we access or process information, but they have also given rise to new modes of cultural evolution. Specifically, temporal cultural changes happen in various settings where humans and intelligent machines interact. This human-machine interaction becomes particularly evident in the context of recommender systems (RS), which constitute increasingly important tools to navigate the ever growing and seemingly infinite space of cultural items online. For instance, RSs help to choose the right movies, songs, or commercial products and therefore represent -- in addition to social learning -- a novel mode of cultural acquirement. Despite their widespread use, the impact of algorithmically personalized recommendations on the evolution of culture has not been investigated systematically. Here, we aim to close this gap by means of agent-based modelling. In particular, we shed light on how recommendations -- that alter the decision-making of individual agents -- lead to drastic changes in the evolution of culture as a whole, i.e. on a population level. Crucially, we find that the type of recommender system strongly impacts both the diversity and the complexity of cultural items. Furthermore, changes in the system size (with respect to agents and/or cultural items) may not only lead to quantitative but, more importantly, to qualitative changes in the cultural dynamics. Taken together, our results suggest that the effects of RSs should not be overlooked in order to fully understand the evolution of culture in the digital age.

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Posters 2, Navy Room: 17:45 to 19:30 Wednesday 20.04.2022

Transmission of alien algorithmic strategies through social learning

Humans rely on a host of strategies (e.g., avoiding large losses) to effectively navigate their complex world. These human strategies may be adaptive in some environments, but maladaptive in others. In a digital society, algorithms may present novel strategies that can adaptively complement human strategies. The present work explores whether humans can learn algorithmic strategies that go against human bias (e.g., accepting large losses for a long-term benefit) and under what conditions alien adaptive strategies might be transmitted among humans. We showed that superior algorithmic solutions are reproduced by human participants (N = 177) and further transmitted through social learning. However, content bias against alien algorithmic solutions hindered sustained transmission among humans. We are now testing whether sustained transmission can be achieved when humans encounter superior algorithmic strategies that generalise in their adaptiveness across a class of novel environments.

We use a combination of behavioural methods and agent-based simulation to test the performance of transmission chains with human and algorithmic agents. More precisely, agents search for a sequence of paths through a network that maximises their payoff. While each network-path is associated with rewards and losses, the path maximising the total payoff requires agents to accept a large loss initially. Our findings suggest that algorithms can improve human performance, but human bias hinders algorithmic solutions from being preserved. We are currently collecting more data that will allow us to speak about conditions under which hybrid social learning can overcome limitations in cultural evolution that are imposed by human biases.

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Posters 2, Purple Room: 17:45 to 19:30 Wednesday 20.04.2022

Participation in antenatal classes and the level of anxiety and depression before and after childbirth during COVID-19 pandemic

Background: Prenatal education in the form of birth school is one of the forms of preparing a pregnant woman for childbirth, puerperium and future motherhood, reducing the feeling of anxiety and stress in the future mother.

Aim: The aim of the study is to assess the impact of childbirth classes on the perception of stress and anxiety before and after childbirth in the context of the COVID-19 pandemic.

Methods: The sample included 488 adult Polish women, pregnant at the time of COVID-19 pandemic (mean = 30.57; SD = 3.95). The levels of state anxiety and depression were measured with the State-Trait Anxiety Inventory (STAI) and the Edinburgh Postnatal Depression Scale (ESDP) via online questionnaires, before and after the child's birth. We compared the differences in state anxiety (STAI X-1) and depression between pregnant women who attended birthing school online, inpatient, and who did not attend any antenatal meetings during the COVID-19 pandemic by t test for dependent groups and general linear model (GLM), with the control for trait anxiety, woman's age, getting COVID-19 during pregnancy, type of childbirth and the occurrence of pregnancy complications.

Results: We did not observe a difference in STAI X-1 ($p=0.321$) and ESDP ($p=0.187$) changes pre to postnatally depending on the participation in the antenatal classes – regardless of whether women participated in the online, onsite or no classes, they did not manifest a change in anxiety between pre and postnatal period.

Conclusions: Our previous studies showed that pregnant women who attend onsite classes cope better with prenatal time, in the context of level of anxiety and depression in comparison to the online and no classes. This analysis shows that the effects of the onsite birth classes are long lasting. Regardless of attendance to antenatal classes, we did not see an increase in neither STAI X-1 nor ESDP levels, which may indicate the legitimacy of attending childbirth school by pregnant women.

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Posters 2, Forest Room: 17:45 to 19:30 Wednesday 20.04.2022

Exploratory preferences explain the cultural success of imaginary worlds in modern societies

Imaginary worlds are present in many of the most culturally successful modern fictions, be it in novels (e.g., Harry Potter), films (e.g., Star Wars), video games (e.g., The Legend of Zelda), graphic novels (e.g., One piece) and TV series (e.g., Game of Thrones). This phenomenon is global and massive (e.g., the worldwide success of Lord of the Ring, the emergence of xuanhuan and xianxia genres in China). Why so much attention devoted to such fictional worlds? Why now, and not before? Why in some societies, and not others? We propose that imaginary worlds are popular because they co-opt evolved exploratory preferences (Dubourg & Baumard, 2022). We make predictions about the cultural distribution and success of fictions with imaginary worlds based on how exploratory preferences adaptively vary. We test synchronic and diachronic predictions with both computational and experimental methods. For computational tests, we leverage three large cultural datasets, namely IMDb (N=85,855 movies), Wikidata (N=96,711 novels), and the Movie Personality Dataset (Nave et al., 2020), which aggregates averaged personality-related and demographic data from the Facebook myPersonality Database (N=3.5 million). For experimental tests, we run an online experiment about movie preferences, with multiple paradigms (N=350). In all, we provide empirical evidence that the appeal for imaginary worlds relies on exploratory preferences.

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Posters 2, Blue Room: 17:45 to 19:30 Wednesday 20.04.2022

How infants teach themselves to speak

Objective: Comparative neuroscience suggests that neural structures commonly associated with human speech have correlates in brains of other hominids. This has led researchers to posit varying accounts of the evolution of human speech. However, such ambitions are confounded by theories of language acquisition, often constituting abstract accounts based on observation rather than deduced from higher-

order principles. Optimally, we as researchers would posit an account of linguistic development (1) rooted in principles of the natural sciences, that also (2) accommodate observations of language acquisition, and (3) makes possible an integration with evolutionary sciences.

Methods: We provide an extensive, synthesizing account of language acquisition, based on innate motor system activity, infants' individual exploration of orosensory space, and neural recycling for the allocation of neural architectures devoted to speechcentric activity.

Results: We posit that babbling – pre-linguistic articulate speech sounds – is the auditory result of infants' explorations of orosensory space and gives rise to emergent pseudosegmental phonetic properties. Invariance in speech-signal attributes further facilitates acquisition of language-specific phonemic vocabularies, and to phonemes proper, defined as invariants in cognitive-orosensory space. Babble is thus replaced by elective values in sound space, selected via interaction with ingroup members. Further, we argue that such adaptation is continually afforded neurological allocation throughout childhood via neural recycling – ontogenetic exploitation of pre-existing circuitry, for learning of some evolutionarily novel activity.

Conclusions: We present an integrative account of language acquisition, rooted in principles of the natural world.

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Posters 2, Navy Room: 17:45 to 19:30 Wednesday 20.04.2022

Is Xenophilia vs. Xenophobia an Evolved Trait?

Only a few subjects polarize as much as migration, insuperable, highly emotional, and often morally loaded positions face each other. The question is why this theme is so polarizing.

We assume that human evolution occurring in small groups may explain at least partly why this is the case. This view is supported by the fact that additive genetics contribute to 20% - 70% to in-group vs. out-group attitudes, depending on which trait has been measured. Furthermore, the profound differences in attitudes between men and women may be a further hint to an evolutionary root of those attitudes. As indicated by archeological and genetic data, particularly men may have been lethally affected by intergroup violence, whereas young women likely have been the least affected group in terms of lethal violence. Rather genetic data suggest that due to female dispersal and by means of marriage networks, young women may have ensured genetic and cultural exchange between groups.

In line with these assumptions, we found on the basis of the European Social Survey that men generally have a more restrictive attitude toward out-group (migrants) than especially young women. Among women, attitudes become more negative with increasing age. Moreover, genetic data from the Wisconsin longitudinal and Brisbane twin sample show that individuals who are ethnocentric have longer and more "Runs of Homozygosity", pointing to a history of in-group marriages and thus the risk of detrimental inbreeding effects.

Both findings support our hypothesis of an evolutionary root of xenophilia vs. xenophobia. According to this hypothesis, both xenophilia as well as xenophobia make sense from an evolutionary point of view: a more restrictive attitude towards strangers may have entailed protection from inter-group violence, whereas a more open attitude may have ensured genetic and cultural exchange between groups.

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Posters 2, Purple Room: 17:45 to 19:30 Wednesday 20.04.2022

Moralizing religions as cognitively compelling tools of social control

Why do many human societies develop beliefs in supernatural entities that punish uncooperative behaviors, such as moralizing gods or karmic forces? Leading evolutionary theories propose that these beliefs are cultural group-adaptations, selected by inter-group competition because they facilitate cooperation. Focusing on within-group interactions, we propose an alternate model, in which individuals strategically endorse supernatural punishment beliefs to manipulate others into cooperating. To achieve their evolved policing strategies, people promote, selectively retain, and cumulatively refine religious beliefs they perceive—based on their intuitive theories of human behavior—as capable of inspiring cooperation in others. This theory's predictions are supported by psychological, cross-cultural, ethnographic, and historical evidence that (i) people intuitively believe that beliefs in moralizing gods can incentivize others' cooperation; (ii) motivations to police others' cooperation increase and predict endorsement of moralistic religious beliefs; (iii) societies with lower tolerance of deviance are more likely to exhibit moralistic religious beliefs; and (iv) people endorse moralistic religious beliefs when their desired level of social control cannot be achieved by secular means. This theory also explains key features of moralizing religions that are not well explained by other accounts, such as (i) their previously underestimated prevalence in small-scale societies, (ii) their association with an enforcement and moralization of religious faith, and (iii) their reliable cultural evolution despite mixed evidence that they actually motivate cooperation. Moralistic religious beliefs are endorsed as long as people believe them likely to incentivize others' cooperation, regardless of their objective effectiveness in doing so.

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Posters 2, Blue Room: 17:45 to 19:30 Wednesday 20.04.2022

Monks bring kin-selected benefits and relax sibling competition over parental resources in Tibetan populations

Why parents in some societies induce some of their sons to become religious celibates is a long-standing evolutionary puzzle. Although long deemed maladaptive, some have speculated that this might be associated with brother competition for family resources. However, the inclusive fitness of religious celibacy remains unexplored and the behavioural ecology of monks has not been investigated anywhere in the world. Here, we present sociodemographic data from Amdo Tibetan agropastoralists in Western China, where parents sometimes sent a son to the monastery, to evaluate the influence of inclusive fitness interests on the evolution of religious celibacy and, if any, what factors determine the probability of becoming a monk and the possible association between wealth and having a monk brother.

We found that men with a monk brother father more children, and grandparents with a monk son have more grandchildren, suggesting that the practice is adaptive. We found that boys with at least one older brother are more likely to be induced by their parents to become celibate monks, whereas patrilocal heads of household, who inherit parental property, are more likely to be first-born sons. Moreover, we find that men with at least one monk brother are wealthier than men who only have non-celibate brothers. Together, these results suggest that inclusive fitness considerations appear to play a key role in shaping adaptation of religious celibacy, and sending a son to the monastery is a way for parents to decrease competition between brothers over family resources.

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Posters 2, Navy Room: 17:45 to 19:30 Wednesday 20.04.2022

Women's WHR, WSR and BMI Predict Their Youthfulness Ratings

Previous research has shown that specific body phenotypes (low waist-to-hip ratio [WHR], low waist-to-stature ratio [WSR], and low to moderate body mass index [BMI]) are judged as attractive because they signal nubility (sexual maturity) and high reproductive value (youthfulness and nulliparity; Lassek & Gaulin, 2019; Andrews et al., 2017; Singh & Young, 1995). The current study aimed to directly test the predictive power of these body characteristics on youthfulness ratings. Participants' (n=43) body characteristics (i.e., waist, hip, height, and weight) were measured to calculate their WHR, WSR, and BMI. Photos of participants' bodies were presented to a separate set of participants (i.e., raters). Raters were asked to judge the stimuli on a youthfulness Likert-like scale (n=15; 1 least youthful – 6 most youthful). Multilevel model analyses showed all body measurements significantly predicted youth ratings: WHR $t(25.94) = -2.28$, $p = 0.031$, WSR $t(12.58) = -3.95$, $p = 0.002$, and BMI $t(19.34) = -4.53$, $p < 0.001$. Specifically, participants with lower WHR, WSR and BMI were rated as more youthful than participants with higher WHR, WSR and BMI. Our findings lend support to the reproductive value hypothesis.

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Posters 2, Purple Room: 17:45 to 19:30 Wednesday 20.04.2022

Evolutionary Sociology, quo vadis?

Unlike evolutionary psychology, which has developed into a significant field of research for over 30 years, evolutionary sociology has a hard time establishing itself, although there were evolutionary sociologists as early as in the 1970s. Since Wilson's Sociobiology in 1975 every few years evolutionary informed sociologists published "wake-up-calls" to convince fellow sociologists and to announce findings which may be relevant to sociological research and theorizing. But for decades that didn't change much – wide parts of sociology remained critical – or unaware – towards evolutionary approaches to human social behavior.

By reference to selected episodes in evolutionary sociology from 1975 to the present this contribution traces the development and milestones of the field, discusses frequent misconceptions about evolutionary behavioral sciences as well as main causes of sociology's 'biophobia', depicts its problem areas, debates significant differences to its neighboring discipline evolutionary psychology, and outlines its potential prospects. Although scientific knowledge points to a stronger integration of evolutionary explanations in many areas of sociology, political reasoning, worldviews, and also a lack of biological training often inhibit that. It also becomes apparent that sociology's creativity in naming concepts and fields prevent evolutionary sociological research to be as visible as evolutionary psychological research. For instance, the unifying term 'evolutionary sociology' only appears in recent years. Also dedicated journals, societies, conferences, or study programs are very scarce, if they exist at all. Beyond that the (small) field is very broad positioned, basically inconsistent, i.e. it incorporates diverse evolutionary (ultimate) and bio-social (proximate) subfields.

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Posters 2, Forest Room: 17:45 to 19:30 Wednesday 20.04.2022

A unified framework of direct and indirect reciprocity

Direct and indirect reciprocity are key mechanisms for human cooperation. Direct reciprocity means that individuals use their own experience to decide whether to cooperate with another person. Indirect reciprocity means that they also consider the experiences of others. Although these two mechanisms are intertwined, they are typically studied in isolation. Here, we introduce a mathematical framework that allows us to explore both kinds of reciprocity simultaneously. We show that the 'generous tit-for-tat' strategy of direct reciprocity has a natural analogue in indirect reciprocity, which we call 'generous scoring'. Using an equilibrium analysis, we characterize the conditions under which either of the two strategies can maintain cooperation. With simulations, we additionally explore which kind of reciprocity evolves when members of a population engage in social learning to adapt to their environment. Our results draw unexpected connections between direct and indirect reciprocity while highlighting important differences regarding their evolvability.

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Posters 2, Blue Room: 17:45 to 19:30 Wednesday 20.04.2022

Changes in ritualized behavior during the menstrual cycle

Recent research started to examine the adaptive value of ritualized behavior and one hypothesized function includes positive effects on individual health and survival. In humans and non-human animals, ritualized behavior has been linked to states of intense stress and anxiety in

a range of studies. Moreover, in humans, the emergence of ritualized behavior has been observed in situations of perceived uncertainty, be it unpredictability of physical danger or a threat to social status. It remains, however, unclear whether ritualized behavior occurs spontaneously also in common, less intense anxiogenic situations. As a suitable naturally occurring condition allowing within-subject comparison, we used state anxiety fluctuations within the menstrual cycle, specifically the luteal phase anxiety and stress changes, as possible sources of spontaneous ritualization. In a pre-registered longitudinal study (36 days), free-cycling female participants regularly marked their day of the cycle, symptoms connected with the menstrual cycle, levels of anxiety, and activities they performed. These activities were evaluated by independent coders on the characteristics of ritualized behavior. The main prediction of the study is that psychological changes, specifically heightened anxiety, and perceived stress during the luteal phase, would lead to both an increase in the types of ritualized behaviors and higher intensity of those behaviors. We report on the results and how they connect to the theorized anxiety management function of ritual systems.

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Posters 2, Navy Room: 17:45 to 19:30 Wednesday 20.04.2022

Perceptions and Reactions to Physical Dominance in Men

Objective: The cost of winning a physical fight can be high due to potential injuries. Therefore, a cognitive mechanism to gauge physical dominance could reduce the risk of engaging in a physical conflict where the fight could be lost, or the cost too high. This study aims to investigate if we can assess when a male is a potential threat, and the likelihood of avoiding them when viewing walking patterns.

Methods: Nineteen males partook in stimuli creation. A baseline and threatening walk (where a scenario was read aloud to provoke them) were recorded using motion capture and exported to animation software to ensure no other characteristics were on display. These were then presented to 156 participants to rate on a 1-7 Likert scale on how physically dominant the males were and the likelihood of crossing the road to avoid them.

Results: There was no significant increase when comparing physical dominance scores on the two walks. When comparing the likelihood of crossing the road, there was found to be a significantly higher increase in crossing the road when viewing the threatening walk compared to the baseline walk, with females being significantly more likely to cross the road on both walks.

Conclusions: The results suggests that males are able to portray some aspect of threat via their movement, however this may not be perceived as physical dominance. Further study into what this trait could be (i.e. threat, intimidation) should be investigated.

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Posters 2, Purple Room: 17:45 to 19:30 Wednesday 20.04.2022

What is cultural evolution anyway?

Objective: The term cultural evolution has become popular in the evolutionary human sciences, but it is often unclear what is meant by it. This is generating confusion and misconceptions that are hindering progress in the field. These include the claim that behavioural ecology disregards culture. We argue that these misunderstandings are caused by the unhelpful use of term cultural evolution to identify both a phenomenon – culture changing through time – and a theory to explain it – the potential role of cultural transmission biases in driving this change.

Methods: We illustrate this point by considering recently published influential studies and opinion pieces. We give examples from papers where the words 'cultural evolution' and sometimes 'human behavioural ecology' or 'function' as set up as opposing forces, when describing human cultural behaviour, generating confusion about what is actually meant by 'cultural evolution'.

Results: If we are to avoid confusion, the term cultural evolution is best reserved to identify the phenomenon of cultural change. This helps clarify that human behavioural ecologists do not disregard culture, but instead have studied its evolution from the very beginning. Different approaches to the study of human behaviour can coexist and complement each other in the framework offered by Tinbergen's four evolutionary questions.

Conclusions: Clarifying key terms is key to achieving this synthesis.

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Posters 2, Forest Room: 17:45 to 19:30 Wednesday 20.04.2022

Discerning organisational and activation hormonal effects – the case of mood variations across the menstrual cycle

Objective: The organizational–activation hypothesis is a viewpoint that shapes the way we research the brain behaviour relationship (Hampson, 2019). Not all the methods used to measure the fluctuating activation effects and the mostly permanent organisational effects assume the dependency of these effects. Latent state-trait theory (LST; Steyer et al., 2014) is a psychometric and methodological framework that allows us to decompose a measure of behaviour into a component mostly permanent across situations (trait) and another characteristic for a situation (state residual). Hormones are expected to have activation effects on the state residual and organisational effects on the trait and the state residual (as moderators).

Methods: The sample consisted of 301 female students who participated in the research on two occasions on average five months apart. The LST was used to analyse the mood measures (PANAS-X-O), and different components were then related to menstrual cycle phase as an indicator of activation effects and 2D:4D as an indicator of organisational effects.

Results: The positive and negative affect subscales proved to have a trait component, that explained about 20% of the variance, and a situation-specific component, that explained 30% of the variance. The situation-specific components were not related to the menstrual cycle phase, and 2D:4D was not related to the situation-specific nor to the trait component.

Conclusions: The non-significant results may be a result of validity of measures of activational and organizational effects, but using LST to discern organisational and activational hormonal effects shows potential to aid further research on the brain behaviour relationship.

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Posters 2, Blue Room: 17:45 to 19:30 Wednesday 20.04.2022

Who punishes moral transgressions? Aggressive, but not prosocial, individuals are more motivated to punish

Prior theoretical and empirical accounts have emphasized distinct functions of punishment. According to some accounts, punishment represents a key mechanism to promote cooperation and norm compliance. Other accounts, however, emphasize that punishment can be used to compete for status and resources. Based on these potential cooperative and competitive functions, we forward alternative hypotheses regarding the personality antecedents of second- and third-party punishment. To investigate who punishes moral transgressions, we examine how prosocial traits (i.e., prosocial value orientation, high Honesty-Humility) and aggressive traits (i.e., trait aggression, low Agreeableness) relate to direct and indirect punishment. Attempting to reconcile approaches emphasizing the prosocial versus competitive use of punishment, we further examine whether distinct personality traits differentially relate to second- and third-party punishment. We use data from four studies (Ns = 284, 257, 347, 233) conducted with diverse samples (community members, online workers, and students) in the US and the Netherlands. We assess motivations to punish hypothetical and real-life offenses, as well as punishment behaviors in economic games and in field settings. Across contexts, we find no evidence that prosocial traits positively relate to punishment motivations. Aggressive traits consistently relate with stronger motivations to engage in both second- and third-party punishment, via both direct and indirect tactics. However, we find little evidence that either prosocial or aggressive traits relate with actual punishment behaviors in the lab and the field. We conclude that future research should consider how situational factors strengthen or weaken the links between aggressive traits and the punishment of moral transgressions.

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Posters 2, Navy Room: 17:45 to 19:30 Wednesday 20.04.2022

Machine Impostors Impede the Evolution of Successful Conventions

Objective: Artificial language experiments in the past have shown that humans can evolve novel conventions from scratch by interacting repeatedly. Especially online, we are increasingly also interacting with algorithms impersonating humans, often with malicious intent. This danger might be particularly pronounced if machines learn from the past interactions many people freely share online. Will access to the interaction history make impostor bots more deceptive and affect the evolution of shared conventions between humans?

Method: We address these questions in an online experiment combining an artificial language game with a Turing test setup. 200 humans were presented with two simple shapes within a 2D space. Their goal was to interact by moving their shape to find out about their partner's identity, who was either another human or an impostor bot. Bot behavior was manipulated: Bots imitated either a participant's human partner or another human from a different dyad.

Results: Our analyses concerned the performance of dyads and how conventional their interactions became, assessed by the Earth Mover's distance. We predicted and found that access to the interaction history made bot impostors more deceptive. Furthermore, pairs acting more conventionally under this condition were less successful, while the opposite was found when bots imitated a human from another pair.

Conclusions: We show that machine impostors can impede the evolution of successful conventions by accessing the interaction history of humans. This implies increased risk if bots access personal information online and, in this scenario, increased adaptiveness for flexibly updated conventions.

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Posters 2, Forest Room: 17:45 to 19:30 Wednesday 20.04.2022

The evolution of gossip

Objective: Gossip, the exchange of personal information about absent third parties, has long been a significant part of human life. Though numerous theories have elucidated the benefit of gossip for the group, the evolution of gossip remains puzzling from an individual's perspective. With an agent-based model that is tailored to the features and functions of gossip, this paper examines the mechanism of the evolution of gossip.

Methods: We build an agent-based model based on an evolutionary game theoretic framework. In the model, we simulate the interactions among gossipers, non-gossipers, and individuals with different cooperation strategies. By adding and removing different processes from the model step-by-step, we examine the mechanism that is sufficient and necessary for the evolution of gossipers.

Results: We find that gossipers evolve under the joint effect of its information and influence functions. Specifically, gossip makes reputation information more accessible and, thus, leads more people to take into account others' reputation when making cooperation decisions toward them (i.e., the information function of gossip). As more people become reputation sensitive, people are motivated to manage their reputation by behaving more altruistically in front of gossipers (i.e., the influence function of gossip). As a result, gossipers gain an economic advantage over non-gossipers, which leads to the evolution of gossipers. The evolution of gossipers further facilitates the information and influence functions of gossip and sustains the cycle.

Conclusions: The information and influence functions of gossip jointly lead to the evolution of gossip.

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Posters 2, Blue Room: 17:45 to 19:30 Wednesday 20.04.2022

Characterizing the differences in cultural dynamics when the rules for production are transmitted instead of the artifact produced itself

Different types of social learning can underlie cultural transmission. We here seek to characterize the different cultural dynamics that emerge when the rules for producing an artifact are transmitted (« rule-copying »), compared to when the artifact itself is transmitted (« product-copying »). Copying the rule for production should be distinguished from mere imitation, which involves copying the exact sequence of actions. Indeed, the rule should be seen as a simple constraint on the successive actions that an individual should perform in order to produce a new instance of the artifact. Using a transmission chain design, we experimentally compare the effects of these two types of social learning, namely rule-copying and product-copying. Each individual in the chain will have to produce a visually complex artifact whose structure presents a particular statistical property. We expect rule-copying to lead to the production of visually different artifacts, but that share the same statistical property. On the contrary, when the artifact itself is transmitted (product-copying), we expect that visual similarity will slowly decrease along the chain, and that the statistical property will quickly be lost. Additionally, we will use these empirical results to inform a computational model of these two types of social learning. Preliminary results will be presented.

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Posters 2, Navy Room: 17:45 to 19:30 Wednesday 20.04.2022

Why evolutionary psychology should abandon modularity

A debate surrounding modularity—the notion that the mind may be exclusively composed of distinct systems or modules—has held philosophers and psychologists captive for nearly 40 years. Concern about this thesis—which has come to be known as the massive modularity debate—serves as the primary grounds for skepticism of evolutionary psychology's claims about the mind. In this talk we argue that the entirety of this debate, and the very notion of massive modularity itself, is ill-posed and confused. In particular, it is based on a confusion about the level of analysis (or reduction) at which one is approaching the mind. Here we provide a framework for clarifying at what level of analysis one is approaching the mind and explain how a systemic failure to distinguish between different levels of analysis has led to profound misunderstandings of not only evolutionary psychology but also of the entire cognitivist enterprise of approaching the mind at the level of the mechanism. We furthermore suggest that confusions between different levels of analysis are endemic, and that evolutionary social scientists have a unique opportunity to sit at the vanguard of a new movement to resolve these confusions throughout the behavioral sciences.

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Posters 2, Purple Room: 17:45 to 19:30 Wednesday 20.04.2022

Importance of ecological knowledge for foraging success in Pemba, Tanzania

Childhood is a period of life unique to humans. Childhood may have evolved through the need to acquire knowledge and subsistence skills. In an attempt to evaluate the importance of learning for the evolution of childhood, previous research examined the increase with age of returns to foraging. Any increase could be due to increases in knowledge or other factors such as body size and strength. In a population of part time children foragers on the island of Pemba, Tanzania, we collected anthropometric measures such as size and strength (N: 202), estimated knowledge (N: 93) and observed a variety of foraging trips (N trips: 310, N individuals: 116). We estimate the contribution of the different elements to foraging success. Clearly defining the relative importance of somatic versus cognitive traits of forager children provides clues about the selective pressures leading to the evolution of early life history traits in our species.

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Posters 2, Forest Room: 17:45 to 19:30 Wednesday 20.04.2022

Perinatal Testosterone, Estradiol, and Vitamin D as Predictors of Human Handedness

Objective: Although hand preference is a salient characteristic of homo sapiens, its developmental origins and evolutionary functions remain elusive. As handedness emerges in early development, it is plausible that differences in the intrauterine environment play an important role. The Geschwind-Beahan-Galaburda theory and Sexual Differentiation hypothesis predict that elevated prenatal testosterone levels promote left-handedness, whereas the Callosal hypothesis predicts the opposite. Others have proposed roles for estradiol and Vitamin D. As findings and interpretations remain equivocal, this study tested these hypotheses using the largest dataset yet analysed.

Methods: We examined n=767 offspring from a population-based pregnancy cohort (The Raine Study) for whom early biological variables and childhood/adolescent handedness were recorded. We tested whether 18-week maternal circulatory Vitamin D and umbilical cord testosterone and estradiol measured at birth predicted the direction of hand preference (right or left), along with right- and left-hand speed, and the strength and direction of relative hand skill measured by a finger-tapping task at ages 10- and/or 16-years.

Results: Higher concentrations of Vitamin D were associated with reduced rightward relative hand skill at 10-years and with reduced strength of relative hand skill (regardless of direction) at 16-years, though neither effect replicated across time-points. Higher concentrations of estradiol predicted more lateralised relative hand skill at 10-years (but not 16-years), with the effect present for males but not females.

Conclusions: Our results suggest perinatal Vitamin D, testosterone, and estradiol contribute minimally, if at all, to variance in handedness.

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Posters 2, Blue Room: 17:45 to 19:30 Wednesday 20.04.2022

Do Animals Have a Sense of Fairness? A Meta-Analysis

Objective: The notion that animals have a sense of fairness is one of the most influential and hotly contested themes in the behavioral sciences. Beginning with Brosnan and de Waal's (2003) seminal report of inequity aversion in capuchins - operationalized as a refusal to accept a less valuable reward than a conspecific in exchange for the same labor - researchers have published a series of studies attempting to replicate and build on their findings. In species from corvids to chimps, findings seemed to support the idea that a sense of fairness is an ancestral trait. At the same time, the evolutionary logic behind these claims has been questioned, alternative explanations for the observed behavior put forth, and a series of failed replications has been published. Importantly, the studies underlying this literature invariably tested small samples, making meta-analytic integration particularly valuable. The aim of the present study is to provide a definitive account of the current state of the evidence.

Methods: The study is an individual participant data meta-analysis. After creating a list of eligible papers based on preregistered criteria, we requested trial-level data from their authors. The data will be analyzed using a series of mixed-effects logistic regressions.

Conclusion: So far, we have collected trial-level data from 24 primary papers, covering 321 subjects across 18 species - making this study one of the largest animal behavior meta-analyses ever conducted. We are currently finishing compiling the data and will soon run our analyses.

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Posters 2, Navy Room: 17:45 to 19:30 Wednesday 20.04.2022

Effects of sex and appearance on trust and trustworthiness

Objective: The ability to trust other individuals, including strangers, is a prerequisite for human cooperation. However, trust is not indiscriminate. Apparently, humans use subtle cues for judging partners' trustworthiness. Here we report on an experiment addressing the impact of sex and appearance on trust and trustworthiness.

Methods: The subjects in our experiment played a Trust Game with strangers. Under "anonymous" conditions, they did not receive any information on their interaction partner; while under "personalized" conditions they were shown a brief silent video of their partner. The features of facial appearance were investigated with geometric morphometrics based on full-face photographs.

Results: Under anonymous conditions males and females did neither differ in their degree of trust nor trustworthiness. In personalized conditions, females elicited a higher degree of trust in partners of both sexes. The degree of facial femininity increased trust in individuals of both sexes. It turned out that males from the stimulus videos were less trustworthy than the stimulus females, and those stimulus males scoring high on facial femininity were more trustworthy than the other stimulus males. Accordingly, the degree of trust elicited by our stimulus group was justified by the actual trustworthiness of these subjects in the Trust Game.

Conclusions: Our results demonstrate that sex and appearance have a strong effect on eliciting trust in strangers. Femininity, both at the level of sex and facial traits, seems to be an important trust stimulus, which, to a certain extent, is a reliable indicator of trustworthiness.

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Posters 2, Purple Room: 17:45 to 19:30 Wednesday 20.04.2022

How natural disaster affects future discounting: exploring insights from a catastrophic flood among the Tsimane of lowland Bolivia

Individuals subjectively discount future rewards relative to immediate ones, and the rate at which they do so can be expressed by the discount parameter k . Theory predicts that discounting rates should evolve to reflect species-typical life history, as well as be sensitive to current ecology and state of the organism (e.g. age, or nutritional status).

Here we test the prediction that k responds to current conditions among the Tsimane by leveraging an unfortunate naturalistic experiment in the form of a catastrophic flood, which affected >60 villages and resulted in major losses of crops and goods, as well as serious health impacts. Soon after the flood, surveys were conducted in affected villages along with humanitarian relief efforts. To measure k , participants (739 men, 215 women) were given a series of choices between (hypothetical) smaller, immediate rewards and larger, delayed rewards, each choice consistent with different values of k , and subjects' overall k was inferred as the indifference point between immediate and delayed rewards. Overall, subjects' choices were highly consistent (mean consistency >80%), suggesting good internal validity. We found that discount rate was higher (i.e. larger k) among people who were more severely impacted by the flood, as measured by the loss of crops and goods, displacement and family illness. These results suggest that discounting does respond to current conditions and that a greater focus on immediate rewards may be an adaptive response to adversity.

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Posters 2, Forest Room: 17:45 to 19:30 Wednesday 20.04.2022

Do Biased Sex Ratios Enhance the Wish for A Strong Leader?

Objective: An evolutionary perspective to leadership assumes that humans have adaptive followership psychology that responds to specific ecological threats by selecting appropriate leaders to mitigate these risks. Here we investigate the impact of local sex-ratios on leadership

preferences. We hypothesize that a biased sex ratio within a population – having either more men or more women – upregulates the wish for a strong, dominant leader, because followers perceive such situations as threatening as sexual conflict increases.

Methods: In three online scenario studies (total N = 1076), we asked people to vote for a national leader in a mock election of a fictitious country and manipulated the operational sex ratios within the country. We then showed participants the dominant and non-dominant versions of the leader's face and asked them to choose between the two faces. Participants also indicated their explicit preference for a strong leader. Finally, they rated how competitive/threatening/dangerous they thought the situation in the country was.

Results: The results showed that having a male-biased sex ratio increased the wish for a strong leader and this effect was robust across three studies (as indicated by our mini meta-analysis). Participants in the male-biased sex ratio also perceived the country's situation as more dangerous. Although having a female-biased sex ratio also increased the wish for a strong leader in some studies, this effect was not robust and participants did not perceive this situation as more dangerous.

Conclusions: Overall, our studies suggest that in societies with a male-biased operational sex ratio, citizens more strongly support a strong, dominant leader to offer protection and safety as sexual competition between men intensifies.

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Posters 2, Blue Room: 17:45 to 19:30 Wednesday 20.04.2022

Do you need to exercise more if you've been large at birth?

Objectives: Size at birth is influenced by conditions in the womb and genetic factors. These factors are also important predictors of a long-term health. Furthermore, there is an association of size at birth with adult sex hormone levels. It is also known that physical activity affects the level of sex hormones in women. The aim of this study was to investigate whether birth weight, birth length, and the ponderal index (PI) are associated with the change in progesterone levels due to exercise in adult women.

Methods: The participants were 48 healthy, regularly menstruating women aged 21-36 (mean=27.3, SD=4.2). Progesterone was measured in daily saliva samples over two menstrual cycles. During the first menstrual cycle, participants did not perform any physical activity, while during the second menstrual cycle, they participated in aerobic exercise for at least 3 hours a week. Birth measurements were taken from personal birth records. General linear models were used, controlling for participants' age and change of body fat percentage between the two cycles.

Results: Birth weight was associated with the difference in progesterone levels between physically active and non-active menstrual cycles. The higher the birth weight, the greater increase in progesterone levels due to exercise was observed. No other statistically significant relationships were revealed.

Conclusions: Our results indicate that prenatal conditions might be a confounding factors in the relation between change in adult reproductive hormones in response to exercise in women.

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Posters 2, Navy Room: 17:45 to 19:30 Wednesday 20.04.2022

Stability and contingency in folk tune complexity

Objectives: The extent to which cultural change is predictable versus contingent is a controversial question, yet an empirical one. Here I investigate the direction and predictability of recent changes in complexity in Irish folk session tunes, testing three alternative predictions: 1. Tune complexity increases over time due to ratcheting effects 2. Tune complexity decreases over time due to increasing compressibility pressure and 3. Tune complexity remains stable over time due to persistent preferences for intermediate complexity.

Methods: I investigate the extent to which measures of folk tune complexity (e.g. entropy, novelty) are predictable based on the date tunes were uploaded to a large online folk tune collection, thesession.org (n=34,757, spanning the years 2001-2021).

Results: I find no strong, consistent evidence for recent directional trends in complexity among folk session tunes. Although date has 'significant' effects on some complexity measures, effects are generally materially negligible and vary in directionality.

Conclusions: Two main conclusions can be drawn from this work. First, the average complexity of folk tunes uploaded to a large online collection has changed little over the last 20 years despite substantial changes in the dissemination and transmission of music, suggesting that some artistic traditions may be relatively immune to the effects of the digital age. Second, the relationship between tune complexity and time is very noisy, suggesting that even where statistical trends are identifiable, the complexity of a tune is largely determined by contingent factors rather than strong temporal trends.

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Posters 2, Purple Room: 17:45 to 19:30 Wednesday 20.04.2022

The case of metal music: An application of the sexual selection or a cognitive byproduct?

There has been much debate about the ultimate explanation behind cultural displays such as music and art. There are two competing hypotheses that have often been applied to explain the function of music: sexual selection or simply a byproduct of a complex brain. While there is some evidence to suggest that playing music increases male attractiveness, the sexual selection explanation may not be applicable to all types of music. Death metal music is a genre that is heavily male-biased not only among the individuals that play this style of music but also among the fans of the genre. Therefore, it is unlikely that death metal musicians are trying to increase their mating success through their music. However, musicians in this genre heavily invest their time in building technical skills (e.g. dexterity, coordination, timing), which leaves open the question of the purpose behind such investment. One possible explanation is that men engage in this genre for intrasexual competition purposes: to intimidate other males with their technical skills and speed and thus gain status. To examine the competing

hypotheses of sexual selection and intrasexual competition, a sample of metal guitarists were recruited and surveyed on their guitar playing habits, sexual behavior, and feelings of competitiveness toward the same sex. Technical skill negatively predicted both sociosexuality and the lifetime number of sexual partners. However, technical skills also negatively predicted intrasexual competitiveness. This preliminary result did not support either the sexual selection or the intrasexual competition hypothesis. Limitations, alternative explanations, and directions for future research are highlighted.

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Posters 2, Forest Room: 17:45 to 19:30 Wednesday 20.04.2022

COVID-19 and memory: Is there a contamination effect in memory?

Objective: Memory is a cognitive component of the Behavioral Immune System (BIS, Schaller & Park, 2011), a defense system whose function is to protect from pathogen exposure. Recent evidence suggest that contaminated things are remembered better than non-contaminated things (Bonin et al., 2019; Nairne, 2014; Fernandes et al., 2017, 2021). In the present studies, we investigated contamination effects in memory in relation with the COVID-19. A key issue that the studies explored was whether a contamination effect in memory can be obtained when there is no visible signs of disgust.

Methods: In Study 1, photographs of everyday objects were shown in the hands of either a healthy or a contagious person who had contracted the SARS-CoV-2. A free recall task required the participants to write down on a sheet of paper the names of all the objects they remembered. In Study 2, we explored emotions underpinning the COVID-19. We asked participants to estimate the degree to which they felt four primary emotions (fear, disgust, sadness, and anger) when they had to think about the COVID-19.

Results and Conclusion: We found that “contaminated objects” were recalled better than “non contaminated objects” in Study 1, and in Study 2 that thinking about the COVID-19 yielded to a higher (perceived) level of fear than disgust—and also to high levels of anger and sadness—suggesting that disgust is not the sole emotion involved in the BIS, and perhaps, in contamination effects in memory.

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Posters 2, Blue Room: 17:45 to 19:30 Wednesday 20.04.2022

Your white blood cells like you to be physically active

Objective: Physical activity is one of the major modifiable protective factors for many diseases. The immune system protects against pathogens and plays a key role in maintaining health. Previous studies show that the activity of the immune system may depend on the level of physical activity. The aim of the study was to investigate whether increased physical activity is associated with an improvement in the immune system's response among young reproductive age women.

Methods: This study was conducted among 81 urban Polish women aged 21-36 (mean=27.4, SD=4.2). Participants received gym membership cards and took part in fitness classes at least 3 times a week. Physical activity data was collected using FitBit trackers. Blood tests were performed twice (before increasing physical activity and after 2 physically active months).

Results: There was a statistically significant difference ($p=0.028$) in the total number of white blood cells between inactive and active months. In particular, the number of neutrophils increased significantly ($p=0.027$).

Conclusions: In our study, women who increased their physical activity for 2 months improved their immune response as well. Physical activity stimulates the body to produce more white blood cells. This suggest that increased physical activity might result in improved immunity and therefore reduced the risk of developing many diseases.

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Posters 2, Navy Room: 17:45 to 19:30 Wednesday 20.04.2022

Love Language: A Comparison Between Dating App Biographies as an Indicator for Important Criteria in Mate Choice

This research focussed on the biographies of dating apps which are space-limited and self-written and therefor give insight in what users find crucial to mention about themselves and their wishes for possible partners. The research investigated what information men and women aged 18 to 38 looking for a heterogeneous partner shared in their biographies on dating applications Tinder, Bumble and Lexa.

The method relied on biographies which were collected in The Netherlands in 2021 from the three dating apps. The texts were analysed on mentions of intention, lifestyle, personality, and appearance and on usage of person point of view, negation, emojis, and opening greeting. Comparisons were made between men and women, between age 18 to 28 and 28 to 38, and between dating applications.

Results give insight in how content of biographies varies by gender, age, and dating application. One result shows that users of Lexa mention more often that they are looking for a serious relationship, similar to older users overall. The serious intention of Lexa users is also reflected by a majority of extensive biographies. In contrast, around half of Tinder users left their biography empty. Moreover, while all investigated dating applications allowed for photos, Tinder users often put their Instagram handle in their biography, indicating a more appearance-oriented mate preference.

In conclusion, the collected data gives insight in how description of identity and projected partner in dating app biographies can be used as indicator for mate choice criteria.

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Covid 19 and mask-wearing: Direct behavioral observations confirm evolutionary predictions

Objective: The Covid 19 pandemic changed public and private life in many ways. Mask-wearing is among the preventive measures that were governmentally enforced. The current study aimed to quantify to which extent urban pedestrians adhere to mask-wearing. Trade-offs following asymmetric obligatory parental investment (higher risk-taking in men; more risk aversion in women), and human life-history, let us assume that women and elderly persons wear masks more frequently.

Methods: Direct behavioral observations of single persons were conducted in public places, where mask-wearing was mandatory. Data were collected in November and December 2020 in Austria (mostly Vienna). The final sample consisted of 1377 women and 1223 men. Gender and age were categorized by the behavioral coder and associated with mask-wearing via cross-tabulations.

Results: We found that 88.3% of the observed women and 85.8% of the men had covered the nose and mouth with a mask. These percentages differed significantly (Fisher's exact test, $p = 0.031$). Age affected mask-wearing behavior in men only (Chi-Square = 8.96, $p = 0.015$), with men younger than 19 years being more likely (SR = 2.7) not to wear a mask (correctly).

Conclusions: Overall, there was high compliance in places of obligatory mask-wearing in Austria in winter 2020. Consistent with evolutionary predictions, men were less likely to adhere to mask-wearing, especially at a young age. Practical and social implications might include targeted awareness campaigns, such as role models and social media performances.

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Posters 2, Forest Room: 17:45 to 19:30 Wednesday 20.04.2022

Hidden Talents in Context: Cognitive performance with abstract versus ecological stimuli among adversity-exposed youth

Objective: Youth living in adverse conditions tend to score lower on cognitive tests compared to youth in supportive conditions. However, the hidden talents approach proposes that some abilities may be enhanced by adversity, especially under ecologically relevant conditions. For example, adversity-exposed youth might score higher on tests with ecological stimuli relative to those with abstract stimuli. Here, we examine this question by comparing performance across abstract and ecological versions of attention-shifting and working memory tasks.

Methods: We sampled 618 youth (M age = 13.62, SD age = 0.81; 48.22% female; 64.56% White) and administered two versions of an attention shifting and working memory updating task—one with abstract stimuli, one with ecological stimuli. We measured environmental unpredictability, violence, and poverty, and tested adversity $\times$ task version interactions using a multiverse analysis approach.

Results: For attention shifting, all youth performed better with abstract stimuli and exposure to unpredictability and violence predicted better overall performance; however, there were no interactions. For working memory updating, all youth performed better with ecological stimuli but this increase was largest for violence- and poverty- exposed youth. These youth scored lower than their peers with abstract stimuli but just as well with ecological stimuli. We did not find interactions for unpredictability.

Conclusions: These results are striking compared to the backdrop of developmental science, which almost exclusively reports lowered performance among adversity-exposed youth. Although more work is needed, our data suggest ecologically relevant testing materials could help "even the playing field" in education, at least for some abilities.

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Posters 2, Purple Room: 17:45 to 19:30 Wednesday 20.04.2022

Fertility-related changes in the chemical signature of body odour in women

The debate about cyclic variation in physical attractiveness in women and whether this evolved to advertise female fertility is still ongoing. Previous studies showed that the perceived attractiveness of women's body odour varies over the ovulatory cycle, with odours collected during the fertile phase being rated as particularly attractive by men. However, the chemical underpinnings of male preference for the odours of ovulating women remain unknown. We therefore sampled axillary and genital body odour of 15 healthy, naturally cycling women on 10 days within their ovulatory cycle covering their fertile and non-fertile period. The fertile state was hormonally confirmed by urinary ovulation tests as well as salivary estradiol and progesterone determination. Odour samples were collected with thermal desorption tubes, a state-of-the-art method for volatile sampling in mammals, and analysed using gas-chromatography mass-spectrometry. We fitted linear mixed models for each body part with four fertility predictors (calculated per sampling day) i) conception risk, ii&iii) cyclic progesterone and estradiol as well as iv) temporal distance to ovulation. Results show that the chemical composition of odour in both body parts was significantly affected by all fertility predictors, indicating variation in odour substances depending on fertile state and cyclic hormonal fluctuations. This is the first study demonstrating chemical variation over the ovulatory cycle in women, providing valuable evidence to disentangle the function of olfactory cues to female fertility in humans.

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Posters 2, Blue Room: 17:45 to 19:30 Wednesday 20.04.2022

Social support during pregnancy and postpartum depression

Objective: Postpartum depression (PPD) is a relatively common health problem that affects approximately 10-15% of women who gave birth. The symptoms of PPD have severe consequences for child development, woman's health, and family functioning. Social and partner support has been proposed as one of the protective factors moderating the risk of maternal PPD. The aim of this study was to examine the association between social support during pregnancy and the risk of postpartum depression.

Methods: This was a web-based study conducted among population of Polish pregnant women who gave birth during the COVID pandemics. Data from 1073 women aged 18-43 (mean 30.9; SD 3.79), collected both during and after pregnancy were analysed using multinomial logistic regression. The risk of postpartum depression was assessed using the Edinburgh Postpartum Depression Scale (EPDS) and the levels of social support were measured using the Berlin Social Support Scales (BSSS).

Results: Significant associations were found between several dimensions of social support and the score on the EPDS scale after controlling for woman's age, being multiparous, pregnancy complications, and child's Apgar score. These included perceived available emotional support ($p < 0.001$), perceived available instrumental support ($p < 0.001$), currently received emotional support ($p < 0.001$), currently received instrumental support ($p < 0.001$), and currently received informational support ($p < 0.001$).

Conclusions: Our results suggest that the more social support the pregnant woman receives, the lower is the risk of postpartum depression. It may be explained by the fact that humans evolved as cooperative breeders and are inherently reliant on social support to raise children.

STUDENT PRESENTERS

Talks

Alami S

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Single 5: 14:40 Thursday 21.04.2022

What motivates intermarriage in small-scale societies, and what consequences for social cohesion?

Ammar MP, Fogarty L, Kandler A

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Parallel 3A: 11:40 Thursday 21.04.2022

Memory in cultural evolution: Implications for social learning in fluctuating environments

Chen ZC, Kilgallen JA, Urassa MU, Schaffnit SB, Kumogola YK, Lawson DL

University of California, Santa Barbara, zhian@ucsb.edu

Single 7: 15:40 Friday 22.04.2022

What other men want: the accuracy of gender norm perception among Tanzanian men

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Parallel 4A: 11:40 Friday 22.04.2022

Introspection dynamics: A simple model of counterfactual learning in asymmetric games

Engelbert LH, Van Elk M, Kandrik M, Theeuwes J, Van Vugt M

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Single 4: 11:00 Thursday 21.04.2022

The effect of leaders' charismatic signaling on information processing outcomes: Combining cognitive science and evolutionary leadership theory

Faessler L, Efferson C

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Parallel 3A: 12:00 Thursday 21.04.2022

Using successful people as examples of how to behave and how not to behave

Fedorova N, McElreath R

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Single 3: 14:00 Wednesday 20.04.2022

Ger or house: Drivers of household shelter choices and investments in Ulaanbaatar, Mongolia

Foster D, Mesoudi A, Postma E, Lamba S

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Single 1: 16:00 Tuesday 19.04.2022

Systematic review and meta-analysis of the effects of predictors of cooperation on loan repayment in microfinance groups

Haux LM, Engelmann JM, Arslan R, Hertwig R, Herrmann E

Max Planck Institute for Human Development, haux@mpib-berlin.mpg.de

Single 3: 13:40 Wednesday 20.04.2022

Chimpanzee and human risk preferences show key similarities

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Parallel 2B: 15:20 Wednesday 20.04.2022

Dynamics of risk regulates cooperation in an incomplete-information game with bandit payoff feedback

Kandza VH, Jang H, Kiabiya-Ntamboundila F, Boyette AH

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Single 2: 11:00 Wednesday 20.04.2022

Insights into inter-group cooperation and inequality from shotgun hunting among the BaYaka and Yambe in Republic of the Congo

Lidborg LH, Graham R, Brewer G, Boothroyd LG

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Single 1: 15:20 Tuesday 19.04.2022

Do more masculine men have better quality partners?

Martin JS, Beheim B, Gurven M, Kaplan H, Stieglitz J, Trumble B, Jaeggi A

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Single 5: 14:00 Thursday 21.04.2022

Social and ecological effects on the evolvability of human fertility

Minocher R, Ross CT

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Parallel 2A: 15:20 Wednesday 20.04.2022

Sexual conflict and spousal age gaps: a conceptual replication of Lawson et al. (2020) using data from network-structured games in four communities in rural Colombia

Schimmelpfennig R, Razek L, Schnell E, Muthukrishna M

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Parallel 3A: 12:20 Thursday 21.04.2022

Paradox of diversity in the collective brain

Stevens R, Alvergne A, Vitzthum VJ

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Single 6: 10:20 Friday 22.04.2022

Does haemoglobin modulate the trade-off between current and future reproductive investment in the Bolivian altiplano?

Walasek N, Frankenhuys WE, Panchanathan K

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Single 7: 15:20 Friday 22.04.2022

Sensitive periods, but not critical periods, evolve in a fluctuating environment: A model of incremental development

Speedtalks

Hagen R

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Speedtalk 1: 16:08 Thursday 21.04.2022

Understanding women's support for patriarchal norms in East Asia

Kilgallen JA, Lawson DL, Schaffnit SB, Kumogola YK, Chen ZC, Urassa MU

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Speedtalk 1: 16:14 Thursday 21.04.2022

Norm internalization or mismeasurement: Are women truly more accepting of intimate partner violence than men?

Kretschmer LDW, Bann D, Chaudhary N, Salali GD

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Speedtalk 2: 13:52 Friday 22.04.2022

Physical activity and energetic trade-offs in the BaYaka hunter-gatherers

Romani T, Hicks TC

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Speedtalk 2: 14:22 Friday 22.04.2022

The art of nesting: ageing nests of Eastern chimpanzee at Bugoma Central Forest Reserve, Uganda

Shilton D, Passmore S, Savage PE

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Speedtalk 1: 15:44 Thursday 21.04.2022

Group singing is globally dominant, but solo singing reflects social complexity

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Speedtalk 2: 14:16 Friday 22.04.2022

Attention styles in harsh and unpredictable environments

Posters

Basava K, Francois P, Whitehouse H

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Posters 1, Red Room: 17:00 to 19:00 Tuesday 19.04.2022

A cross-cultural study of social complexity and levels of violence in war

Boon-Falleur M, André J-B, Baumard N

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Posters 1, Orange Room: 17:00 to 19:00 Tuesday 19.04.2022

The human capital assessment mechanism and its behavioral implications

Chen Y, Mace R

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Posters 1, Yellow Room: 17:00 to 19:00 Tuesday 19.04.2022

Adult Sex Ratio and dispersal state lead to sex inequality in work burden

Ciochoń A, Marcinkowska MU, Danel D, Galbarczyk A, Apanasewicz-Grzegorzczak A, Klimek M, Mijas M, Ziomkiewicz-Wichary A

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Posters 2, Purple Room: 17:45 to 19:30 Wednesday 20.04.2022

Participation in antenatal classes and the level of anxiety and depression before and after childbirth during COVID-19 pandemic

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Posters 1, Lime Room: 17:00 to 19:00 Tuesday 19.04.2022

The Psychological Profile of Incels (involuntary celibates)

Dubourg E, Thouzeau, V, de Dampierre C, Baumard N

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Posters 2, Forest Room: 17:45 to 19:30 Wednesday 20.04.2022

Exploratory preferences explain the cultural success of imaginary worlds in modern societies

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Posters 1, Red Room: 17:00 to 19:00 Tuesday 19.04.2022

Sound of Success in the Early Paleolithic: Experimental Archaeology of Lithoacoustics

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Posters 2, Blue Room: 17:45 to 19:30 Wednesday 20.04.2022

How infants teach themselves to speak

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Posters 1, Orange Room: 17:00 to 19:00 Tuesday 19.04.2022

Investigating the differential role of early life unpredictability and volatility on human psychopathological development

Fišerová A, Havlíček J, Štěrbová Z

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Posters 1, Yellow Room: 17:00 to 19:00 Tuesday 19.04.2022

Concordance and discordance in intervention behaviour within mate choice reported by parent-offspring dyads

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Posters 2, Purple Room: 17:45 to 19:30 Wednesday 20.04.2022

Moralizing religions as cognitively compelling tools of social control

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Posters 1, Red Room: 17:00 to 19:00 Tuesday 19.04.2022

Creativity across the ovulatory cycle - a within subject study

Zhou L, Ge E, Micheletti AJC, Chen Y, Du J, Mace R

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Posters 2, Blue Room: 17:45 to 19:30 Wednesday 20.04.2022

Monks bring kin-selected benefits and relax sibling competition over parental resources in Tibetan populations

Golden Eddy A, Avalos K, Da Silva Colaco K, Taguchi N, Peissig JJ

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Posters 2, Navy Room: 17:45 to 19:30 Wednesday 20.04.2022

Women's WHR, WSR and BMI Predict Their Youthfulness Ratings

Goodman HCA, Coall D, Fay N, Walker B, Grueter CC

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Posters 1, Yellow Room: 17:00 to 19:00 Tuesday 19.04.2022

Health status moderates preferences for male risk takers as mates

Leslie C, Newman AV, McCarty K

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Posters 2, Navy Room: 17:45 to 19:30 Wednesday 20.04.2022

Perceptions and Reactions to Physical Dominance in Men

Lie-Panis J, Hoffman M, André J-B

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Posters 1, Yellow Room: 17:00 to 19:00 Tuesday 19.04.2022

Cross-contextual reputation and the extent of human cooperation

Malaei F, Sohrabi A, Jahanitabesh A

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Posters 1, Lime Room: 17:00 to 19:00 Tuesday 19.04.2022

An Evolutionary Perspective on Emotion-Cognition Interactions

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Posters 1, Red Room: 17:00 to 19:00 Tuesday 19.04.2022

Building Better Than We Know: Maintenance, Eye Movements, and Social Trust.

Mokos J, Scheuring I

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Posters 1, Orange Room: 17:00 to 19:00 Tuesday 19.04.2022

Running for reputation - how to use your altruistic act to become cool and sexy

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Posters 1, Red Room: 17:00 to 19:00 Tuesday 19.04.2022

Reputation facilitates the co-evolution of cooperation and social rewarding

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Posters 2, Forest Room: 17:45 to 19:30 Wednesday 20.04.2022

The evolution of gossip

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Posters 2, Blue Room: 17:45 to 19:30 Wednesday 20.04.2022

Characterizing the differences in cultural dynamics when the rules for production are transmitted instead of the artifact produced itself

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Posters 2, Purple Room: 17:45 to 19:30 Wednesday 20.04.2022

Importance of ecological knowledge for foraging success in Pemba, Tanzania

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Posters 2, Blue Room: 17:45 to 19:30 Wednesday 20.04.2022

Do Animals Have a Sense of Fairness? A Meta-Analysis

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Posters 1, Orange Room: 17:00 to 19:00 Tuesday 19.04.2022

Cooperation in Simultaneous Interactions

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Posters 2, Navy Room: 17:45 to 19:30 Wednesday 20.04.2022

Effects of sex and appearance on trust and trustworthiness

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Posters 1, Yellow Room: 17:00 to 19:00 Tuesday 19.04.2022

Individual Differences in Content Biases for Danger and Edibility in the Food Domain

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Posters 2, Forest Room: 17:45 to 19:30 Wednesday 20.04.2022

Do Biased Sex Ratios Enhance the Wish for A Strong Leader?

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Posters 1, Red Room: 17:00 to 19:00 Tuesday 19.04.2022

Traces of cooperation in early childhood. Cultural developmental perspective embedded in a processual-relational paradigm

Słojewska K, Galbarczyk A, Klimek M, Miłkowska K, Tubek-Krokosz A, Ścibor M, Jasienska G

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Posters 2, Blue Room: 17:45 to 19:30 Wednesday 20.04.2022

Do you need to exercise more if you've been large at birth?

Stucky K, Gardner A

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Posters 1, Orange Room: 17:00 to 19:00 Tuesday 19.04.2022

The evolution of religiosity by kin selection

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Posters 1, Yellow Room: 17:00 to 19:00 Tuesday 19.04.2022

Women's serial monogamy as an adaptive strategy -A case study of a population in Botswana

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Posters 1, Lime Room: 17:00 to 19:00 Tuesday 19.04.2022

Emergence of word-like phonology from iterated imitations

Tubek-Krokosz A, Galbarczyk A, Klimek M, Miłkowska K, Słojewska K, Ścibor M, Jasienska G

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Posters 2, Blue Room: 17:45 to 19:30 Wednesday 20.04.2022

Your white blood cells like you to be physically active

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Posters 1, Red Room: 17:00 to 19:00 Tuesday 19.04.2022

The role of resource dynamics in variations of the life cycle among female humans

Visser MS

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Posters 2, Navy Room: 17:45 to 19:30 Wednesday 20.04.2022

Love Language: A Comparison Between Dating App Biographies as an Indicator for Important Criteria in Mate Choice

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Posters 1, Yellow Room: 17:00 to 19:00 Tuesday 19.04.2022

Heterogeneity of aggression: The evolution of human aggression types

Zetzsche M, Weiß BM, Kücklich M, Stern J, Birkemeyer C, Penke L, Widdig A

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Posters 2, Purple Room: 12:20 Friday 22.04.2022

Fertility-related changes in the chemical signature of body odour in women

Żyrek J, Klimek M, Apanasewicz A, Ciochoń A, Danel DP, Marcinkowska UM, Mijas M, Ziomkiewicz A, Galbarczyk A

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Posters 2, Blue Room: 17:45 to 19:30 Wednesday 20.04.2022

Social support during pregnancy and postpartum depression